Templates

Application Build

- Backend/API (API)
- Background Jobs & Scheduling (JBS)
- Eventing & Webhooks (EVT)
- Rate Limits & Abuse Controls (RLIM)
- Feature Flags & Config (FFCFG)
- API Pagination/Filtering/Sorting (PFS)
- File Processing & Media Pipelines (FPMP)
- Admin & Internal Tools APIs (ADMIN)
- Frontend/UI Implementation (FE)
- State Management & Data Fetching (SMD)
- Client Performance & Rendering (CPR)
- Forms & Validation (FORM)
- Client Routing & Deep Links (ROUTE)
- UI Composition & Layout Patterns (UICP)
- Client Error Handling & Recovery (CER)
- Client Security (CSec)
- Mobile Implementation (MOB)
- Mobile Device Capabilities (MDC)
- Offline & Sync (OFS)
- Mobile Performance & Battery (MBAT)
- Mobile Deep Links & Universal Links (MDL)
- Push Notifications (MPUSH)
- App Store Release & Signing (SIGN)

Application Build (Definition)

What it is

The Application Build group is the **implementation-facing specification layer**: the concrete contracts, module plans, and build rules needed to implement the product in code across backend, frontend, mobile, and supporting runtime systems (jobs, events, flags, rate limits, media pipelines). It translates architecture + data + design into **build-ready specs**.

What it produces

- API and backend execution specs (endpoints, handlers, services, jobs, events)

- Client implementation specs (routes, state, rendering/perf, forms, error recovery, security)

- Mobile implementation specs (navigation, offline, device features, push, deep links, signing)
- Runtime controls specs (rate limits, feature flags, pagination/filter/sort, webhook/event contracts)
- Admin/internal tool specs (privileged APIs, moderation/support tooling boundaries)

What it is used for (exactly)

- Driving code generation and implementation planning (IMP slices) with deterministic mappings
- Enforcing consistent patterns across modules (platform rules, error models, authz checks)
- Ensuring cross-cutting runtime behaviors (rate limits, flags, retries, validation) are implemented consistently
- Producing testable, verifiable build outputs (contracts and acceptance hooks)

What it is not

- Not product definition (Group 1) and not design deliverables (Group 2)
- Not architecture boundary decisions (Group 3) — assumes ARC is locked
- Not data governance/lifecycle (Group 4) — consumes DATA/DLR/DGL/DQV outputs
- Not ops/reliability runbooks (Group 8) — though it references them

Scope rules

- Must reference upstream truth: PRD feature IDs, DMG terms, ARC boundaries, DATA entities, ERR reason codes, PMAD authz policies
- Must define enforcement points (where validation/authz/rate limits happen in code)
- Must specify “shape” not “implementation code” (code comes later)
- Must include unknown handling and completeness gates per template

Subcategory meaning (one-line each)

- **API**: backend endpoint catalogs/specs, auth/errors, jobs + WS contracts pointer
- **JBS**: background jobs definitions, schedules, retries/DLQ, idempotency
- **EVT**: event schemas, producer/consumer rules, webhooks, delivery guarantees
- **RLIM**: rate limiting + abuse controls across API/realtime/search
- **FFCFG**: feature flag model, config matrix, rollout + kill-switch behavior
- **PFS**: pagination/filtering/sorting contract rules for list/search endpoints
- **FPMP**: file upload/processing, media pipeline stages, storage/transcode rules
- **ADMIN**: privileged/admin/internal APIs, moderation/support tooling endpoints
- **FE**: web client route/layout/component implementation surfaces
- **SMD**: client state architecture + data fetching/caching rules
- **CPR**: client performance budgets, rendering strategy, profiling hooks
- **FORM**: forms, input schemas, validation UX + mapping to DQV/ERR
- **ROUTE**: routing + deep links, route params, navigation contracts
- **UICP**: layout/composition patterns, reusable page structures, UI assembly rules
- **CER**: client error recovery patterns, retries, offline/error UX integration
- **CSec**: client-side security rules (storage, tokens, anti-tamper, CSP where applicable)
- **MOB**: mobile app implementation surfaces (navigation + screens + platform glue)
- **MDC**: device capability usage rules (camera/files/gps/push)
- **OFS**: offline-first rules, sync engine, conflict resolution
- **MBAT**: mobile perf/battery constraints, background work rules
- **MDL**: deep link/universal link rules on mobile

- **MPUSH**: push notification types, routing, permissions, delivery rules
- **SIGN**: app signing, build/release artifacts, store submission constraints

Template Sets

UI Composition & Layout Patterns (UICP)
UICP-01 Page Composition Patterns (shells, sections)
UICP-02 Component Composition Rules (props, slots, variants)
UICP-03 Responsive Layout Patterns (breakpoints usage)
UICP-04 Content Hierarchy Rules (typography, spacing)
UICP-05 Interaction Patterns (modals, drawers, toasts)

Client Error Handling & Recovery (CER)
CER-01 Error Boundary Strategy (global vs local)
CER-02 Retry & Recovery Patterns (per error class)
CER-03 Offline/Error Mode UX (degraded experiences)
CER-04 Session Expiry Handling (re-auth prompts, restore state)
CER-05 Client Logging & Crash Reporting (fields, redaction)

Client Security (CSec)
CSec-01 Token Storage Policy (secure storage, rotation)
CSec-02 Client Data Protection (PII redaction, screenshots, clipboard)
CSec-03 Web Security Headers/CSP (if web)
CSec-04 Anti-Tamper & Integrity Checks (basic)
CSec-05 Secure Deep Link Handling (validation, allowlists)

Mobile Implementation (MOB)
MOB-01 Navigation Map (stacks/tabs/modals)
MOB-02 Screen Implementation Spec (per screen binding)
MOB-03 Native Integration Map (bridges, permissions)
MOB-04 App Lifecycle + State (foreground/background)
MOB-05 Release & Signing (stores, builds)

Mobile Device Capabilities (MDC)
MDC-01 Capabilities Inventory (camera/gps/files/push)
MDC-02 Permissions UX Rules (prompts, denial handling)
MDC-03 Capability Security Rules (least privilege)
MDC-04 Capability Failure Handling (fallbacks)
MDC-05 Telemetry for Capabilities (errors, latency)

Offline & Sync (OFS)
OFS-01 Offline Scope (what works offline)
OFS-02 Sync Model (queues, conflict resolution)
OFS-03 Local Storage Schema (entities cached locally)
OFS-04 Reconciliation Rules (merge, LWW, prompts)
OFS-05 Sync Observability (stuck queues, retries, metrics)

Mobile Performance & Battery (MBAT)
MBAT-01 Background Work Rules (limits, schedules)
MBAT-02 Network Usage Policy (batching, retries, metered)
MBAT-03 Battery Budget & Constraints (targets)
MBAT-04 Perf Profiling Plan (mobile tooling)

Mobile Deep Links & Universal Links (MDL)
MDL-01 Link Scheme & Domains (app links, universal links)
MDL-02 Routing Rules (link → screen/action mapping)
MDL-03 Auth Gating & Safety (signed links, validation)
MDL-04 Fallback Behavior (not installed, unknown routes)

Push Notifications (MPUSH)
MPUSH-01 Notification Types Catalog (by notif_id)
MPUSH-02 Payload Contract (fields, routing, localization keys)
MPUSH-03 Permission & Opt-In Rules (platform specific)
MPUSH-04 Delivery/Retry Rules (quiet hours, collapse keys)
MPUSH-05 Deep Link Routing (notif tap behavior)
MPUSH-06 Abuse/Spam Controls for Notifications

App Store Release & Signing (SIGN)
SIGN-01 Build Artifact Types (debug/beta/prod)
SIGN-02 Signing Keys & Rotation Policy
SIGN-03 Store Submission Checklist (iOS/Android)
SIGN-04 Versioning Rules (build numbers, semver mapping)
SIGN-05 Release Channel Policy (internal/beta/GA)

UI Composition & Layout Patterns (UICP)

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UICP-01 Page Composition Patterns (shells, sections)
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UICP-03 Responsive Layout Patterns (breakpoints usage)
UICP-04 Content Hierarchy Rules (typography, spacing)
UICP-05 Interaction Patterns (modals, drawers, toasts)

UICP-01

UICP-01 — Page Composition Patterns (shells, sections)
Header Block
template_id: UICP-01
title: Page Composition Patterns (shells, sections)
type: ui_composition_page_patterns
template_version: 1.0.0
output_path: 10_app/ui_composition/UICP-01_Page_Composition_Patterns.md
compliance_gate_id: TMP-05.PRIMARY.UICP
upstream_dependencies: ["FE-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FE-01", "UICP-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical page composition patterns used across the app: layout shells, section structures, standard regions, and how pages compose components into consistent, reusable patterns. This template must be consistent with route/layout specs and must not invent page pattern IDs not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FE-01 Route Map + Layout Spec: {{fe.route_layout}} | OPTIONAL
UICP-04 Content Hierarchy Rules: {{ui.content_hierarchy}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Page pattern registry (pattern_id list)
Shell types supported (app/auth/admin/modal)
Standard regions per shell (header/nav/content/footer)
Section types (hero, list, details, form, empty/error sections)
Composition rules (ordering, nesting)
Spacing/padding rules binding (UICP-04)
Responsive composition rules
Accessibility considerations for layout regions
Examples/pattern usage rules

Optional Fields
Per-domain pattern variants | OPTIONAL
Animation/motion guidance | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not introduce new pattern_ids; use only {{spec.ui_patterns_by_id}} if present, else mark UNKNOWN.
Patterns MUST be reusable and consistent with FE-01 shells.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Shell Types
   shells_supported: {{shells.supported}}
   regions_by_shell: {{shells.regions_by_shell}} | OPTIONAL
2. Page Pattern Registry
   Pattern
   pattern_id: {{patterns[0].pattern_id}}
   name: {{patterns[0].name}}
   shell_id: {{patterns[0].shell_id}} | OPTIONAL
   description: {{patterns[0].description}}
   regions_used: {{patterns[0].regions_used}}
   sections (ordered):
   {{patterns[0].sections[0]}}
   {{patterns[0].sections[1]}} | OPTIONAL
   responsive_rules: {{patterns[0].responsive_rules}} | OPTIONAL
   a11y_notes: {{patterns[0].a11y_notes}} | OPTIONAL
   usage_rules: {{patterns[0].usage_rules}} | OPTIONAL
   open_questions:
   {{patterns[0].open_questions[0]}} | OPTIONAL

(Repeat per pattern_id.)

3. Section Types
   section_types: {{sections.types}}
   section_nesting_rules: {{sections.nesting_rules}} | OPTIONAL
4. Composition Rules
   ordering_rules: {{compose.ordering_rules}}
   nesting_rules: {{compose.nesting_rules}} | OPTIONAL
5. Spacing & Responsive Binding
   spacing_ref: {{spacing.spacing_ref}} (expected: {{xref:UICP-04}}) | OPTIONAL
   padding_rules: {{spacing.padding_rules}} | OPTIONAL
6. References
   Route/layout shells: {{xref:FE-01}} | OPTIONAL
   Content hierarchy: {{xref:UICP-04}} | OPTIONAL

Cross-References
Upstream: {{xref:FE-01}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:UICP-02}}, {{xref:UICP-03}}, {{xref:UICP-05}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for pattern registry if not provided; define shell/region baseline.
intermediate: Required. Define sections and ordering rules and responsive guidelines.
advanced: Required. Add a11y notes and per-domain pattern variants where applicable.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, regions_by_shell, shell_id, section examples, responsive rules, a11y notes, usage rules, section nesting, compose nesting, spacing/padding rules, per-domain variants, motion guidance, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If shells.supported is UNKNOWN → block Completeness Gate.
If patterns registry is UNKNOWN → block Completeness Gate.
If compose.ordering_rules is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.UICP
Pass conditions:
required_fields_present == true
shells_and_regions_defined == true
pattern_registry_defined == true
composition_rules_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

UICP-02

UICP-02 — Component Composition Rules (props, slots, variants)
Header Block
template_id: UICP-02
title: Component Composition Rules (props, slots, variants)
type: ui_composition_component_rules
template_version: 1.0.0
output_path: 10_app/ui_composition/UICP-02_Component_Composition_Rules.md
compliance_gate_id: TMP-05.PRIMARY.UICP
upstream_dependencies: ["UICP-01", "FE-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "UICP-01", "FE-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical rules for composing UI components: how props are passed, how slots/children are used, how variants are selected, how to avoid prop drilling, and how composition patterns enforce consistency across the UI. This template must be consistent with component implementation specs and must not invent composition primitives not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
UICP-01 Page Composition Patterns: {{ui.page_patterns}} | OPTIONAL
FE-02 Component Implementation Specs: {{fe.component_specs}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Composition principles (reusability, separation of concerns)
Props rules (required/optional/default)
Slots/children rules (when used, constraints)
Variant selection rules (how variants chosen)
State ownership rules (where state lives)
Data dependency boundaries (components vs containers)
Styling boundaries (token usage, class composition)
Event/callback patterns (naming, payload)
Anti-patterns list (what not to do)
Testing considerations (component tests vs integration)

Optional Fields
Performance notes (memoization) | OPTIONAL

Accessibility composition notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Component composition MUST not break a11y requirements.
Variants MUST be selected deterministically (no “magic” branching).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Principles
   principles:
   {{principles[0]}}
   {{principles[1]}}
   {{principles[2]}}
2. Props Rules
   required_props_rule: {{props.required_rule}}
   defaults_rule: {{props.defaults_rule}} | OPTIONAL
   prop_naming_rule: {{props.naming_rule}} | OPTIONAL
3. Slots / Children Rules
   slots_supported: {{slots.supported}}
   slot_contract_rule: {{slots.contract_rule}}
   children_usage_rule: {{slots.children_usage_rule}} | OPTIONAL
4. Variant Selection
   variant_selection_rule: {{variants.selection_rule}}
   variant_inputs: {{variants.inputs}} | OPTIONAL
   fallback_variant_rule: {{variants.fallback_rule}} | OPTIONAL
5. State Ownership
   state_in_container_rule: {{state.container_rule}}
   state_in_component_rule: {{state.component_rule}} | OPTIONAL
   derived_state_rule: {{state.derived_rule}} | OPTIONAL
6. Data Dependency Boundaries
   container_components_rule: {{data.container_rule}}
   presentational_components_rule: {{data.presentational_rule}}
   hook_usage_rule: {{data.hook_usage_rule}} | OPTIONAL
7. Styling Boundaries
   token_only_rule: {{style.token_only_rule}}
   class_composition_rule: {{style.class_composition_rule}} | OPTIONAL
   no_inline_styles_rule: {{style.no_inline_styles_rule}} | OPTIONAL
8. Events / Callbacks
   event_naming_rule: {{events.naming_rule}}
   payload_shape_rule: {{events.payload_rule}} | OPTIONAL

9. Anti-Patterns
   anti_patterns:
   {{anti[0]}}
   {{anti[1]}} | OPTIONAL
10. Testing
    unit_test_rule: {{tests.unit_test_rule}}
    integration_test_rule: {{tests.integration_test_rule}} | OPTIONAL
11. References
    Page patterns: {{xref:UICP-01}} | OPTIONAL
    Component specs: {{xref:FE-02}} | OPTIONAL
    Theming/tokens: {{xref:FE-06}} | OPTIONAL

Cross-References
Upstream: {{xref:UICP-01}} | OPTIONAL, {{xref:FE-02}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:UICP-03}}, {{xref:UICP-05}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-A11Y]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define principles + prop/variant basics; use UNKNOWN for perf notes.
intermediate: Required. Define state/data boundaries and anti-patterns.
advanced: Required. Add memoization rules and accessibility composition notes.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, defaults/naming rules, children usage, variant inputs/fallback, state component/derived rules, hook usage rule, class composition/no inline styles, payload shape, integration tests, performance notes, a11y notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If principles list is UNKNOWN → block Completeness Gate.
If variants.selection_rule is UNKNOWN → block Completeness Gate.
If style.token_only_rule is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.UICP
Pass conditions:
required_fields_present == true
principles_defined == true
variant_selection_defined == true
styling_boundaries_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

UICP-03

UICP-03 — Responsive Layout Patterns (breakpoints usage)
Header Block
template_id: UICP-03
title: Responsive Layout Patterns (breakpoints usage)
type: ui_composition_responsive_layout_patterns
template_version: 1.0.0
output_path: 10_app/ui_composition/UICP-03_Responsive_Layout_Patterns.md
compliance_gate_id: TMP-05.PRIMARY.UICP
upstream_dependencies: ["UICP-01", "FE-06"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "UICP-01", "FE-06"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical responsive layout patterns used across the app, including breakpoint definitions, grid rules, container widths, and responsive behavior for common components and page sections. This template must be consistent with theming/tokens and page composition patterns and must not invent breakpoints not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
UICP-01 Page Composition Patterns: {{ui.page_patterns}} | OPTIONAL
FE-06 Theming/Tokens Integration: {{ui.theming}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Breakpoint set (names + widths)
Layout primitives (grid/flex/stack)
Container width rules (max width by breakpoint)
Spacing scaling rules (token binding)
Responsive typography rules (token binding)
Responsive navigation rules (sidebar → drawer, etc.)
Component responsive patterns (cards, lists, tables, forms)
Testing guidance (how to verify responsiveness)

Optional Fields
Orientation rules (portrait/landscape) | OPTIONAL
Platform differences (web vs mobile) | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
All responsive rules MUST use design tokens and avoid raw values unless allowed.
Breakpoints MUST be a fixed, shared set across the app.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Breakpoints
   breakpoints:
- name: {{bp[0].name}}
  min_width_px: {{bp[0].min_width_px}}
- name: {{bp[1].name}} | OPTIONAL
  min_width_px: {{bp[1].min_width_px}} | OPTIONAL
2. Layout Primitives
   primitives: {{layout.primitives}}
   grid_rules: {{layout.grid_rules}} | OPTIONAL
   stack_rules: {{layout.stack_rules}} | OPTIONAL
3. Container Rules
   max_width_by_bp: {{container.max_width_by_bp}}
   gutter_rules: {{container.gutter_rules}} | OPTIONAL
4. Spacing Scaling
   spacing_tokens_ref: {{spacing.tokens_ref}} (expected: {{xref:FE-06}}) | OPTIONAL
   padding_scaling_rule: {{spacing.padding_scaling_rule}}
   gap_scaling_rule: {{spacing.gap_scaling_rule}} | OPTIONAL
5. Typography Scaling
   typography_tokens_ref: {{type.tokens_ref}} (expected: {{xref:FE-06}}) | OPTIONAL
   heading_scaling_rule: {{type.heading_scaling_rule}}
   body_scaling_rule: {{type.body_scaling_rule}} | OPTIONAL
6. Responsive Navigation
   nav_responsive_rules: {{nav.rules}}
   sidebar_to_drawer_rule: {{nav.sidebar_to_drawer_rule}} | OPTIONAL
7. Component Patterns
   cards: {{components.cards}}
   lists: {{components.lists}}
   tables: {{components.tables}} | OPTIONAL
   forms: {{components.forms}} | OPTIONAL
8. Testing Guidance
   test_checklist:
   {{tests.checklist[0]}}
   {{tests.checklist[1]}} | OPTIONAL
9. References
   Page patterns: {{xref:UICP-01}} | OPTIONAL
   Theming/tokens: {{xref:FE-06}} | OPTIONAL

Cross-References
Upstream: {{xref:UICP-01}} | OPTIONAL, {{xref:FE-06}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:FE-02}}, {{xref:FE-05}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define breakpoint set and baseline layout primitives; use UNKNOWN for tables/forms if not present.
intermediate: Required. Define container + spacing + nav responsive rules.
advanced: Required. Add platform/orientation differences and robust testing guidance.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, grid/stack rules, gutter rules, token refs, gap/body scaling, sidebar/drawer, tables/forms patterns, test checklist items, orientation/platform notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If breakpoint set is UNKNOWN → block Completeness Gate.
If padding_scaling_rule is UNKNOWN → block Completeness Gate.
If tests.checklist is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.UICP
Pass conditions:
required_fields_present == true
breakpoints_defined == true
token_bound_spacing_and_typography_defined == true
responsive_nav_rules_defined == true
testing_guidance_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

UICP-04

UICP-04 — Content Hierarchy Rules (typography, spacing)
Header Block
template_id: UICP-04
title: Content Hierarchy Rules (typography, spacing)
type: ui_content_hierarchy_rules
template_version: 1.0.0
output_path: 10_app/ui_composition/UICP-04_Content_Hierarchy_Rules.md
compliance_gate_id: TMP-05.PRIMARY.UICP
upstream_dependencies: ["FE-06", "DSYS-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FE-06", "DSYS-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical content hierarchy rules for UI: typography levels, spacing rhythm, density rules, and how to enforce consistent visual hierarchy across screens and components. This template must be consistent with design tokens/theming and must not invent token names not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FE-06 Theming/Tokens Integration: {{ui.theming}} | OPTIONAL
DSYS-01 Tokens Catalog: {{dsys.tokens_catalog}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Typography scale levels (H1/H2/H3/body/caption)
Token mapping per level (font size/weight/line height)
Spacing scale (token-based)
Vertical rhythm rules (stack spacing)
Section hierarchy rules (headline → content → actions)
Density rules (compact/comfortable)
Iconography sizing rules (if applicable)
Accessibility considerations (readability, line length)
Examples/usage rules

Optional Fields
Per-platform typography adjustments | OPTIONAL
Localization considerations (long strings) | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Use tokens only; no raw typography/spacing values unless explicitly allowed.
Hierarchy rules MUST be applied consistently across UICP patterns and FE components.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Typography Levels
   Level
   name: {{type.levels[0].name}} (H1/H2/body/UNKNOWN)
   token_font_size: {{type.levels[0].token_font_size}}
   token_font_weight: {{type.levels[0].token_font_weight}} | OPTIONAL
   token_line_height: {{type.levels[0].token_line_height}} | OPTIONAL
   usage: {{type.levels[0].usage}} | OPTIONAL

(Repeat per level.)

2. Spacing Scale
   spacing_tokens: {{spacing.tokens}}
   stack_spacing_rules: {{spacing.stack_rules}}
   section_padding_rules: {{spacing.section_padding_rules}} | OPTIONAL
3. Vertical Rhythm
   default_gap_token: {{rhythm.default_gap_token}}
   heading_to_body_gap: {{rhythm.heading_to_body_gap}} | OPTIONAL
   body_to_actions_gap: {{rhythm.body_to_actions_gap}} | OPTIONAL
4. Section Hierarchy
   headline_rule: {{sections.headline_rule}}
   content_rule: {{sections.content_rule}}
   actions_rule: {{sections.actions_rule}} | OPTIONAL
5. Density Rules
   modes: {{density.modes}} (compact/comfortable/UNKNOWN)
   mode_rules: {{density.rules}} | OPTIONAL
6. Iconography
   icon_size_tokens: {{icons.size_tokens}} | OPTIONAL
   alignment_rules: {{icons.alignment_rules}} | OPTIONAL
7. Accessibility Considerations
   line_length_rule: {{a11y.line_length_rule}} | OPTIONAL
   min_font_size_rule: {{a11y.min_font_size_rule}} | OPTIONAL
8. Examples / Usage
   examples:
   {{examples[0]}}
   {{examples[1]}} | OPTIONAL

9. References
   Theming/tokens integration: {{xref:FE-06}} | OPTIONAL
   Tokens catalog: {{xref:DSYS-01}} | OPTIONAL

Cross-References
Upstream: {{xref:FE-06}} | OPTIONAL, {{xref:DSYS-01}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:UICP-01}}, {{xref:FE-02}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-A11Y]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define typography + spacing scales; use UNKNOWN for optional iconography rules.
intermediate: Required. Bind levels to tokens and define rhythm rules.
advanced: Required. Add density modes and localization adjustments.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, usage, font weight/line height tokens, section padding rules, rhythm gaps, actions rule, density rules, icon tokens/alignment, a11y rules, examples, platform/localization notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If type.levels list is UNKNOWN → block Completeness Gate.
If spacing.tokens is UNKNOWN → block Completeness Gate.
If spacing.stack_rules is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.UICP
Pass conditions:
required_fields_present == true
typography_levels_defined == true
spacing_scale_defined == true
rhythm_rules_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

UICP-05

UICP-05 — Interaction Patterns (modals, drawers, toasts)
Header Block
template_id: UICP-05
title: Interaction Patterns (modals, drawers, toasts)
type: ui_interaction_patterns
template_version: 1.0.0
output_path: 10_app/ui_composition/UICP-05_Interaction_Patterns.md
compliance_gate_id: TMP-05.PRIMARY.UICP
upstream_dependencies: ["UICP-01", "FE-03", "FE-07", "ROUTE-05"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "UICP-01", "FE-03", "FE-07", "ROUTE-05", "FE-05"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical interaction patterns used across the UI: modals, drawers, toasts, tooltips, confirmations, and other transient UI. Includes behavior rules, accessibility rules, back/escape handling, and when to use each pattern. This template must be consistent with state/error models and routing back/history rules and must not invent interaction primitives not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
UICP-01 Page Composition Patterns: {{ui.page_patterns}} | OPTIONAL
FE-03 UI State Model: {{fe.state_model}} | OPTIONAL
FE-07 Error Handling UX: {{fe.error_ux}} | OPTIONAL
ROUTE-05 Back/History Rules: {{route.back_rules}} | OPTIONAL
FE-05 Accessibility Notes: {{fe.a11y_notes}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Pattern registry (pattern types used)
When to use each (modal vs drawer vs page)
Open/close rules (escape/back, click outside)
Focus management rules (trap, return focus)
Stacking rules (multiple modals)
Toast rules (duration, max stack, placement)
Confirmation patterns (destructive actions)
Error surface integration (bind to FE-07)
Accessibility rules for each pattern
Telemetry requirements (optional)

Optional Fields
Animation/motion rules | OPTIONAL
Mobile-specific gesture rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Interaction patterns MUST follow back/history rules ({{xref:ROUTE-05}}).
Focus trapping/return MUST follow a11y rules ({{xref:FE-05}}).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Pattern Registry
   patterns_used: {{patterns.used}} (modal/drawer/toast/tooltip/confirm/UNKNOWN)
2. Usage Rules
   modal_when: {{usage.modal_when}}
   drawer_when: {{usage.drawer_when}}
   toast_when: {{usage.toast_when}}
   page_when: {{usage.page_when}} | OPTIONAL
3. Open/Close Rules
   escape_closes: {{openclose.escape_closes}}
   back_closes: {{openclose.back_closes}} | OPTIONAL
   click_outside_closes: {{openclose.click_outside_closes}} | OPTIONAL
   close_button_required: {{openclose.close_button_required}} | OPTIONAL
4. Focus Management
   focus_trap_required: {{focus.trap_required}}
   initial_focus_rule: {{focus.initial_focus_rule}} | OPTIONAL
   return_focus_rule: {{focus.return_focus_rule}}
5. Stacking Rules
   multiple_modals_allowed: {{stacking.multiple_modals_allowed}}
   stacking_limit: {{stacking.limit}} | OPTIONAL
   z_index_rules: {{stacking.z_index_rules}} | OPTIONAL
6. Toast Rules
   max_toasts: {{toast.max}}
   duration_ms: {{toast.duration_ms}}
   placement: {{toast.placement}} | OPTIONAL
   dedupe_rule: {{toast.dedupe_rule}} | OPTIONAL
7. Confirmation Patterns
   confirm_required_for: {{confirm.required_for}}
   confirm_copy_policy: {{confirm.copy_policy}} | OPTIONAL
   danger_style_rule: {{confirm.danger_style_rule}} | OPTIONAL
8. Error Surface Integration
   error_surface_ref: {{errors.surface_ref}} (expected: {{xref:FE-07}}) | OPTIONAL

9. Accessibility Rules
   aria_roles: {{a11y.aria_roles}}
   announce_toasts: {{a11y.announce_toasts}} | OPTIONAL
   reduced_motion_support: {{a11y.reduced_motion_support}} | OPTIONAL
10. Telemetry (Optional)
    interaction_event_metric: {{telemetry.event_metric}} | OPTIONAL
    fields: {{telemetry.fields}} | OPTIONAL
11. References
    Page patterns: {{xref:UICP-01}} | OPTIONAL
    UI state model: {{xref:FE-03}} | OPTIONAL
    Error UX: {{xref:FE-07}} | OPTIONAL
    Back/history rules: {{xref:ROUTE-05}} | OPTIONAL
    Accessibility notes: {{xref:FE-05}} | OPTIONAL

Cross-References
Upstream: {{xref:UICP-01}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:FE-02}}, {{xref:ROUTE-06}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-A11Y]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define when-to-use and close rules; use UNKNOWN for telemetry/animations.
intermediate: Required. Define focus/stacking/toast limits and error integration.
advanced: Required. Add reduced-motion and mobile gesture rules with measurable constraints.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, page_when, back/click-outside/close button rules, focus initial rule, stacking limits/z-index, toast placement/dedupe, confirm copy/danger style, error surface ref, a11y announce/reduced motion, telemetry fields, animation/gesture notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If patterns.used is UNKNOWN → block Completeness Gate.
If focus.trap_required is UNKNOWN → block Completeness Gate.
If toast.max or toast.duration_ms is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.UICP
Pass conditions:
required_fields_present == true
patterns_and_usage_defined == true
open_close_rules_defined == true
a11y_rules_defined == true

placeholder_resolution == true
no_unapproved_unknowns == true

Client Error Handling & Recovery (CER)

Client Error Handling & Recovery (CER)
CER-01 Error Boundary Strategy (global vs local)
CER-02 Retry & Recovery Patterns (per error class)
CER-03 Offline/Error Mode UX (degraded experiences)
CER-04 Session Expiry Handling (re-auth prompts, restore state)
CER-05 Client Logging & Crash Reporting (fields, redaction)

CER-01

CER-01 — Error Boundary Strategy (global vs local)
Header Block
template_id: CER-01
title: Error Boundary Strategy (global vs local)
type: client_error_boundary_strategy
template_version: 1.0.0
output_path: 10_app/client_error/CER-01_Error_Boundary_Strategy.md
compliance_gate_id: TMP-05.PRIMARY.CER
upstream_dependencies: ["FE-01", "FE-07"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FE-01", "FE-07"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical error boundary strategy for the client: where boundaries exist (global vs per-route vs per-component), what fallback UIs are used, what gets logged, and how recovery/reset works. This template must be consistent with the error handling UX rules and must not invent boundary scopes not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FE-01 Route Map + Layout Spec: {{fe.route_layout}} | OPTIONAL
FE-07 Error Handling UX: {{fe.error_ux}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Boundary layers (global/app, shell, route, component)
Placement rules (where boundaries must be applied)
Fallback UI rules (what is shown at each layer)
Reset/retry rules (how user recovers)
Logging rules (what to log on boundary hit)
User messaging/copy policy (safe messaging)
Telemetry requirements (boundary hit metrics)
Escalation/reporting pathway (optional)

Optional Fields
Per-route override rules | OPTIONAL
Dev-mode diagnostics | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Error boundaries MUST not leak sensitive details in UI.
Fallback UIs MUST align with {{xref:FE-07}} policies.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Boundary Layers
   global_boundary: {{layers.global_boundary}}
   shell_boundaries: {{layers.shell_boundaries}} | OPTIONAL
   route_boundaries: {{layers.route_boundaries}} | OPTIONAL
   component_boundaries: {{layers.component_boundaries}} | OPTIONAL
2. Placement Rules
   global_placement_rule: {{placement.global_rule}}
   route_placement_rule: {{placement.route_rule}}
   component_placement_rule: {{placement.component_rule}} | OPTIONAL
3. Fallback UI Rules
   global_fallback_ui: {{fallback.global_ui}}
   route_fallback_ui: {{fallback.route_ui}} | OPTIONAL
   component_fallback_ui: {{fallback.component_ui}} | OPTIONAL
4. Reset / Recovery
   reset_supported: {{recovery.reset_supported}}
   reset_mechanism: {{recovery.reset_mechanism}}
   (reload/reset_state/navigate_home/UNKNOWN)
   user_actions: {{recovery.user_actions}} | OPTIONAL
5. Logging Rules
   log_required_fields: {{logs.required_fields}}
   redaction_policy: {{logs.redaction_policy}} | OPTIONAL
6. Messaging / Copy Policy
   copy_policy: {{copy.policy}}
   no_stacktrace_rule: {{copy.no_stacktrace_rule}} | OPTIONAL
7. Telemetry
   boundary_hit_metric: {{telemetry.boundary_hit_metric}}
   fields: {{telemetry.fields}} | OPTIONAL
8. Reporting / Escalation (Optional)
   report_button_supported: {{report.supported}} | OPTIONAL
   report_payload_fields: {{report.payload_fields}} | OPTIONAL
9. References
   Error UX: {{xref:FE-07}} | OPTIONAL
   Retry/recovery patterns: {{xref:CER-02}} | OPTIONAL
   Client logging/crash reporting: {{xref:CER-05}} | OPTIONAL

Cross-References
Upstream: {{xref:FE-01}} | OPTIONAL, {{xref:FE-07}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:CER-02}}, {{xref:CER-05}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-SECURITY]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define global boundary + fallback UI; use UNKNOWN for reporting/diagnostics.
intermediate: Required. Define placement rules and reset mechanism and telemetry metric.
advanced: Required. Add per-route overrides and strict redaction/logging policy.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, shell/route/component boundaries, component placement, route/component fallback, user actions, redaction policy, no stacktrace rule, telemetry fields, reporting fields, per-route overrides, dev diagnostics, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If layers.global_boundary is UNKNOWN → block Completeness Gate.
If fallback.global_ui is UNKNOWN → block Completeness Gate.
If telemetry.boundary_hit_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.CER
Pass conditions:
required_fields_present == true
boundary_layers_defined == true
fallback_rules_defined == true
logging_and_telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

CER-02

CER-02 — Retry & Recovery Patterns (per error class)
Header Block
template_id: CER-02
title: Retry & Recovery Patterns (per error class)
type: client_retry_recovery_patterns
template_version: 1.0.0
output_path: 10_app/client_error/CER-02_Retry_Recovery_Patterns.md
compliance_gate_id: TMP-05.PRIMARY.CER
upstream_dependencies: ["API-03", "FE-07", "RLIM-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "API-03", "FE-07", "RLIM-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical retry and recovery patterns used by the client, mapped per error class (network, 5xx, 429, timeouts, auth expiry, conflicts). Includes backoff rules, retry limits, user prompts, and safe recovery actions. This template must be consistent with API error policy and rate limit guidance and must not invent retry behaviors not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
API-03 Error & Status Code Policy: {{api.error_policy}} | OPTIONAL
FE-07 Error UX: {{fe.error_ux}} | OPTIONAL
RLIM-01 Rate Limit Policy: {{rlim.policy}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Error classes covered (network, timeout, 5xx, 429, 401, 403, 409)
Retry eligibility rules (which errors retry)
Backoff policy (exponential/jitter/UNKNOWN)
Max retry attempts (by class)
Retry-after handling (429)
User prompt rules (when to show retry vs auto)
Recovery actions (re-auth, refresh, navigate away)
Circuit breaker rules (stop retrying)
Telemetry requirements (retry counts, recovery outcomes)

Optional Fields
Per-endpoint overrides | OPTIONAL
Background retry rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Rate limit handling MUST follow {{xref:RLIM-01}} guidance.
Auth expiry handling MUST bind to {{xref:CER-04}} when present.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Error Classes
   classes:
   network: {{classes.network}}
   timeout: {{classes.timeout}}
   server_5xx: {{classes.server_5xx}}
   rate_limit_429: {{classes.rate_limit_429}}
   auth_401: {{classes.auth_401}}
   forbidden_403: {{classes.forbidden_403}}
   conflict_409: {{classes.conflict_409}}
2. Retry Eligibility
   retryable_classes: {{retry.retryable_classes}}
   non_retryable_classes: {{retry.non_retryable_classes}} | OPTIONAL
   retry_conditions: {{retry.conditions}} | OPTIONAL
3. Backoff Policy
   policy: {{backoff.policy}} (exponential/jitter/UNKNOWN)
   base_delay_ms: {{backoff.base_delay_ms}} | OPTIONAL
   max_delay_ms: {{backoff.max_delay_ms}} | OPTIONAL
4. Max Attempts (by Class)
   network_max_attempts: {{attempts.network}}
   timeout_max_attempts: {{attempts.timeout}} | OPTIONAL
   server_5xx_max_attempts: {{attempts.server_5xx}}
   rate_limit_429_max_attempts: {{attempts.rate_limit_429}}
   auth_401_max_attempts: {{attempts.auth_401}} | OPTIONAL
5. 429 Retry-After Handling
   retry_after_respected: {{rate.retry_after_respected}}
   header_name: {{rate.header_name}} | OPTIONAL
   fallback_delay_ms: {{rate.fallback_delay_ms}} | OPTIONAL
6. User Prompts
   auto_retry_rule: {{ux.auto_retry_rule}}
   show_retry_cta_rule: {{ux.show_retry_cta_rule}}
   show_offline_ui_rule: {{ux.show_offline_ui_rule}} | OPTIONAL
7. Recovery Actions
   re_auth_action: {{recovery.re_auth_action}}
   refresh_action: {{recovery.refresh_action}} | OPTIONAL
   navigate_away_action: {{recovery.navigate_away_action}} | OPTIONAL

8. Circuit Breaker
   breaker_supported: {{breaker.supported}}
   breaker_trigger: {{breaker.trigger}} | OPTIONAL
   cooldown_ms: {{breaker.cooldown_ms}} | OPTIONAL
9. Telemetry
   retry_attempt_metric: {{telemetry.retry_attempt_metric}}
   recovery_outcome_metric: {{telemetry.recovery_outcome_metric}} | OPTIONAL
   fields: {{telemetry.fields}} | OPTIONAL
10. References
    API error policy: {{xref:API-03}} | OPTIONAL
    Error UX: {{xref:FE-07}} | OPTIONAL
    Rate limit policy: {{xref:RLIM-01}} | OPTIONAL
    Session expiry handling: {{xref:CER-04}} | OPTIONAL

Cross-References
Upstream: {{xref:API-03}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:CER-03}}, {{xref:CER-04}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define retryable classes and max attempts; use UNKNOWN for backoff numeric values.
intermediate: Required. Define backoff policy and UX prompt rules including 429 handling.
advanced: Required. Add circuit breaker and per-endpoint override strategy.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, retry conditions, base/max delay, timeout/auth attempts, 429 header name/fallback delay, offline UI rule, refresh/navigate actions, breaker details, telemetry fields, overrides/background rules, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If backoff.policy is UNKNOWN → block Completeness Gate.
If attempts.network is UNKNOWN → block Completeness Gate.
If rate.retry_after_respected is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.CER
Pass conditions:
required_fields_present == true
retry_and_backoff_defined == true
attempt_limits_defined == true
rate_limit_handling_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

CER-03

CER-03 — Offline/Error Mode UX (degraded experiences)
Header Block
template_id: CER-03
title: Offline/Error Mode UX (degraded experiences)
type: client_offline_error_mode_ux
template_version: 1.0.0
output_path: 10_app/client_error/CER-03_Offline_Error_Mode_UX.md
compliance_gate_id: TMP-05.PRIMARY.CER
upstream_dependencies: ["FE-03", "FE-07", "SMD-05"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FE-03", "FE-07", "SMD-05"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical user experience for offline and degraded-error modes: what UI is shown when offline, how cached data is indicated, what actions are disabled, how queued operations are communicated, and how the client recovers when connectivity returns. This template must be consistent with UI state model and offline state handling and must not invent UX states not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FE-03 UI State Model: {{fe.state_model}}
FE-07 Error Handling UX: {{fe.error_ux}} | OPTIONAL
SMD-05 Offline Handling: {{smd.offline_handling}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Offline detection UX trigger
Global offline banner/indicator rules
Per-screen offline affordances (read-only vs disabled)
Cached/stale data labeling rules
Queued write indicator rules
Disabled action rules (what must be blocked)
Copy policy (what to say)
Recovery UX (reconnect, retry, refresh)
Accessibility rules (announce offline status)

Optional Fields
Airplane mode differences | OPTIONAL

Per-route exceptions | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Offline UX MUST match offline handling model ({{xref:SMD-05}}).
Offline status changes SHOULD be announced for accessibility ({{xref:FE-05}}) | OPTIONAL.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Offline Trigger
   trigger_rule: {{offline.trigger_rule}}
   notes: {{offline.notes}} | OPTIONAL
2. Global Indicator
   banner_enabled: {{ui.banner_enabled}}
   banner_placement: {{ui.banner_placement}} | OPTIONAL
   status_icon_rule: {{ui.status_icon_rule}} | OPTIONAL
3. Per-Screen Behavior
   read_only_when_offline: {{screen.read_only_when_offline}}
   disabled_actions: {{screen.disabled_actions}} | OPTIONAL
   allowed_actions: {{screen.allowed_actions}} | OPTIONAL
4. Cached/Stale Labeling
   show_stale_badge: {{stale.show_badge}}
   stale_copy_policy: {{stale.copy_policy}} | OPTIONAL
5. Queued Writes
   show_queue_indicator: {{queue.show_indicator}}
   queue_copy_policy: {{queue.copy_policy}} | OPTIONAL
6. Disabled Action Rules
   must_block_operations: {{block.must_block_operations}}
   soft_disable_operations: {{block.soft_disable_operations}} | OPTIONAL
7. Copy Policy
   tone: {{copy.tone}}
   safe_messaging: {{copy.safe_messaging}} | OPTIONAL
8. Recovery UX
   on_reconnect_behavior: {{recovery.on_reconnect_behavior}}
   manual_retry_cta: {{recovery.manual_retry_cta}} | OPTIONAL
   refresh_cta: {{recovery.refresh_cta}} | OPTIONAL
9. Accessibility
   announce_offline: {{a11y.announce_offline}}
   announce_reconnect: {{a11y.announce_reconnect}} | OPTIONAL
10. References
    UI state model: {{xref:FE-03}}
    Offline handling: {{xref:SMD-05}} | OPTIONAL

Error UX: {{xref:FE-07}} | OPTIONAL
A11y notes: {{xref:FE-05}} | OPTIONAL

Cross-References
Upstream: {{xref:FE-03}}, {{xref:SMD-05}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:CER-04}}, {{xref:CER-05}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-A11Y]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define banner + basic offline copy; use UNKNOWN for per-route nuance.
intermediate: Required. Define stale labeling and queued write communication.
advanced: Required. Add accessibility announcements and per-screen exception handling.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, banner placement/icon rule, disabled/allowed actions, stale/queue copy policies, soft disable ops, safe messaging, CTAs, announce reconnect, airplane/per-route exceptions, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If offline.trigger_rule is UNKNOWN → block Completeness Gate.
If ui.banner_enabled is UNKNOWN → block Completeness Gate.
If recovery.on_reconnect_behavior is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.CER
Pass conditions:
required_fields_present == true
offline_indicator_defined == true
stale_and_queue_rules_defined == true
recovery_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

CER-04

CER-04 — Session Expiry Handling (re-auth prompts, restore state)
Header Block
template_id: CER-04
title: Session Expiry Handling (re-auth prompts, restore state)
type: client_session_expiry_handling
template_version: 1.0.0
output_path: 10_app/client_error/CER-04_Session_Expiry_Handling.md
compliance_gate_id: TMP-05.PRIMARY.CER
upstream_dependencies: ["API-04", "CSec-01", "FE-07", "ROUTE-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "API-04", "CSec-01", "FE-07", "ROUTE-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical client behavior when sessions expire or authentication becomes invalid: how 401s are handled, when to prompt re-auth, how to preserve/restore navigation state, and how to avoid loops. This template must be consistent with AuthZ/token storage policies and route guards and must not invent auth flows not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
API-04 AuthZ Rules: {{api.authz_rules}} | OPTIONAL
CSec-01 Token Storage Policy: {{csec.token_policy}} | OPTIONAL
FE-07 Error UX: {{fe.error_ux}} | OPTIONAL
ROUTE-04 Guard Rules: {{route.guard_rules}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Session invalidation triggers (401/expired token/user revoked)
Detection rules (how client detects expiry)
Re-auth prompt UX (modal/screen)
Return-to behavior (restore previous route/action)
State preservation rules (forms, drafts)
Refresh token policy (if any) and failure behavior
Loop prevention rules (avoid repeated prompts)
Logout fallback rule (when re-auth fails)
Telemetry requirements (session expiry events)
Security constraints (do not persist sensitive auth data)

Optional Fields
Multi-device session invalidation behavior | OPTIONAL

Grace period policy | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Re-auth behavior MUST align with route guards ({{xref:ROUTE-04}}).
Token handling MUST align with {{xref:CSec-01}}.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Invalidation Triggers
   triggers: {{session.triggers}}
   notes: {{session.notes}} | OPTIONAL
2. Detection
   detection_rule: {{detect.rule}}
   signals: {{detect.signals}} | OPTIONAL
3. Re-Auth UX
   prompt_type: {{ux.prompt_type}} (modal/screen/UNKNOWN)
   prompt_copy_policy: {{ux.copy_policy}} | OPTIONAL
   primary_action: {{ux.primary_action}}
   secondary_action: {{ux.secondary_action}} | OPTIONAL
4. Return-To Behavior
   return_to_enabled: {{return.enabled}}
   return_to_storage: {{return.storage}} | OPTIONAL
   return_to_restore_rule: {{return.restore_rule}}
5. State Preservation
   preserve_forms: {{preserve.forms}} | OPTIONAL
   preserve_drafts: {{preserve.drafts}} | OPTIONAL
   clear_sensitive_state_rule: {{preserve.clear_sensitive_state_rule}}
6. Refresh Policy (if applicable)
   refresh_supported: {{refresh.supported}}
   refresh_attempt_rule: {{refresh.attempt_rule}} | OPTIONAL
   refresh_fail_behavior: {{refresh.fail_behavior}}
7. Loop Prevention
   max_prompts_per_window: {{loops.max_prompts_per_window}}
   cooldown_ms: {{loops.cooldown_ms}} | OPTIONAL
8. Logout Fallback
   force_logout_when: {{logout.force_when}}
   post_logout_route: {{logout.post_logout_route}} | OPTIONAL
9. Telemetry
   session_expiry_metric: {{telemetry.session_expiry_metric}}
   fields: {{telemetry.fields}} | OPTIONAL

10. Security Constraints
    no_token_in_logs: {{security.no_token_in_logs}}
    secure_storage_rule: {{security.secure_storage_rule}} | OPTIONAL
11. References
    AuthZ rules: {{xref:API-04}} | OPTIONAL
    Token storage: {{xref:CSec-01}} | OPTIONAL
    Error UX: {{xref:FE-07}} | OPTIONAL
    Guard rules: {{xref:ROUTE-04}} | OPTIONAL

Cross-References
Upstream: {{xref:API-04}} | OPTIONAL, {{xref:CSec-01}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:CER-05}} | OPTIONAL
Standards: {{standards.rules[STD-SECURITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define 401 handling + re-auth prompt; use UNKNOWN for refresh tokens if not used.
intermediate: Required. Define return-to + loop prevention + logout fallback.
advanced: Required. Add multi-device invalidation and state preservation rules with privacy constraints.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, detect signals, prompt copy policy, secondary action, return storage, preserve forms/drafts, refresh attempt rule, cooldown, post logout route, telemetry fields, multi-device/grace period, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If detect.rule is UNKNOWN → block Completeness Gate.
If ux.primary_action is UNKNOWN → block Completeness Gate.
If return.restore_rule is UNKNOWN → block Completeness Gate.
If telemetry.session_expiry_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.CER
Pass conditions:
required_fields_present == true
expiry_detection_defined == true
re_auth_ux_defined == true
return_to_defined == true
logout_fallback_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

CER-05

CER-05 — Client Logging & Crash Reporting (fields, redaction)
Header Block
template_id: CER-05
title: Client Logging & Crash Reporting (fields, redaction)
type: client_logging_crash_reporting
template_version: 1.0.0
output_path: 10_app/client_error/CER-05_Client_Logging_Crash_Reporting.md
compliance_gate_id: TMP-05.PRIMARY.CER
upstream_dependencies: ["FE-07", "CER-01", "ADMIN-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "FE-07", "CER-01", "ADMIN-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical client logging and crash reporting requirements: what events are logged, required fields, redaction rules (PII/secrets), sampling, and how crash reports are linked to server requests and user sessions. This template must be consistent with error UX and audit/redaction standards and must not invent logging systems not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
FE-07 Error Handling UX: {{fe.error_ux}} | OPTIONAL
CER-01 Error Boundary Strategy: {{cer.boundaries}} | OPTIONAL
ADMIN-03 Audit Trail Spec: {{admin.audit_trail}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Logging scope (what is logged)
Event types (error, warning, info, crash)
Required log fields (timestamp, route_id, request_id, error_code)
Correlation rules (link client logs to server traces)
Redaction policy (PII, secrets, tokens)
Crash reporting enabled policy
Sampling policy (volume control)
Storage/transport policy (where logs go)
User privacy policy (opt-out if applicable)
Alerting integration notes (optional)

Optional Fields
Breadcrumbs (user actions) | OPTIONAL

Offline log buffering | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not log secrets/tokens or raw PII.
Crash reports MUST be redacted and privacy-safe.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Logging Scope
   scope_statement: {{scope.statement}}
   included_events: {{scope.included_events}}
   excluded_events: {{scope.excluded_events}} | OPTIONAL
2. Event Types
   types: {{events.types}} (error/warn/info/crash/UNKNOWN)
   severity_rules: {{events.severity_rules}} | OPTIONAL
3. Required Fields
   timestamp: {{fields.timestamp}}
   event_type: {{fields.event_type}}
   route_id: {{fields.route_id}} | OPTIONAL
   screen_id: {{fields.screen_id}} | OPTIONAL
   request_id: {{fields.request_id}} | OPTIONAL
   trace_id: {{fields.trace_id}} | OPTIONAL
   session_id: {{fields.session_id}} | OPTIONAL
   user_id_hash: {{fields.user_id_hash}} | OPTIONAL
   device_id_hash: {{fields.device_id_hash}} | OPTIONAL
   app_version: {{fields.app_version}} | OPTIONAL
   error_code: {{fields.error_code}} | OPTIONAL
   status_code: {{fields.status_code}} | OPTIONAL
   message: {{fields.message}} | OPTIONAL
4. Correlation Rules
   correlate_with_server: {{corr.correlate_with_server}}
   correlation_id_sources: {{corr.id_sources}} | OPTIONAL
5. Redaction Policy
   pii_redaction: {{redact.pii_redaction}}
   secrets_redaction: {{redact.secrets_redaction}}
   token_redaction: {{redact.token_redaction}}
   field_allowlist: {{redact.field_allowlist}} | OPTIONAL
6. Crash Reporting
   enabled: {{crash.enabled}}
   crash_tooling: {{crash.tooling}} | OPTIONAL
   crash_upload_rules: {{crash.upload_rules}} | OPTIONAL

7. Sampling / Volume Controls
   sampling_enabled: {{sampling.enabled}}
   sampling_rules: {{sampling.rules}} | OPTIONAL
8. Storage / Transport
   destination: {{storage.destination}}
   buffering_supported: {{storage.buffering_supported}} | OPTIONAL
   offline_buffering_rules: {{storage.offline_buffering_rules}} | OPTIONAL
9. Privacy
   user_opt_out_supported: {{privacy.opt_out_supported}}
   data_retention_policy: {{privacy.retention_policy}} | OPTIONAL
10. References
    Error UX: {{xref:FE-07}} | OPTIONAL
    Error boundaries: {{xref:CER-01}} | OPTIONAL
    Admin audit/redaction: {{xref:ADMIN-03}} | OPTIONAL

Cross-References
Upstream: {{xref:FE-07}}, {{xref:CER-01}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ADMIN-02}} | OPTIONAL
Standards: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define required fields + redaction; use UNKNOWN for tooling/destination if missing.
intermediate: Required. Define correlation rules and sampling policy.
advanced: Required. Add offline buffering, privacy opt-out, and retention policy rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, excluded events, severity rules, optional fields, correlation id sources, field allowlist, crash tooling/upload rules, sampling rules, buffering/offline rules, retention policy, alert integration, breadcrumbs, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If scope.statement is UNKNOWN → block Completeness Gate.
If redact.token_redaction is UNKNOWN → block Completeness Gate.
If storage.destination is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.CER
Pass conditions:
required_fields_present == true
redaction_policy_defined == true
storage_destination_defined == true
sampling_policy_defined == true

placeholder_resolution == true
no_unapproved_unknowns == true

Client Security (CSec)

Client Security (CSec)
CSec-01 Token Storage Policy (secure storage, rotation)
CSec-02 Client Data Protection (PII redaction, screenshots, clipboard)
CSec-03 Web Security Headers/CSP (if web)
CSec-04 Anti-Tamper & Integrity Checks (basic)
CSec-05 Secure Deep Link Handling (validation, allowlists)

CSEC-01

CSec-01 — Token Storage Policy (secure storage, rotation)
Header Block
template_id: CSec-01
title: Token Storage Policy (secure storage, rotation)
type: client_token_storage_policy
template_version: 1.0.0
output_path: 10_app/client_security/CSec-01_Token_Storage_Policy.md
compliance_gate_id: TMP-05.PRIMARY.CSec
upstream_dependencies: ["API-04", "CER-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "API-04", "CER-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical client policy for storing and using auth tokens/credentials, including secure storage location, access rules, rotation/refresh handling, logout clearing, and platform-specific constraints. This template must be consistent with AuthZ/session expiry rules and must not invent auth mechanisms not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
API-04 AuthZ Rules: {{api.authz_rules}} | OPTIONAL
CER-04 Session Expiry Handling: {{cer.session_expiry}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Token types stored (access/refresh/session/UNKNOWN)
Storage location by platform (web/mobile)
Secure storage requirement statement
Encryption-at-rest rule (if applicable)
Read/write access rules (who/what code paths)
Token rotation/refresh policy
Token lifetime handling (expiry detection)
Logout clearing rules (clear all auth artifacts)
XSS/CSRF considerations (web)
Logging rules (never log tokens)

Optional Fields
Multi-account support | OPTIONAL
Device binding rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Tokens MUST NOT be stored in insecure locations unless explicitly allowed.
Token handling MUST align with {{xref:CER-04}} session expiry behavior.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Token Types
   token_types: {{tokens.types}} (access/refresh/session/UNKNOWN)
   notes: {{tokens.notes}} | OPTIONAL
2. Storage Locations
   web_storage: {{storage.web}} (httpOnly_cookie/memory/localStorage/UNKNOWN)
   mobile_storage: {{storage.mobile}} (keychain/keystore/secure_store/UNKNOWN)
   encryption_required: {{storage.encryption_required}} | OPTIONAL
3. Secure Storage Rules
   must_use_secure_storage: {{rules.must_use_secure_storage}}
   no_plaintext_rule: {{rules.no_plaintext_rule}}
4. Access Rules
   read_access: {{access.read_access}}
   write_access: {{access.write_access}}
   centralized_auth_module_rule: {{access.centralized_auth_module_rule}} | OPTIONAL
5. Rotation / Refresh
   refresh_supported: {{rotation.refresh_supported}}
   refresh_trigger_rule: {{rotation.refresh_trigger_rule}} | OPTIONAL
   rotation_policy: {{rotation.policy}} | OPTIONAL
   on_refresh_fail: {{rotation.on_refresh_fail}}
6. Expiry Handling
   expiry_detection_rule: {{expiry.detection_rule}}
   grace_period_policy: {{expiry.grace_period_policy}} | OPTIONAL
7. Logout Clearing
   clear_on_logout: {{logout.clear_on_logout}}
   also_clear_cache: {{logout.also_clear_cache}} | OPTIONAL
8. Web Threat Considerations
   xss_mitigation_notes: {{web.xss_mitigation_notes}} | OPTIONAL
   csrf_mitigation_notes: {{web.csrf_mitigation_notes}} | OPTIONAL
9. Logging Rules
   never_log_tokens_rule: {{logging.never_log_tokens_rule}}
   redaction_policy_ref: {{logging.redaction_policy_ref}} | OPTIONAL
10. References
    AuthZ rules: {{xref:API-04}} | OPTIONAL
    Session expiry handling: {{xref:CER-04}} | OPTIONAL
    Client logging/redaction: {{xref:CER-05}} | OPTIONAL

Cross-References
Upstream: {{xref:API-04}} | OPTIONAL, {{xref:CER-04}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:CSec-02}}, {{xref:CSec-05}} | OPTIONAL
Standards: {{standards.rules[STD-SECURITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define storage locations and “never log tokens”; use UNKNOWN for refresh specifics.
intermediate: Required. Define refresh/rotation and logout clearing behavior.
advanced: Required. Add web XSS/CSRF constraints and centralized module enforcement.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, encryption_required, centralized auth module rule, refresh trigger/rotation policy, grace period, cache clear, xss/csrf notes, redaction policy ref, multi-account/device binding, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If storage.web is UNKNOWN → block Completeness Gate (for web clients).
If storage.mobile is UNKNOWN → block Completeness Gate (for mobile clients).
If logging.never_log_tokens_rule is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.CSec
Pass conditions:
required_fields_present == true
storage_locations_defined == true
logout_clearing_defined == true
no_token_logging_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

CSEC-02

CSec-02 — Client Data Protection (PII redaction, screenshots, clipboard)
Header Block
template_id: CSec-02
title: Client Data Protection (PII redaction, screenshots, clipboard)
type: client_data_protection_policy
template_version: 1.0.0
output_path: 10_app/client_security/CSec-02_Client_Data_Protection.md
compliance_gate_id: TMP-05.PRIMARY.CSec
upstream_dependencies: ["CSec-01", "CER-05", "FPMP-06"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "CSec-01", "CER-05", "FPMP-06"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical client-side data protection rules for PII and sensitive content, including redaction in logs, screenshot/recording policies (mobile), clipboard handling, on-device storage constraints, and safe display rules. This template must be consistent with logging/redaction and file security policies and must not invent data classes not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
CSec-01 Token Storage Policy: {{csec.token_storage}}
CER-05 Client Logging/Crash Reporting: {{cer.logging}} | OPTIONAL
FPMP-06 File Security/Compliance: {{fpmp.security_compliance}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Sensitive data classes (PII, secrets, financial, etc.)
Redaction rules (logs, crash reports, analytics)
Screenshot/screen recording policy (allow/block/UNKNOWN)
Clipboard policy (copy allowed, auto-clear)
Local storage policy for sensitive data (avoid or encrypt)
UI display rules (masking, reveal)
Session privacy rules (lock on background)
Data export/share rules (if applicable)
Telemetry constraints (no sensitive fields)

Optional Fields
Per-screen “secure view” flags | OPTIONAL
OS-level protections (FLAG_SECURE on Android) | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
No secrets/tokens in logs or analytics.
If screenshots are blocked, policy must specify which screens/data classes trigger block.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Sensitive Data Classes
   classes: {{data.classes}}
   definitions_ref: {{data.definitions_ref}} | OPTIONAL
2. Redaction Rules
   logs_redaction: {{redaction.logs}}
   crash_redaction: {{redaction.crash}} | OPTIONAL
   analytics_redaction: {{redaction.analytics}} | OPTIONAL
   field_allowlist: {{redaction.field_allowlist}} | OPTIONAL
3. Screenshot / Recording Policy
   screenshots_allowed: {{screenshots.allowed}}
   recording_allowed: {{screenshots.recording_allowed}} | OPTIONAL
   block_when: {{screenshots.block_when}} | OPTIONAL
   platform_mechanisms: {{screenshots.platform_mechanisms}} | OPTIONAL
4. Clipboard Policy
   copy_allowed: {{clipboard.copy_allowed}}
   sensitive_copy_allowed: {{clipboard.sensitive_copy_allowed}} | OPTIONAL
   auto_clear_seconds: {{clipboard.auto_clear_seconds}} | OPTIONAL
   paste_handling_rules: {{clipboard.paste_handling_rules}} | OPTIONAL
5. Local Storage
   store_sensitive_locally: {{storage.store_sensitive_locally}}
   encryption_required: {{storage.encryption_required}} | OPTIONAL
   allowed_storage_locations: {{storage.allowed_locations}} | OPTIONAL
6. UI Display Rules
   masking_rule: {{ui.masking_rule}}
   reveal_interaction_rule: {{ui.reveal_interaction_rule}} | OPTIONAL
   timeout_on_reveal: {{ui.timeout_on_reveal}} | OPTIONAL
7. Session Privacy
   lock_on_background: {{session.lock_on_background}}
   privacy_screen_on_switcher: {{session.privacy_screen_on_switcher}} | OPTIONAL
8. Export / Share
   share_allowed: {{share.allowed}}
   share_constraints: {{share.constraints}} | OPTIONAL
9. Telemetry Constraints
   no_sensitive_fields_rule: {{telemetry.no_sensitive_fields_rule}}
   event_filtering_rules: {{telemetry.event_filtering_rules}} | OPTIONAL

10. References
    Token storage: {{xref:CSec-01}}
    Client logging/redaction: {{xref:CER-05}} | OPTIONAL
    File security/compliance: {{xref:FPMP-06}} | OPTIONAL

Cross-References
Upstream: {{xref:CSec-01}}, {{xref:CER-05}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:CSec-05}} | OPTIONAL
Standards: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL,
{{standards.rules[STD-SECURITY]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define redaction + screenshot/clipboard baseline; use UNKNOWN for OS mechanisms.
intermediate: Required. Define local storage + UI masking/reveal rules.
advanced: Required. Add per-screen secure view and session privacy rigor with platform controls.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, definitions ref, crash/analytics redaction, field allowlist, recording allowed, block_when, platform mechanisms, clipboard options, storage encryption/locations, reveal rules/timeouts, privacy switcher, share constraints, telemetry filtering rules, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If data.classes is UNKNOWN → block Completeness Gate.
If redaction.logs is UNKNOWN → block Completeness Gate.
If screenshots.allowed is UNKNOWN → block Completeness Gate.
If telemetry.no_sensitive_fields_rule is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.CSec
Pass conditions:
required_fields_present == true
sensitive_classes_defined == true
redaction_defined == true
screenshot_and_clipboard_policy_defined == true
telemetry_constraints_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

CSEC-03

CSec-03 — Web Security Headers/CSP (if web)
Header Block
template_id: CSec-03
title: Web Security Headers/CSP (if web)
type: web_security_headers_csp
template_version: 1.0.0
output_path: 10_app/client_security/CSec-03_Web_Security_Headers_CSP.md
compliance_gate_id: TMP-05.PRIMARY.CSec
upstream_dependencies: ["CSec-01", "CSec-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "CSec-01", "CSec-02"]
required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose
Define the canonical web security headers and CSP policy for the web client, including required headers, CSP directives, allowed origins, and reporting. This template must be consistent with token storage and data protection policies and must not invent security controls not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
CSec-01 Token Storage Policy: {{csec.token_storage}} | OPTIONAL
CSec-02 Data Protection: {{csec.data_protection}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Headers required list (header_name → value policy)
CSP enabled (true/false/UNKNOWN)
CSP directives (default-src, script-src, style-src, img-src, connect-src, frame-ancestors, etc.)
Allowed origins (CDN, API domains)
Inline script/style policy (nonce/hash)
Reporting policy (report-uri/report-to)
HSTS policy (includeSubDomains, preload)
Clickjacking protection (frame-ancestors/XFO)
Referrer policy
Permissions policy (camera/mic/geolocation)

Optional Fields
COOP/COEP/CORP policies | OPTIONAL
CSP violation monitoring | OPTIONAL

Per-environment differences | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Headers MUST be explicit and consistent across environments unless documented.
CSP MUST minimize allowed sources and avoid unsafe-inline unless explicitly required.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Required Security Headers
   Header
   name: {{headers[0].name}}
   value_policy: {{headers[0].value_policy}}
   applies_to: {{headers[0].applies_to}} | OPTIONAL
   notes: {{headers[0].notes}} | OPTIONAL

(Repeat per header.)

2. CSP Policy
   csp_enabled: {{csp.enabled}}
   default_src: {{csp.default_src}}
   script_src: {{csp.script_src}}
   style_src: {{csp.style_src}}
   img_src: {{csp.img_src}} | OPTIONAL
   connect_src: {{csp.connect_src}}
   font_src: {{csp.font_src}} | OPTIONAL
   frame_ancestors: {{csp.frame_ancestors}}
   object_src: {{csp.object_src}} | OPTIONAL
   base_uri: {{csp.base_uri}} | OPTIONAL

inline_policy:
script_inline: {{csp.inline.script_inline}} (nonce/hash/blocked/UNKNOWN)
style_inline: {{csp.inline.style_inline}} (nonce/hash/blocked/UNKNOWN) | OPTIONAL

reporting:
report_to: {{csp.reporting.report_to}} | OPTIONAL
report_uri: {{csp.reporting.report_uri}} | OPTIONAL

3. HSTS
   enabled: {{hsts.enabled}}
   max_age_seconds: {{hsts.max_age_seconds}}
   include_subdomains: {{hsts.include_subdomains}} | OPTIONAL
   preload: {{hsts.preload}} | OPTIONAL

4. Clickjacking Protection
   frame_ancestors_policy: {{clickjacking.frame_ancestors_policy}}
   x_frame_options: {{clickjacking.x_frame_options}} | OPTIONAL
5. Referrer Policy
   referrer_policy: {{referrer.policy}}
6. Permissions Policy
   permissions_policy: {{perm.policy}}
7. Optional Isolation (COOP/COEP/CORP)
   coop: {{isolation.coop}} | OPTIONAL
   coep: {{isolation.coep}} | OPTIONAL
   corp: {{isolation.corp}} | OPTIONAL
8. References
   Token storage: {{xref:CSec-01}} | OPTIONAL
   Data protection: {{xref:CSec-02}} | OPTIONAL

Cross-References
Upstream: {{xref:CSec-01}} | OPTIONAL, {{xref:CSec-02}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:CSec-04}} | OPTIONAL
Standards: {{standards.rules[STD-SECURITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Not required.
intermediate: Required. Define headers + CSP directives + allowed origins.
advanced: Required. Add reporting + isolation policies and per-env documentation.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, applies_to, notes, img/font sources, optional directives, style inline, reporting destinations, include subdomains/preload, xfo, isolation policies, per-env differences, monitoring, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If csp.enabled is UNKNOWN → block Completeness Gate (for web).
If csp.script_src is UNKNOWN → block Completeness Gate.
If hsts.max_age_seconds is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.CSec
Pass conditions:
required_fields_present == true
security_headers_defined == true
csp_defined == true
clickjacking_and_hsts_defined == true

placeholder_resolution == true
no_unapproved_unknowns == true

CSEC-04

CSec-04 — Anti-Tamper & Integrity Checks (basic)
Header Block
template_id: CSec-04
title: Anti-Tamper & Integrity Checks (basic)
type: client_anti_tamper_integrity_checks
template_version: 1.0.0
output_path: 10_app/client_security/CSec-04_Anti_Tamper_Integrity_Checks.md
compliance_gate_id: TMP-05.PRIMARY.CSec
upstream_dependencies: ["CSec-01", "CSec-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "CSec-01", "CSec-02", "CSec-03"]
required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose
Define the canonical baseline anti-tamper and integrity check policy for the client, including detection signals (root/jailbreak, debugger, modified binaries), integrity verification approaches, and response behavior. This template must be consistent with data protection and token storage policies and must not invent security claims not supported by upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
CSec-01 Token Storage: {{csec.token_storage}} | OPTIONAL
CSec-02 Data Protection: {{csec.data_protection}} | OPTIONAL
CSec-03 Web Security (if web): {{csec.web_security}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Threat model summary (tampering risks)
Integrity checks supported list
Device compromise detection (root/jailbreak) policy (mobile)
Debug/Hook detection policy (optional)
Web integrity approach (SRI / build integrity) if web
Response actions (warn, block, degrade)
Telemetry for integrity events
False positive handling policy
Privacy constraints (don’t collect sensitive signals)

Optional Fields
App attestation integration notes | OPTIONAL
Per-risk-tier responses | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not claim perfect prevention; define detection + response.
Responses MUST not leak sensitive details or create bypass instructions.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Threat Model
   tamper_threats: {{threats.tamper_threats}}
   notes: {{threats.notes}} | OPTIONAL
2. Integrity Checks
   checks_supported: {{checks.supported}}
   check_frequency: {{checks.frequency}} | OPTIONAL
3. Device Compromise Detection (Mobile)
   root_jailbreak_detection: {{mobile.root_jailbreak_detection}}
   hook_detection: {{mobile.hook_detection}} | OPTIONAL
   debugger_detection: {{mobile.debugger_detection}} | OPTIONAL
4. Web Integrity (if web)
   sri_supported: {{web.sri_supported}} | OPTIONAL
   build_integrity_checks: {{web.build_integrity_checks}} | OPTIONAL
5. Response Actions
   on_detect_warn: {{response.warn}}
   on_detect_block: {{response.block}} | OPTIONAL
   on_detect_degrade: {{response.degrade}} | OPTIONAL
6. Telemetry
   integrity_event_metric: {{telemetry.integrity_event_metric}}
   fields: {{telemetry.fields}} | OPTIONAL
7. False Positives
   appeal_or_retry_policy: {{fp.policy}}
   support_flow_ref: {{fp.support_flow_ref}} | OPTIONAL
8. Privacy Constraints
   no_sensitive_signal_collection: {{privacy.no_sensitive_signal_collection}}
   minimal_data_rule: {{privacy.minimal_data_rule}} | OPTIONAL
9. References
   Token storage: {{xref:CSec-01}} | OPTIONAL
   Data protection: {{xref:CSec-02}} | OPTIONAL
   Web security headers/CSP: {{xref:CSec-03}} | OPTIONAL

Cross-References
Upstream: {{xref:CSec-01}} | OPTIONAL, {{xref:CSec-02}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:CSec-05}} | OPTIONAL

Standards: {{standards.rules[STD-SECURITY]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Not required.
intermediate: Required. Define supported checks and response actions and telemetry.
advanced: Required. Add per-risk-tier responses and attestation notes.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, check frequency, hook/debugger detection, web integrity, optional response actions, telemetry fields, support flow ref, minimal data rule, attestation notes, risk tier responses, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If checks.supported is UNKNOWN → block Completeness Gate.
If response.warn is UNKNOWN → block Completeness Gate.
If telemetry.integrity_event_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.CSec
Pass conditions:
required_fields_present == true
integrity_checks_defined == true
response_actions_defined == true
telemetry_defined == true
privacy_constraints_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

CSEC-05

CSec-05 — Secure Deep Link Handling (validation, allowlists)
Header Block
template_id: CSec-05
title: Secure Deep Link Handling (validation, allowlists)
type: client_secure_deep_link_handling
template_version: 1.0.0
output_path: 10_app/client_security/CSec-05_Secure_Deep_Link_Handling.md
compliance_gate_id: TMP-05.PRIMARY.CSec
upstream_dependencies: ["ROUTE-03", "ROUTE-06", "CSec-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "ROUTE-03", "ROUTE-06", "CSec-01", "MDL-01", "MDL-03"]
required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose
Define the canonical security rules for deep link handling: allowlists, strict param parsing, signature verification if used, safe redirect prevention, auth gating, and telemetry for abuse detection. This template must be consistent with route validation policies and must not invent security mechanics not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
ROUTE-03 Deep Link Map: {{route.deep_link_map}} | OPTIONAL
ROUTE-06 Link Validation & Unknown Handling: {{route.unknown_handling}} | OPTIONAL
CSec-01 Token Storage Policy: {{csec.token_storage}} | OPTIONAL
MDL-01 Link Scheme & Domains: {{mobile.links}} | OPTIONAL
MDL-03 Auth Gating & Safety: {{mobile.link_safety}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Allowed schemes/domains allowlist
Strict parsing rules (reject unknown keys)
Signature validation (if supported)
Auth gating rules for deep link targets
Sensitive action protection (confirmations)
Open redirect prevention rules
Telemetry requirements (link abuse attempts)
Error handling (invalid link UX)

Optional Fields
Link expiry rules | OPTIONAL

One-time links policy | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Deep links MUST be allowlisted and validated before navigation.
No open redirects.
If signature validation is used, failures MUST not leak details.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Allowlist
   schemes: {{allowlist.schemes}}
   domains: {{allowlist.domains}} | OPTIONAL
2. Parsing Rules
   reject_unknown_keys: {{parse.reject_unknown_keys}}
   param_type_validation: {{parse.param_type_validation}}
   decode_rules: {{parse.decode_rules}} | OPTIONAL
3. Signature Validation (Optional)
   signed_links_supported: {{sig.supported}}
   signature_scheme: {{sig.scheme}} | OPTIONAL
   verification_rules: {{sig.verification_rules}} | OPTIONAL
   failure_behavior: {{sig.failure_behavior}} | OPTIONAL
4. Auth Gating
   auth_required_for_targets: {{auth.targets_require_auth}}
   behavior_when_logged_out: {{auth.behavior_logged_out}}
   return_to_rule: {{auth.return_to_rule}} | OPTIONAL
5. Sensitive Actions
   confirm_required_for_actions: {{sensitive.confirm_required_for_actions}} | OPTIONAL
   rate_limit_ref: {{sensitive.rate_limit_ref}} | OPTIONAL
6. Open Redirect Prevention
   no_open_redirects_rule: {{security.no_open_redirects_rule}}
   external_nav_policy: {{security.external_nav_policy}} | OPTIONAL
7. Telemetry
   invalid_link_metric: {{telemetry.invalid_link_metric}}
   abuse_attempt_metric: {{telemetry.abuse_attempt_metric}} | OPTIONAL
   fields: {{telemetry.fields}} | OPTIONAL
8. Error Handling UX
   invalid_link_ux: {{ux.invalid_link_ux}}
   support_contact_policy: {{ux.support_contact_policy}} | OPTIONAL
9. References
   Deep link map: {{xref:ROUTE-03}} | OPTIONAL
   Unknown handling: {{xref:ROUTE-06}} | OPTIONAL

Mobile link scheme/domains: {{xref:MDL-01}} | OPTIONAL
Mobile auth gating/safety: {{xref:MDL-03}} | OPTIONAL

Cross-References
Upstream: {{xref:ROUTE-03}} | OPTIONAL, {{xref:ROUTE-06}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:CER-04}} | OPTIONAL
Standards: {{standards.rules[STD-SECURITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Not required.
intermediate: Required. Define allowlists, parsing rules, and invalid link UX.
advanced: Required. Add signature/expiry/one-time link policies and abuse telemetry rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, domains allowlist, decode rules, signature details, return-to rule, sensitive action confirms/ratelimits, external nav policy, telemetry fields, support contact, expiry/one-time links, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If allowlist.schemes is UNKNOWN → block Completeness Gate.
If parse.reject_unknown_keys is UNKNOWN → block Completeness Gate.
If security.no_open_redirects_rule is UNKNOWN → block Completeness Gate.
If ux.invalid_link_ux is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.CSec
Pass conditions:
required_fields_present == true
allowlist_defined == true
parsing_and_redirect_rules_defined == true
telemetry_defined == true
invalid_link_ux_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

Mobile Implementation (MOB)

Mobile Implementation (MOB)
MOB-01 Navigation Map (stacks/tabs/modals)
MOB-02 Screen Implementation Spec (per screen binding)
MOB-03 Native Integration Map (bridges, permissions)
MOB-04 App Lifecycle + State (foreground/background)
MOB-05 Release & Signing (stores, builds)

MOB-01

MOB-01 — Navigation Map (stacks/tabs/modals)
Header Block
template_id: MOB-01
title: Navigation Map (stacks/tabs/modals)
type: mobile_navigation_map
template_version: 1.0.0
output_path: 10_app/mobile/MOB-01_Navigation_Map.md
compliance_gate_id: TMP-05.PRIMARY.MOB
upstream_dependencies: ["ROUTE-01", "ROUTE-02", "ROUTE-05"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "ROUTE-01", "ROUTE-02", "ROUTE-03", "ROUTE-05", "MDL-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical mobile navigation structure (React Native / native-style): stacks, tabs, modals, and their relationships. Includes screen registry, navigator hierarchy, entry points, modal policy, and back behavior alignment. This template must be consistent with the route/navigation contracts and must not invent screen_ids/route_ids not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
ROUTE-01 Route Contract: {{route.contract}}
ROUTE-02 Navigation Map: {{route.nav_map}}
ROUTE-03 Deep Link Map: {{route.deep_link_map}} | OPTIONAL
ROUTE-05 Back/History Rules: {{route.back_rules}} | OPTIONAL
MDL-02 Routing Rules (mobile links): {{mobile.routing_rules}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Navigator registry (navigator_id list)
Navigator types (stack/tab/modal)
Navigator hierarchy (parent → children)
Screen registry (screen_id list)
Screen → route_id bindings (if used)
Entry points (initial route/screen per navigator)
Modal policy (which screens are modal)
Back behavior rules binding (ROUTE-05)
Auth/role gating notes (ROUTE-04 binding)
Deep link handling binding (ROUTE-03/MDL-02)

Optional Fields
Shared screen options (headers, gestures) | OPTIONAL
Per-role navigator variants | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not introduce new screen_ids/route_ids; use only existing IDs from SPEC_INDEX/ROUTE docs.
Navigator hierarchy MUST be deterministic and acyclic.
Back behavior MUST align with {{xref:ROUTE-05}}.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Navigator Registry
   Navigator
   navigator_id: {{navs[0].navigator_id}}
   type: {{navs[0].type}} (stack/tab/modal/UNKNOWN)
   parent_navigator_id: {{navs[0].parent_navigator_id}} | OPTIONAL
   initial_screen_id: {{navs[0].initial_screen_id}}
   screens: {{navs[0].screens}}
   options: {{navs[0].options}} | OPTIONAL
   notes: {{navs[0].notes}} | OPTIONAL

(Repeat per navigator_id.)

2. Screen Registry (Mobile)
   Screen
   screen_id: {{screens[0].screen_id}}
   name: {{screens[0].name}} | OPTIONAL
   route_id: {{screens[0].route_id}} | OPTIONAL
   navigator_id: {{screens[0].navigator_id}}
   presentation: {{screens[0].presentation}} (push/modal/transparent_modal/UNKNOWN)
   header_shown: {{screens[0].header_shown}} | OPTIONAL
   gesture_enabled: {{screens[0].gesture_enabled}} | OPTIONAL
   guards:
   auth_required: {{screens[0].guards.auth_required}} | OPTIONAL
   role_gates: {{screens[0].guards.role_gates}} | OPTIONAL
   feature_flag_gate: {{screens[0].guards.feature_flag_gate}} | OPTIONAL
   open_questions:
   {{screens[0].open_questions[0]}} | OPTIONAL

(Repeat per screen_id.)

3. Entry Points
   app_initial_navigator: {{entry.app_initial_navigator}}
   app_initial_screen: {{entry.app_initial_screen}}
   post_login_landing: {{entry.post_login_landing}} | OPTIONAL
4. Modal Policy
   modal_screens: {{modal.screens}}
   modal_close_rules_ref: {{modal.close_rules_ref}} (expected: {{xref:ROUTE-05}}) | OPTIONAL
5. Back Behavior Binding
   back_rules_ref: {{back.rules_ref}} (expected: {{xref:ROUTE-05}})
   android_back_overrides: {{back.android_overrides}} | OPTIONAL
6. Deep Links Binding
   deep_link_map_ref: {{deep.ref}} (expected: {{xref:ROUTE-03}}) | OPTIONAL
   mobile_routing_rules_ref: {{deep.mobile_rules_ref}} (expected: {{xref:MDL-02}}) | OPTIONAL
7. References
   Route contract: {{xref:ROUTE-01}}
   Navigation map: {{xref:ROUTE-02}}
   Back/history rules: {{xref:ROUTE-05}} | OPTIONAL
   Deep link map: {{xref:ROUTE-03}} | OPTIONAL
   Guard rules: {{xref:ROUTE-04}} | OPTIONAL
   Mobile deep link routing: {{xref:MDL-02}} | OPTIONAL

Cross-References
Upstream: {{xref:ROUTE-01}}, {{xref:ROUTE-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:MOB-02}}, {{xref:MDL-02}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define navigator hierarchy and screen registry; use UNKNOWN for options/guards if missing.
intermediate: Required. Bind modal/back behavior and deep link routing.
advanced: Required. Add per-role variants and deterministic guard bindings for all screens.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, options/notes, screen name, route_id, presentation, header/gesture flags, guards, post-login landing, modal close rules ref, android overrides, deep link refs, per-role variants, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If navs list is UNKNOWN → block Completeness Gate.
If screens list is UNKNOWN → block Completeness Gate.
If back.rules_ref is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.MOB
Pass conditions:
required_fields_present == true
navigator_registry_defined == true
screen_registry_defined == true
back_behavior_bound == true
placeholder_resolution == true
no_unapproved_unknowns == true

MOB-02

MOB-02 — Screen Implementation Spec (per screen binding)
Header Block
template_id: MOB-02
title: Screen Implementation Spec (per screen binding)
type: mobile_screen_implementation_spec
template_version: 1.0.0
output_path: 10_app/mobile/MOB-02_Screen_Implementation_Spec.md
compliance_gate_id: TMP-05.PRIMARY.MOB
upstream_dependencies: ["MOB-01", "FE-03", "FE-04", "FE-05"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "MOB-01", "FE-03", "FE-04", "FE-05", "FORM-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical per-screen implementation binding for mobile: route/params, data dependencies, UI state handling, forms, permissions, and navigation actions. This template must be consistent with mobile navigation map and frontend state/data binding rules and must not invent screen_ids not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
MOB-01 Navigation Map: {{mob.nav_map}}
FE-03 UI State Model: {{fe.state_model}} | OPTIONAL
FE-04 Data Binding Rules: {{fe.data_binding}} | OPTIONAL
FE-05 Accessibility Notes: {{fe.a11y_notes}} | OPTIONAL
FORM-01 Forms Inventory: {{forms.inventory}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
screen_id (existing)
route_id binding (if any)
params schema ref (ROUTE-01 types)
layout shell/presentation (from MOB-01)
data bindings (endpoint_ids and timing)
UI state handling (loading/empty/error/success)
primary user actions (buttons/gestures)
navigation actions (where it goes next)
form bindings (form_id if present)
analytics/telemetry hooks (optional)
accessibility notes (screen-specific)

Optional Fields
Offline behavior notes | OPTIONAL
Realtime subscription notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not introduce new screen_ids; must exist in {{xref:MOB-01}}/SPEC_INDEX.
Data bindings MUST reference existing endpoint_ids (API-01/02).
State handling MUST conform to {{xref:FE-03}}.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Screen Identity
   screen_id: {{screen.screen_id}}
   name: {{screen.name}} | OPTIONAL
   route_id: {{screen.route_id}} | OPTIONAL
   presentation: {{screen.presentation}}
   params_schema_ref: {{screen.params_schema_ref}} | OPTIONAL
2. Layout & Navigation
   navigator_id: {{screen.navigator_id}}
   header_shown: {{screen.header_shown}} | OPTIONAL
   gesture_enabled: {{screen.gesture_enabled}} | OPTIONAL
   back_behavior_ref: {{screen.back_behavior_ref}} | OPTIONAL
3. Data Bindings
   Binding
   endpoint_id: {{bindings[0].endpoint_id}}
   call_timing: {{bindings[0].call_timing}}
   params_source: {{bindings[0].params_source}} | OPTIONAL
   cache_policy: {{bindings[0].cache_policy}} | OPTIONAL

(Repeat per binding.)

4. UI State Handling
   state_model_ref: {{ui.state_model_ref}} (expected: {{xref:FE-03}}) | OPTIONAL
   loading_ui: {{ui.loading_ui}}
   empty_ui: {{ui.empty_ui}} | OPTIONAL
   error_ui: {{ui.error_ui}}
   success_ui: {{ui.success_ui}} | OPTIONAL
5. Primary Actions
   actions:
   {{actions[0]}}
   {{actions[1]}} | OPTIONAL

6. Navigation Actions
   nav_actions:
   {{nav_actions[0]}}
   {{nav_actions[1]}} | OPTIONAL
7. Form Bindings (Optional)
   form_ids: {{forms.form_ids}} | OPTIONAL
8. Accessibility Notes
   a11y_notes: {{a11y.notes}}
9. Telemetry (Optional)
   analytics_id: {{telemetry.analytics_id}} | OPTIONAL
   events: {{telemetry.events}} | OPTIONAL
10. References
    Mobile nav map: {{xref:MOB-01}}
    UI state model: {{xref:FE-03}} | OPTIONAL
    Data bindings: {{xref:FE-04}} | OPTIONAL
    A11y notes: {{xref:FE-05}} | OPTIONAL
    Forms inventory: {{xref:FORM-01}} | OPTIONAL

Cross-References
Upstream: {{xref:MOB-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:MOB-03}}, {{xref:OFS-01}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define screen identity + state handling; use UNKNOWN for telemetry.
intermediate: Required. Bind endpoints and define navigation actions and form bindings.
advanced: Required. Add offline/realtime notes and consistent params schema refs.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, screen name, route id, params schema ref, header/gesture/back ref, binding params/cache, empty/success UI, secondary actions, form ids, telemetry, offline/realtime notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If screen.screen_id is UNKNOWN → block Completeness Gate.
If ui.error_ui is UNKNOWN → block Completeness Gate.
If bindings list is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.MOB
Pass conditions:
required_fields_present == true
screen_ids_reference_existing == true
data_bindings_defined == true
ui_state_defined == true

placeholder_resolution == true
no_unapproved_unknowns == true

MOB-03

MOB-03 — Native Integration Map (bridges, permissions)
Header Block
template_id: MOB-03
title: Native Integration Map (bridges, permissions)
type: mobile_native_integration_map
template_version: 1.0.0
output_path: 10_app/mobile/MOB-03_Native_Integration_Map.md
compliance_gate_id: TMP-05.PRIMARY.MOB
upstream_dependencies: ["MDC-01", "MDC-02", "MOB-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "MDC-01", "MDC-02", "MDC-03", "MOB-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical map of mobile native integrations: which device capabilities are used, what bridges/modules are required, what permissions are needed, and which screens/features depend on them. This template must be consistent with capability and permission rules and must not invent capabilities not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
MDC-01 Capabilities Inventory: {{mdc.capabilities}}
MDC-02 Permissions UX Rules: {{mdc.permissions_ux}} | OPTIONAL
MDC-03 Capability Security Rules: {{mdc.cap_security}} | OPTIONAL
MOB-02 Screen Implementation Specs: {{mob.screens}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Integration registry (integration_id list)
Capability binding (capability_id)
Native module/bridge name
Permissions required (platform-specific)
Screens/features using the integration (screen_id list)
Permission request timing (on action vs upfront)
Fallback behavior when denied
Security constraints (least privilege)
Telemetry requirements (permission denied rates)

Optional Fields
OS version constraints | OPTIONAL

Background mode dependencies | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not introduce new capability_ids; use only those in {{xref:MDC-01}}.
Permission UX MUST align with {{xref:MDC-02}}.
Least privilege MUST align with {{xref:MDC-03}}.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Integration Registry (by integration_id)
   Integration
   integration_id: {{integrations[0].integration_id}}
   capability_id: {{integrations[0].capability_id}}
   native_module: {{integrations[0].native_module}}
   permissions:
   ios: {{integrations[0].permissions.ios}} | OPTIONAL
   android: {{integrations[0].permissions.android}} | OPTIONAL
   screens: {{integrations[0].screens}}
   request_timing: {{integrations[0].request_timing}}
   fallback_on_denied: {{integrations[0].fallback_on_denied}}
   security_notes: {{integrations[0].security_notes}} | OPTIONAL
   telemetry: {{integrations[0].telemetry}} | OPTIONAL
   open_questions:
   {{integrations[0].open_questions[0]}} | OPTIONAL

(Repeat per integration_id.)

2. Global Permission Timing Rules
   default_request_timing: {{global.default_request_timing}}
   never_request_upfront_for: {{global.never_request_upfront_for}} | OPTIONAL
3. References
   Capabilities inventory: {{xref:MDC-01}}
   Permissions UX: {{xref:MDC-02}} | OPTIONAL
   Capability security: {{xref:MDC-03}} | OPTIONAL
   Screen specs: {{xref:MOB-02}} | OPTIONAL

Cross-References
Upstream: {{xref:MDC-01}}, {{xref:MOB-02}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:MDC-04}}, {{xref:MDC-05}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL,
{{standards.rules[STD-SECURITY]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. List integrations and basic permission timing; use UNKNOWN for OS constraints.
intermediate: Required. Define fallbacks and least privilege notes.
advanced: Required. Add background mode constraints and telemetry rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, ios/android permission lists, security notes, telemetry, never request upfront list, OS/background constraints, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If integrations list is UNKNOWN → block Completeness Gate.
If capability_id is UNKNOWN → block Completeness Gate.
If fallback_on_denied is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.MOB
Pass conditions:
required_fields_present == true
capability_bindings_defined == true
permission_and_fallback_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

MOB-04

MOB-04 — App Lifecycle + State (foreground/background)
Header Block
template_id: MOB-04
title: App Lifecycle + State (foreground/background)
type: mobile_app_lifecycle_state
template_version: 1.0.0
output_path: 10_app/mobile/MOB-04_App_Lifecycle_State.md
compliance_gate_id: TMP-05.PRIMARY.MOB
upstream_dependencies: ["MOB-01", "CSec-02", "SMD-05"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "MOB-01", "CSec-02", "SMD-05", "MBAT-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical mobile app lifecycle behavior: what happens on foreground/background transitions, app resume, cold start, memory pressure, and how state is saved/restored. Includes security/privacy behavior when backgrounded and offline queue behavior. This template must be consistent with data protection and offline handling and must not invent lifecycle behaviors not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
MOB-01 Navigation Map: {{mob.nav_map}} | OPTIONAL
CSec-02 Client Data Protection: {{csec.data_protection}} | OPTIONAL
SMD-05 Offline Handling: {{smd.offline_handling}} | OPTIONAL
MBAT-01 Background Work Rules: {{mbat.bg_work_rules}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Lifecycle events handled (cold start, resume, background, terminate)
State persistence model (what is saved)
Resume refresh rules (what refetches on resume)
Offline queue behavior on resume
Security/privacy rules on background (lock, blur, screenshot)
Session expiry checks on resume (CER-04 binding)
Push/deeplink behavior on cold start vs resume (MDL/MPUSH binding)
Telemetry requirements (resume counts, crashes)

Optional Fields
Memory pressure handling | OPTIONAL

App update migration rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Background behavior MUST protect sensitive data per {{xref:CSec-02}}.
Offline queue handling MUST align with {{xref:SMD-05}}.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Lifecycle Events
   events_handled: {{lifecycle.events_handled}}
   notes: {{lifecycle.notes}} | OPTIONAL
2. State Persistence
   persisted_state: {{state.persisted_state}}
   persistence_location: {{state.location}} | OPTIONAL
   hydration_order: {{state.hydration_order}} | OPTIONAL
3. Resume Refresh
   refetch_on_resume: {{resume.refetch_on_resume}}
   refresh_targets: {{resume.refresh_targets}} | OPTIONAL
   throttle_rules: {{resume.throttle_rules}} | OPTIONAL
4. Offline Queue on Resume
   drain_queue_on_resume: {{offline.drain_on_resume}}
   drain_policy_ref: {{offline.policy_ref}} (expected: {{xref:SMD-05}}) | OPTIONAL
5. Security/Privacy on Background
   lock_on_background: {{security.lock_on_background}}
   blur_in_app_switcher: {{security.blur_in_app_switcher}} | OPTIONAL
   disable_screenshots: {{security.disable_screenshots}} | OPTIONAL
6. Session Checks
   check_session_on_resume: {{session.check_on_resume}}
   session_expiry_ref: {{session.expiry_ref}} (expected: {{xref:CER-04}}) | OPTIONAL
7. Deep Links / Push Interaction
   cold_start_deeplink_rule: {{links.cold_start_rule}}
   resume_deeplink_rule: {{links.resume_rule}} | OPTIONAL
8. Telemetry
   resume_metric: {{telemetry.resume_metric}}
   cold_start_metric: {{telemetry.cold_start_metric}} | OPTIONAL
   crash_metric: {{telemetry.crash_metric}} | OPTIONAL
9. References
   Data protection: {{xref:CSec-02}} | OPTIONAL
   Offline handling: {{xref:SMD-05}} | OPTIONAL
   Session expiry: {{xref:CER-04}} | OPTIONAL
   Background work rules: {{xref:MBAT-01}} | OPTIONAL

Deep link routing: {{xref:MDL-02}} | OPTIONAL
Push notifications: {{xref:MPUSH-05}} | OPTIONAL

Cross-References
Upstream: {{xref:MOB-01}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:MOB-05}}, {{xref:MBAT-01}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-SECURITY]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define lifecycle events + basic persistence; use UNKNOWN for push/deeplink specifics.
intermediate: Required. Define resume refresh, offline drain, and background privacy rules.
advanced: Required. Add memory pressure handling and migration rules.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, persistence location/hydration order, refresh targets/throttle rules, drain policy ref, blur/screenshot rules, session expiry ref, deeplink rules, telemetry metrics, memory/migration rules, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If lifecycle.events_handled is UNKNOWN → block Completeness Gate.
If state.persisted_state is UNKNOWN → block Completeness Gate.
If security.lock_on_background is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.MOB
Pass conditions:
required_fields_present == true
lifecycle_events_defined == true
state_persistence_defined == true
security_background_rules_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

MOB-05

MOB-05 — Release & Signing (stores, builds)
Header Block
template_id: MOB-05
title: Release & Signing (stores, builds)
type: mobile_release_signing
template_version: 1.0.0
output_path: 10_app/mobile/MOB-05_Release_Signing.md
compliance_gate_id: TMP-05.PRIMARY.MOB
upstream_dependencies: ["SIGN-01", "SIGN-02", "SIGN-03", "SIGN-04", "SIGN-05"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "SIGN-01", "SIGN-02", "SIGN-03", "SIGN-04", "SIGN-05"]
required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose
Define the canonical mobile release and signing process: build types, signing keys, CI/release channels, store submission steps, and versioning rules. This template must be consistent with signing standards and must not invent store policies not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
SIGN-01 Build Artifact Types: {{sign.artifacts}}
SIGN-02 Signing Keys & Rotation: {{sign.keys}} | OPTIONAL
SIGN-03 Store Submission Checklist: {{sign.submission}} | OPTIONAL
SIGN-04 Versioning Rules: {{sign.versioning}} | OPTIONAL
SIGN-05 Release Channel Policy: {{sign.channels}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Build artifact types (debug/beta/prod)
Signing key storage policy
Key rotation policy
Store submission process (iOS/Android)
Versioning mapping (build numbers, semver)
Release channel flow (internal → beta → GA)
Rollback/hotfix policy
Telemetry/release checks (crash rate gates)

Optional Fields
Automated store upload notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Signing keys MUST be protected and never committed.
Versioning MUST follow SIGN-04.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Build Artifact Types
   artifacts: {{build.artifacts}}
   mapping_rules: {{build.mapping_rules}} | OPTIONAL
2. Signing Key Policy
   key_storage: {{keys.storage}}
   key_access_rules: {{keys.access_rules}} | OPTIONAL
   rotation_policy_ref: {{keys.rotation_policy_ref}} (expected: {{xref:SIGN-02}}) | OPTIONAL
3. Store Submission
   ios_steps: {{store.ios_steps}}
   android_steps: {{store.android_steps}}
   checklist_ref: {{store.checklist_ref}} (expected: {{xref:SIGN-03}}) | OPTIONAL
4. Versioning
   versioning_ref: {{version.ref}} (expected: {{xref:SIGN-04}})
   build_number_rule: {{version.build_number_rule}}
   semver_rule: {{version.semver_rule}} | OPTIONAL
5. Release Channels
   channel_policy_ref: {{channels.ref}} (expected: {{xref:SIGN-05}}) | OPTIONAL
   flow: {{channels.flow}}
   promotion_rules: {{channels.promotion_rules}} | OPTIONAL
6. Rollback / Hotfix
   hotfix_policy: {{rollback.hotfix_policy}}
   rollback_triggers: {{rollback.triggers}} | OPTIONAL
7. Release Health Checks
   crash_rate_gate: {{health.crash_rate_gate}}
   anr_rate_gate: {{health.anr_rate_gate}} | OPTIONAL
   monitoring_window: {{health.monitoring_window}} | OPTIONAL
8. References
   Build artifacts: {{xref:SIGN-01}}
   Signing keys: {{xref:SIGN-02}} | OPTIONAL
   Submission checklist: {{xref:SIGN-03}} | OPTIONAL
   Versioning rules: {{xref:SIGN-04}} | OPTIONAL
   Release channels: {{xref:SIGN-05}} | OPTIONAL

Cross-References
Upstream: {{xref:SIGN-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:RELEASE-GATE}} | OPTIONAL

Standards: {{standards.rules[STD-SECURITY]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Not required.
intermediate: Required. Define signing + submission + versioning.
advanced: Required. Add rotation/hotfix policies and release health gates.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, mapping rules, key access rules, rotation ref, checklist ref, semver rule, channel policy ref, promotion rules, rollback triggers, anr gate, monitoring window, automation notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If keys.storage is UNKNOWN → block Completeness Gate.
If version.build_number_rule is UNKNOWN → block Completeness Gate.
If health.crash_rate_gate is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.MOB
Pass conditions:
required_fields_present == true
signing_and_submission_defined == true
versioning_defined == true
release_health_gates_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

Mobile Device Capabilities (MDC)

Mobile Device Capabilities (MDC)
MDC-01 Capabilities Inventory (camera/gps/files/push)
MDC-02 Permissions UX Rules (prompts, denial handling)
MDC-03 Capability Security Rules (least privilege)
MDC-04 Capability Failure Handling (fallbacks)
MDC-05 Telemetry for Capabilities (errors, latency)

MDC-01

MDC-01 — Capabilities Inventory (camera/gps/files/push)
Header Block
template_id: MDC-01
title: Capabilities Inventory (camera/gps/files/push)
type: mobile_capabilities_inventory
template_version: 1.0.0
output_path: 10_app/mobile_capabilities/MDC-01_Capabilities_Inventory.md
compliance_gate_id: TMP-05.PRIMARY.MDC
upstream_dependencies: ["SPEC_INDEX", "MOB-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "MOB-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Create the canonical inventory of mobile device capabilities used by the app, indexed by capability_id, including what features/screens use them and the high-level purpose. This template must be consistent with mobile integration maps and must not invent capability_ids not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
MOB-03 Native Integration Map: {{mob.native_integrations}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Capability registry (capability_id list)
capability_id (stable identifier)
name (human-readable)
category (camera/location/files/push/etc.)
purpose (what it enables)
screens/features using it (screen_id list)
permissions required pointer (MDC-02)
security constraints pointer (MDC-03)
fallback behavior pointer (MDC-04)

Optional Fields
Platform availability notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL

Do not introduce new capability_ids unless explicitly allowed; otherwise use UNKNOWN and flag.
Each capability MUST bind to at least one screen/feature or be flagged as unused.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Inventory Summary
   total_capabilities: {{caps.total}}
   categories: {{caps.categories}} | OPTIONAL
2. Capability Entries (by capability_id)
   Capability
   capability_id: {{items[0].capability_id}}
   name: {{items[0].name}}
   category: {{items[0].category}}
   purpose: {{items[0].purpose}}
   screens: {{items[0].screens}}
   permissions_ref: {{items[0].permissions_ref}} (expected: {{xref:MDC-02}}) | OPTIONAL
   security_ref: {{items[0].security_ref}} (expected: {{xref:MDC-03}}) | OPTIONAL
   fallback_ref: {{items[0].fallback_ref}} (expected: {{xref:MDC-04}}) | OPTIONAL
   platform_notes: {{items[0].platform_notes}} | OPTIONAL
   open_questions:
   {{items[0].open_questions[0]}} | OPTIONAL

(Repeat per capability_id.)

3. References
   Permissions UX: {{xref:MDC-02}} | OPTIONAL
   Capability security: {{xref:MDC-03}} | OPTIONAL
   Failure handling: {{xref:MDC-04}} | OPTIONAL
   Telemetry: {{xref:MDC-05}} | OPTIONAL

Cross-References
Upstream: {{xref:MOB-03}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:MDC-02}}, {{xref:MDC-03}}, {{xref:MDC-04}}, {{xref:MDC-05}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Use UNKNOWN for optional refs; do not invent capabilities.
intermediate: Required. Populate screens and references to permission/security/fallback.
advanced: Required. Add platform availability nuances and traceability pointers.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, categories summary, refs, platform

notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If capability registry is UNKNOWN → block Completeness Gate.
If items[*].category is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.MDC
Pass conditions:
required_fields_present == true
capability_registry_defined == true
each capability binds to screens/features
placeholder_resolution == true
no_unapproved_unknowns == true

MDC-02

MDC-02 — Permissions UX Rules (prompts, denial handling)
Header Block
template_id: MDC-02
title: Permissions UX Rules (prompts, denial handling)
type: mobile_permissions_ux_rules
template_version: 1.0.0
output_path: 10_app/mobile_capabilities/MDC-02_Permissions_UX_Rules.md
compliance_gate_id: TMP-05.PRIMARY.MDC
upstream_dependencies: ["MDC-01", "MOB-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "MDC-01", "MOB-03", "MDC-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical UX rules for requesting and managing mobile permissions: when to request, how to explain, how to handle denial, and how to route users to settings. This template must be consistent with capability inventory and native integration map and must not invent permissions not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
MDC-01 Capabilities Inventory: {{mdc.capabilities}}
MOB-03 Native Integration Map: {{mob.native_integrations}} | OPTIONAL
MDC-04 Capability Failure Handling: {{mdc.failures}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Request timing rules (on action vs upfront)
Pre-permission rationale rules (what to say)
Permission prompt copy policy (tone)
Denial handling rules (temporary vs permanent)
“Go to settings” routing rules
Retry rules (when to re-ask)
Per-capability permission mapping (capability_id → permission set)
Accessibility considerations (announce prompts)

Optional Fields
Batch permission request rules | OPTIONAL
OS version nuances | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Request least privilege; only ask when needed.
Denial handling MUST provide fallback or clear blocked state per {{xref:MDC-04}} when applicable.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Request Timing
   default_timing: {{timing.default}}
   on_action_rule: {{timing.on_action_rule}}
   never_upfront_for: {{timing.never_upfront_for}} | OPTIONAL
2. Rationale / Copy
   rationale_required: {{copy.rationale_required}}
   copy_tone: {{copy.tone}}
   template_rationale_text: {{copy.template_rationale_text}} | OPTIONAL
3. Denial Handling
   temporary_denial_behavior: {{deny.temporary_behavior}}
   permanent_denial_behavior: {{deny.permanent_behavior}}
   blocked_state_ui_rule: {{deny.blocked_state_ui_rule}} | OPTIONAL
4. Settings Routing
   settings_deeplink_supported: {{settings.deeplink_supported}}
   settings_route_rule: {{settings.route_rule}}
5. Retry Rules
   when_to_retry_prompt: {{retry.when_to_retry_prompt}}
   cooldown_ms: {{retry.cooldown_ms}} | OPTIONAL
6. Per-Capability Permission Map
   Map
   capability_id: {{map[0].capability_id}}
   ios_permissions: {{map[0].ios_permissions}} | OPTIONAL
   android_permissions: {{map[0].android_permissions}} | OPTIONAL
   request_timing_override: {{map[0].timing_override}} | OPTIONAL
   rationale_copy: {{map[0].rationale_copy}} | OPTIONAL

(Repeat per capability_id.)

7. Accessibility
   announce_permission_dialog: {{a11y.announce_permission_dialog}}
   focus_rules: {{a11y.focus_rules}} | OPTIONAL
8. References
   Capabilities inventory: {{xref:MDC-01}}
   Failure handling: {{xref:MDC-04}} | OPTIONAL
   Telemetry: {{xref:MDC-05}} | OPTIONAL

Cross-References
Upstream: {{xref:MDC-01}}, {{xref:MOB-03}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:MDC-04}}, {{xref:MDC-05}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-A11Y]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define timing + denial handling baseline; use UNKNOWN for OS nuances.
intermediate: Required. Populate per-capability permission map and settings routing.
advanced: Required. Add batching policies and per-OS differences with precise rules.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, never upfront list, template rationale, blocked state UI, cooldown, per-capability permission details, timing overrides, rationale copy, focus rules, batching/os nuances, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If timing.on_action_rule is UNKNOWN → block Completeness Gate.
If settings.route_rule is UNKNOWN → block Completeness Gate.
If per-capability map is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.MDC
Pass conditions:
required_fields_present == true
timing_and_denial_rules_defined == true
capability_permission_map_defined == true
settings_routing_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

MDC-03

MDC-03 — Capability Security Rules (least privilege)
Header Block
template_id: MDC-03
title: Capability Security Rules (least privilege)
type: mobile_capability_security_rules
template_version: 1.0.0
output_path: 10_app/mobile_capabilities/MDC-03_Capability_Security_Rules.md
compliance_gate_id: TMP-05.PRIMARY.MDC
upstream_dependencies: ["MDC-01", "CSec-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "MDC-01", "CSec-02", "MOB-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical security rules for using mobile device capabilities, enforcing least privilege, safe data handling, and controlled access. Includes capability-specific constraints, data minimization, and secure-by-default patterns. This template must be consistent with client data protection policies and must not invent capability usage not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
MDC-01 Capabilities Inventory: {{mdc.capabilities}}
CSec-02 Client Data Protection: {{csec.data_protection}} | OPTIONAL
MOB-03 Native Integration Map: {{mob.native_integrations}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Least privilege statement (general)
Per-capability security constraints (capability_id → rules)
Data minimization rules (collect only needed fields)
Sensitive data handling rules (encryption/redaction)
Permission scope rules (foreground only vs background)
Access logging rules (if relevant)
Third-party SDK constraints (if capability uses SDKs)

Optional Fields
Threat considerations per capability | OPTIONAL
App attestation bindings | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Capability use MUST follow CSec-02 protections for sensitive data.
Do not request broader permissions than necessary.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. General Least Privilege
   policy_statement: {{policy.statement}}
   data_minimization_rule: {{policy.data_minimization_rule}} | OPTIONAL
2. Per-Capability Security Constraints
   Capability Rule
   capability_id: {{rules[0].capability_id}}
   allowed_context: {{rules[0].allowed_context}} (foreground/background/UNKNOWN)
   data_collected: {{rules[0].data_collected}} | OPTIONAL
   storage_rules: {{rules[0].storage_rules}} | OPTIONAL
   transmission_rules: {{rules[0].transmission_rules}} | OPTIONAL
   redaction_rules: {{rules[0].redaction_rules}} | OPTIONAL
   third_party_constraints: {{rules[0].third_party_constraints}} | OPTIONAL
   logging_rules: {{rules[0].logging_rules}} | OPTIONAL
   notes: {{rules[0].notes}} | OPTIONAL
   open_questions:
   {{rules[0].open_questions[0]}} | OPTIONAL

(Repeat per capability_id.)

3. Sensitive Data Handling
   encryption_required: {{sensitive.encryption_required}}
   pii_redaction_required: {{sensitive.pii_redaction_required}} | OPTIONAL
4. Third-Party SDK Constraints
   sdk_allowlist_policy: {{sdk.allowlist_policy}} | OPTIONAL
   data_sharing_rules: {{sdk.data_sharing_rules}} | OPTIONAL
5. References
   Capabilities inventory: {{xref:MDC-01}}
   Client data protection: {{xref:CSec-02}} | OPTIONAL
   Native integrations: {{xref:MOB-03}} | OPTIONAL

Cross-References
Upstream: {{xref:MDC-01}}, {{xref:CSec-02}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:MDC-04}}, {{xref:MDC-05}} | OPTIONAL
Standards: {{standards.rules[STD-SECURITY]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define least privilege + per-capability allowed context.
intermediate: Required. Add storage/transmission/redaction constraints.
advanced: Required. Add threat notes and third-party SDK governance.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, data collected/storage/transmission/redaction, third party constraints, logging rules, notes, data minimization rule, pii redaction, sdk rules, attestation/threat notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If policy.statement is UNKNOWN → block Completeness Gate.
If rules[*].capability_id is UNKNOWN → block Completeness Gate.*
*If rules[*].allowed_context is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.MDC
Pass conditions:
required_fields_present == true
least_privilege_defined == true
per_capability_rules_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

MDC-04

MDC-04 — Capability Failure Handling (fallbacks)
Header Block
template_id: MDC-04
title: Capability Failure Handling (fallbacks)
type: mobile_capability_failure_handling
template_version: 1.0.0
output_path: 10_app/mobile_capabilities/MDC-04_Capability_Failure_Handling.md
compliance_gate_id: TMP-05.PRIMARY.MDC
upstream_dependencies: ["MDC-01", "MDC-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "MDC-01", "MDC-02", "CER-03"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical failure handling and fallback behaviors for mobile device capabilities, including permission denial, hardware unavailability, OS restrictions, and transient errors. This template must be consistent with permissions UX rules and offline/degraded mode UX and must not invent fallback behaviors not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
MDC-01 Capabilities Inventory: {{mdc.capabilities}}
MDC-02 Permissions UX Rules: {{mdc.permissions_ux}} | OPTIONAL
CER-03 Offline/Error Mode UX: {{cer.offline_mode}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Failure types taxonomy (denied/unavailable/offline/error)
Per-capability fallback rules (capability_id → fallback behavior)
Permission denied behavior (temporary vs permanent)
Hardware unavailable behavior
Offline behavior (if capability depends on network)
User messaging/copy policy (safe messaging)
Retry rules (when to retry capability access)
Telemetry requirements (denial/unavailable rates)

Optional Fields
Per-screen fallback variants | OPTIONAL
Support contact policy | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Fallbacks MUST be safe and should preserve core app usability when possible.
Permission denial handling MUST align to {{xref:MDC-02}}.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Failure Types
   types:
   denied: {{fail.denied}}
   unavailable: {{fail.unavailable}}
   offline: {{fail.offline}} | OPTIONAL
   error: {{fail.error}}
2. Per-Capability Fallbacks
   Fallback
   capability_id: {{fallbacks[0].capability_id}}
   on_temp_denied: {{fallbacks[0].on_temp_denied}}
   on_perm_denied: {{fallbacks[0].on_perm_denied}} | OPTIONAL
   on_unavailable: {{fallbacks[0].on_unavailable}}
   on_offline: {{fallbacks[0].on_offline}} | OPTIONAL
   retry_rule: {{fallbacks[0].retry_rule}} | OPTIONAL
   copy_policy: {{fallbacks[0].copy_policy}} | OPTIONAL
   notes: {{fallbacks[0].notes}} | OPTIONAL
   open_questions:
   {{fallbacks[0].open_questions[0]}} | OPTIONAL

(Repeat per capability_id.)

3. User Messaging
   tone: {{copy.tone}}
   safe_message_rules: {{copy.safe_message_rules}} | OPTIONAL
4. Retry Rules
   retry_allowed: {{retry.allowed}}
   retry_trigger: {{retry.trigger}} | OPTIONAL
   cooldown_ms: {{retry.cooldown_ms}} | OPTIONAL
5. Telemetry
   denial_metric: {{telemetry.denial_metric}}
   unavailable_metric: {{telemetry.unavailable_metric}} | OPTIONAL
   fields: {{telemetry.fields}} | OPTIONAL
6. References
   Capabilities inventory: {{xref:MDC-01}}
   Permissions UX rules: {{xref:MDC-02}} | OPTIONAL

Offline UX: {{xref:CER-03}} | OPTIONAL
Telemetry: {{xref:MDC-05}} | OPTIONAL

Cross-References
Upstream: {{xref:MDC-01}}, {{xref:MDC-02}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:MDC-05}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define denied/unavailable fallback basics; use UNKNOWN for per-screen nuance.
intermediate: Required. Define retry rules and telemetry mapping.
advanced: Required. Add support escalation and per-screen variants.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, offline type, perm denied behavior, on_offline, retry rules, copy policy, notes, safe message rules, retry trigger/cooldown, telemetry optional metrics/fields, per-screen variants, support policy, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If fail.error is UNKNOWN → block Completeness Gate.
If fallbacks list is UNKNOWN → block Completeness Gate.
If telemetry.denial_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.MDC
Pass conditions:
required_fields_present == true
failure_types_defined == true
per_capability_fallbacks_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

MDC-05

MDC-05 — Telemetry for Capabilities (errors, latency)
Header Block
template_id: MDC-05
title: Telemetry for Capabilities (errors, latency)
type: mobile_capability_telemetry
template_version: 1.0.0
output_path: 10_app/mobile_capabilities/MDC-05_Telemetry_for_Capabilities.md
compliance_gate_id: TMP-05.PRIMARY.MDC
upstream_dependencies: ["MDC-01", "MDC-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "MDC-01", "MDC-04", "SMD-06", "CER-05"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical telemetry requirements for mobile device capabilities: what metrics/logs/traces to collect for capability usage, errors, latency, permission outcomes, and fallback usage. This template must be consistent with client observability and logging/redaction rules and must not invent telemetry that violates privacy constraints.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
MDC-01 Capabilities Inventory: {{mdc.capabilities}}
MDC-04 Failure Handling: {{mdc.failures}} | OPTIONAL
SMD-06 Client Data Layer Observability: {{smd.observability}} | OPTIONAL
CER-05 Client Logging/Crash Reporting: {{cer.logging}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Core capability metrics set (attempts, success, failure)
Latency metrics (time-to-result)
Error taxonomy (denied/unavailable/error)
Permission outcome metrics (granted/denied/permanent)
Fallback usage metrics
Log field requirements (capability_id, error_type)
Redaction/privacy constraints (no PII)
Dashboards minimum panels
Alerts (spikes in denied/unavailable/errors)

Optional Fields
Tracing spans for capability ops | OPTIONAL

Sampling policy | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
No sensitive data in telemetry.
Metrics MUST use stable identifiers (capability_id) and avoid high-cardinality labels.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Core Metrics
   attempts: {{metrics.attempts}}
   success: {{metrics.success}}
   failure: {{metrics.failure}}
2. Latency Metrics
   latency_ms: {{metrics.latency_ms}} | OPTIONAL
   time_to_first_result_ms: {{metrics.ttfr_ms}} | OPTIONAL
3. Error Taxonomy
   denied: {{errors.denied}}
   unavailable: {{errors.unavailable}}
   error: {{errors.error}}
4. Permission Outcomes
   granted: {{perm.granted}}
   denied_temp: {{perm.denied_temp}} | OPTIONAL
   denied_perm: {{perm.denied_perm}} | OPTIONAL
5. Fallback Usage
   fallback_invocations: {{fallback.invocations}} | OPTIONAL
   fallback_types: {{fallback.types}} | OPTIONAL
6. Logs — Required Fields
   timestamp: {{logs.timestamp}}
   capability_id: {{logs.capability_id}}
   result: {{logs.result}}
   error_type: {{logs.error_type}} | OPTIONAL
   latency_ms: {{logs.latency_ms}} | OPTIONAL
   screen_id: {{logs.screen_id}} | OPTIONAL
   app_version: {{logs.app_version}} | OPTIONAL
   redaction_applied: {{logs.redaction_applied}} | OPTIONAL
7. Privacy / Redaction
   no_pii_rule: {{privacy.no_pii_rule}}
   field_allowlist: {{privacy.field_allowlist}} | OPTIONAL
8. Dashboards
   capability_health_dashboard: {{dashboards.health}}
   minimum_panels: {{dashboards.minimum_panels}}

9. Alerts
   denial_spike_alert: {{alerts.denial_spike_alert}}
   unavailable_spike_alert: {{alerts.unavailable_spike_alert}} | OPTIONAL
   error_spike_alert: {{alerts.error_spike_alert}}
   threshold_model: {{alerts.threshold_model}} (static/baseline/UNKNOWN)
   routing_policy: {{alerts.routing_policy}} | OPTIONAL
10. References
   Capabilities inventory: {{xref:MDC-01}}
   Failure handling: {{xref:MDC-04}} | OPTIONAL
   Client observability: {{xref:SMD-06}} | OPTIONAL
   Client logging/redaction: {{xref:CER-05}} | OPTIONAL

Cross-References
Upstream: {{xref:MDC-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:OPS-OBS}} | OPTIONAL
Standards: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define attempts/success/failure and required log fields; use UNKNOWN for tracing.
intermediate: Required. Define dashboards and denial/error alerts.
advanced: Required. Add sampling, tracing spans, and alert threshold rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, latency metrics, denied temp/perm, fallback details, optional log fields, field allowlist, routing policy, sampling/tracing, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If logs.capability_id is UNKNOWN → block Completeness Gate.
If privacy.no_pii_rule is UNKNOWN → block Completeness Gate.
If dashboards.minimum_panels is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.MDC
Pass conditions:
required_fields_present == true
core_metrics_defined == true
log_fields_defined == true
privacy_constraints_defined == true
dashboard_minimum_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

Offline & Sync (OFS)

Offline & Sync (OFS)
OFS-01 Offline Scope (what works offline)
OFS-02 Sync Model (queues, conflict resolution)
OFS-03 Local Storage Schema (entities cached locally)
OFS-04 Reconciliation Rules (merge, LWW, prompts)
OFS-05 Sync Observability (stuck queues, retries, metrics)

OFS-01

OFS-01 — Offline Scope (what works offline)
Header Block
template_id: OFS-01
title: Offline Scope (what works offline)
type: offline_scope_spec
template_version: 1.0.0
output_path: 10_app/offline_sync/OFS-01_Offline_Scope.md
compliance_gate_id: TMP-05.PRIMARY.OFS
upstream_dependencies: ["SMD-05", "CER-03", "PRD-06"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "SMD-05", "CER-03", "PRD-06", "FE-01"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical offline scope: exactly what features/screens/actions work offline, what is read-only, what is blocked, and what is queued for later sync. This template must be consistent with offline handling model and offline UX rules and must not invent offline support not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
SMD-05 Offline State Handling: {{smd.offline_handling}}
CER-03 Offline/Error Mode UX: {{cer.offline_ux}} | OPTIONAL
PRD-06 NFRs: {{prd.nfrs}} | OPTIONAL
FE-01 Route Map + Layout: {{fe.route_layout}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Offline scope statement (overall)
Per-feature offline capability map (feature_id/screen_id → mode)
Modes supported (full/read-only/blocked/queued)
Read behavior rules (cache/stale limits)
Write behavior rules (queued vs blocked)
User messaging rules (which banner/indicators)
Security constraints (what cannot be cached offline)
Open gaps list (what is not supported offline)

Optional Fields
Per-role offline differences | OPTIONAL
Data size constraints | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Offline scope MUST match what is implementable per {{xref:SMD-05}}.
Do not claim offline support for any screen/action not explicitly mapped.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Offline Scope Statement
   statement: {{scope.statement}}
   modes_supported: {{scope.modes_supported}}
   (full/read_only/blocked/queued/UNKNOWN)
2. Per-Feature / Screen Map
   Entry
   feature_id: {{map[0].feature_id}} | OPTIONAL
   screen_id: {{map[0].screen_id}} | OPTIONAL
   route_id: {{map[0].route_id}} | OPTIONAL
   offline_mode: {{map[0].offline_mode}}
   reads: {{map[0].reads}} | OPTIONAL
   writes: {{map[0].writes}} | OPTIONAL
   notes: {{map[0].notes}} | OPTIONAL

(Repeat per mapped item.)

3. Read Rules
   read_source: {{reads.source}}
   stale_limits: {{reads.stale_limits}} | OPTIONAL
4. Write Rules
   queue_supported: {{writes.queue_supported}}
   blocked_operations: {{writes.blocked_operations}} | OPTIONAL
5. Messaging / Indicators
   banner_policy_ref: {{ux.banner_policy_ref}} (expected: {{xref:CER-03}}) | OPTIONAL
   queued_indicator_policy: {{ux.queued_indicator_policy}} | OPTIONAL
6. Security Constraints
   do_not_cache: {{security.do_not_cache}}
   sensitive_data_rules_ref: {{security.sensitive_data_rules_ref}} (expected: {{xref:CSec-02}}) | OPTIONAL
7. Open Gaps
   not_supported_offline: {{gaps.not_supported_offline}}
   future_candidates: {{gaps.future_candidates}} | OPTIONAL
8. References
   Offline handling: {{xref:SMD-05}}
   Offline UX: {{xref:CER-03}} | OPTIONAL

Client data protection: {{xref:CSec-02}} | OPTIONAL
Sync model: {{xref:OFS-02}} | OPTIONAL

Cross-References
Upstream: {{xref:SMD-05}}, {{xref:CER-03}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:OFS-02}}, {{xref:OFS-03}}, {{xref:OFS-04}}, {{xref:OFS-05}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define scope statement and map entries with UNKNOWN where needed.
intermediate: Required. Define read/write rules and security constraints.
advanced: Required. Add per-role differences and data size constraints.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, feature_id, screen/route ids, reads/writes notes, stale limits, blocked ops, banner/queued indicator policies, sensitive data ref, future candidates, role differences, data size constraints, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If scope.statement is UNKNOWN → block Completeness Gate.
If map entries are UNKNOWN → block Completeness Gate.
If security.do_not_cache is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.OFS
Pass conditions:
required_fields_present == true
offline_scope_mapped == true
read_write_rules_defined == true
security_constraints_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

OFS-02

OFS-02 — Sync Model (queues, conflict resolution)
Header Block
template_id: OFS-02
title: Sync Model (queues, conflict resolution)
type: offline_sync_model
template_version: 1.0.0
output_path: 10_app/offline_sync/OFS-02_Sync_Model.md
compliance_gate_id: TMP-05.PRIMARY.OFS
upstream_dependencies: ["SMD-05", "OFS-01", "SMD-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "OFS-01", "SMD-05", "SMD-03", "EVT-04"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical offline sync model: how queued ops are stored and drained, ordering guarantees, conflict detection/resolution, and reconciliation between local state and server truth. This template must be consistent with offline queue and mutation patterns and must not invent sync semantics not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
OFS-01 Offline Scope: {{ofs.scope}}
SMD-05 Offline State Handling: {{smd.offline_handling}}
SMD-03 Mutation Patterns: {{smd.mutation_patterns}} | OPTIONAL
EVT-04 Delivery Semantics: {{evt.delivery_semantics}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Queue drain strategy (when and how)
Ordering guarantees (FIFO per entity, etc.)
Batching policy (if any)
Conflict detection rules (version, etag)
Conflict resolution policy (LWW/merge/prompt)
Retry/backoff policy for sync
Partial failure handling (some ops fail)
Reconciliation strategy after drain (refresh queries)
Telemetry requirements (queue drain success/fail, conflicts)

Optional Fields
Priority lanes (high/low) | OPTIONAL

Background sync constraints | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Sync must preserve user intent and not silently drop ops.
Conflict policy MUST be explicit and deterministic.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Queue Drain
   drain_trigger: {{drain.trigger}} (reconnect/app_resume/manual/UNKNOWN)
   drain_order: {{drain.order}}
   batching_enabled: {{drain.batching_enabled}} | OPTIONAL
   max_batch_size: {{drain.max_batch_size}} | OPTIONAL
2. Ordering Guarantees
   ordering_model: {{ordering.model}} (fifo_per_queue/fifo_per_entity/UNKNOWN)
   dedupe_rule: {{ordering.dedupe_rule}} | OPTIONAL
3. Conflict Detection
   version_field_rule: {{conflict.version_field_rule}}
   etag_rule: {{conflict.etag_rule}} | OPTIONAL
4. Conflict Resolution
   policy: {{conflict.policy}} (LWW/merge/prompt/UNKNOWN)
   user_prompt_rule: {{conflict.user_prompt_rule}} | OPTIONAL
   merge_rules: {{conflict.merge_rules}} | OPTIONAL
5. Retry / Backoff
   backoff_policy: {{retry.backoff_policy}}
   max_attempts: {{retry.max_attempts}} | OPTIONAL
   retryable_errors: {{retry.retryable_errors}} | OPTIONAL
6. Partial Failure Handling
   on_partial_failure: {{partial.on_partial_failure}}
   poison_message_rule: {{partial.poison_message_rule}} | OPTIONAL
7. Reconciliation After Drain
   refresh_rule: {{reconcile.refresh_rule}}
   queries_to_refetch: {{reconcile.queries_to_refetch}} | OPTIONAL
8. Telemetry
   drain_success_metric: {{telemetry.drain_success_metric}}
   drain_failure_metric: {{telemetry.drain_failure_metric}} | OPTIONAL
   conflict_metric: {{telemetry.conflict_metric}} | OPTIONAL
   fields: {{telemetry.fields}} | OPTIONAL
9. References
   Offline scope: {{xref:OFS-01}}
   Offline handling: {{xref:SMD-05}}

Mutation patterns: {{xref:SMD-03}} | OPTIONAL
Retry patterns: {{xref:CER-02}} | OPTIONAL

Cross-References
Upstream: {{xref:OFS-01}}, {{xref:SMD-05}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:OFS-03}}, {{xref:OFS-04}}, {{xref:OFS-05}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define drain trigger + ordering model; use UNKNOWN for batching.
intermediate: Required. Define conflict policy and retry/backoff.
advanced: Required. Add poison/partial failure handling and background constraints.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, batching/max batch size, dedupe rule, etag rule, user prompt/merge rules, max attempts/retryable errors, poison message rule, queries to refetch, telemetry fields, priority lanes, background constraints, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If drain.order is UNKNOWN → block Completeness Gate.
If conflict.version_field_rule is UNKNOWN → block Completeness Gate.
If retry.backoff_policy is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.OFS
Pass conditions:
required_fields_present == true
drain_and_ordering_defined == true
conflict_policy_defined == true
retry_backoff_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

OFS-03

OFS-03 — Local Storage Schema (entities cached locally)
Header Block
template_id: OFS-03
title: Local Storage Schema (entities cached locally)
type: offline_local_storage_schema
template_version: 1.0.0
output_path: 10_app/offline_sync/OFS-03_Local_Storage_Schema.md
compliance_gate_id: TMP-05.PRIMARY.OFS
upstream_dependencies: ["OFS-01", "SMD-02", "CSec-02", "DATA-06"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "OFS-01", "SMD-02", "CSec-02", "DATA-06"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical local storage schema for offline use: which entities are cached on-device, table/collection structure, keys/indexes, retention/eviction rules, and encryption constraints. This template must be consistent with offline scope and data protection policies and must not invent local data storage beyond what is allowed.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
OFS-01 Offline Scope: {{ofs.scope}}
SMD-02 Cache Strategy: {{smd.cache_strategy}}
CSec-02 Data Protection: {{csec.data_protection}} | OPTIONAL
DATA-06 Canonical Data Schemas: {{data.schemas}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Storage engine (sqlite/asyncstorage/realm/UNKNOWN)
Cached entity list (entity_id/schema_ref list)
Per-entity storage shape (table/collection fields)
Primary keys and indexes
Versioning/migrations rules (schema changes)
Retention/eviction rules (TTL, max size)
Encryption requirements (at rest)
PII constraints (do-not-store classes)
Queue storage schema pointer (SMD-05/OFS-02)

Optional Fields
Delta sync fields (last_synced_at) | OPTIONAL

Compression policy | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not store prohibited data classes per {{xref:CSec-02}}.
Schema MUST support migrations without data loss where possible.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Storage Engine
   engine: {{storage.engine}} (sqlite/realm/asyncstorage/UNKNOWN)
   notes: {{storage.notes}} | OPTIONAL
2. Cached Entities
   Entity Cache
   entity_id: {{entities[0].entity_id}}
   schema_ref: {{entities[0].schema_ref}} | OPTIONAL
   storage_object_name: {{entities[0].storage_object_name}}
   primary_key: {{entities[0].primary_key}}
   indexes: {{entities[0].indexes}} | OPTIONAL
   fields_stored: {{entities[0].fields_stored}}
   pii_notes: {{entities[0].pii_notes}} | OPTIONAL
   retention_policy: {{entities[0].retention_policy}} | OPTIONAL
   open_questions:
   {{entities[0].open_questions[0]}} | OPTIONAL

(Repeat per cached entity.)

3. Migrations / Versioning
   schema_version_field: {{migrate.schema_version_field}}
   migration_policy: {{migrate.policy}}
   fallback_on_migration_fail: {{migrate.fallback_on_fail}} | OPTIONAL
4. Retention / Eviction
   default_ttl_ms: {{retention.default_ttl_ms}} | OPTIONAL
   max_storage_mb: {{retention.max_storage_mb}} | OPTIONAL
   eviction_strategy: {{retention.eviction_strategy}} (lru/ttl/UNKNOWN)
5. Encryption & PII Constraints
   encryption_required: {{security.encryption_required}}
   do_not_store_classes: {{security.do_not_store_classes}}
   secure_storage_binding: {{security.secure_storage_binding}} | OPTIONAL
6. Queue Storage Pointer
   queue_schema_ref: {{queue.schema_ref}} (expected: {{xref:SMD-05}}/{{xref:OFS-02}}) | OPTIONAL

7. References
   Offline scope: {{xref:OFS-01}}
   Cache strategy: {{xref:SMD-02}}
   Data protection: {{xref:CSec-02}} | OPTIONAL
   Data schemas: {{xref:DATA-06}} | OPTIONAL
   Sync model: {{xref:OFS-02}} | OPTIONAL

Cross-References
Upstream: {{xref:OFS-01}}, {{xref:SMD-02}}, {{xref:CSec-02}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:OFS-04}}, {{xref:OFS-05}} | OPTIONAL
Standards: {{standards.rules[STD-SECURITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define engine and cached entities list; use UNKNOWN for indexes/TTL.
intermediate: Required. Define migrations and eviction rules.
advanced: Required. Add encryption details and per-entity retention/PII notes.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, schema_ref, indexes, pii notes, retention policy, migration fallback, default ttl/max size, secure storage binding, queue schema ref, delta/compression, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If storage.engine is UNKNOWN → block Completeness Gate.
If entities list is UNKNOWN → block Completeness Gate.
If security.do_not_store_classes is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.OFS
Pass conditions:
required_fields_present == true
storage_engine_defined == true
cached_entities_defined == true
migrations_defined == true
security_constraints_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

OFS-04

OFS-04 — Reconciliation Rules (merge, LWW, prompts)
Header Block
template_id: OFS-04
title: Reconciliation Rules (merge, LWW, prompts)
type: offline_reconciliation_rules
template_version: 1.0.0
output_path: 10_app/offline_sync/OFS-04_Reconciliation_Rules.md
compliance_gate_id: TMP-05.PRIMARY.OFS
upstream_dependencies: ["OFS-02", "SMD-03", "API-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "OFS-02", "SMD-03", "API-03", "FE-07"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical reconciliation rules for resolving divergence between local/offline state and server truth, including merge strategy, LWW rules, user prompts, and how reconciliation errors surface to the user. This template must be consistent with sync/mutation patterns and API error policy and must not invent conflict semantics not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
OFS-02 Sync Model: {{ofs.sync_model}}
SMD-03 Mutation Patterns: {{smd.mutation_patterns}} | OPTIONAL
API-03 Error Policy: {{api.error_policy}} | OPTIONAL
FE-07 Error UX: {{fe.error_ux}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Reconciliation trigger points (after sync, on open, on conflict)
Conflict taxonomy (types of conflicts)
Default resolution policy (LWW/merge/prompt)
Per-entity/per-field resolution rules (if applicable)
User prompt rules (when prompting is required)
Merge rules (field-level precedence)
Audit/telemetry requirements (conflict events)
Error surfacing rules (bind to FE-07)

Optional Fields
“Keep mine / keep theirs” UX copy policy | OPTIONAL
Batch reconciliation rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Resolution policy MUST be deterministic and explicit.
User prompts MUST not be spammy; consolidate when possible.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Triggers
   triggers: {{recon.triggers}}
   notes: {{recon.notes}} | OPTIONAL
2. Conflict Taxonomy
   types: {{conflicts.types}}
   examples: {{conflicts.examples}} | OPTIONAL
3. Default Policy
   default_policy: {{policy.default}} (LWW/merge/prompt/UNKNOWN)
   policy_rationale: {{policy.rationale}} | OPTIONAL
4. Per-Entity / Field Rules
   Rule
   entity_id: {{rules[0].entity_id}}
   field: {{rules[0].field}} | OPTIONAL
   resolution: {{rules[0].resolution}}
   precedence: {{rules[0].precedence}} | OPTIONAL
   prompt_required: {{rules[0].prompt_required}} | OPTIONAL

(Repeat per entity/field rule.)

5. User Prompt Rules
   when_to_prompt: {{prompt.when_to_prompt}}
   prompt_ui_pattern: {{prompt.ui_pattern}} (modal/screen/UNKNOWN)
   copy_policy: {{prompt.copy_policy}} | OPTIONAL
6. Merge Rules
   merge_strategy: {{merge.strategy}} (field_level/object_level/UNKNOWN)
   merge_precedence: {{merge.precedence}} | OPTIONAL
7. Telemetry / Audit
   conflict_metric: {{telemetry.conflict_metric}}
   resolution_metric: {{telemetry.resolution_metric}} | OPTIONAL
   fields: {{telemetry.fields}} | OPTIONAL
8. Error Surfacing
   error_ux_ref: {{errors.ux_ref}} (expected: {{xref:FE-07}}) | OPTIONAL
   unresolved_conflict_behavior: {{errors.unresolved_conflict_behavior}}
9. References
   Sync model: {{xref:OFS-02}}
   Offline handling: {{xref:SMD-05}} | OPTIONAL

Mutation patterns: {{xref:SMD-03}} | OPTIONAL
Error UX: {{xref:FE-07}} | OPTIONAL

Cross-References
Upstream: {{xref:OFS-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:OFS-05}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define default policy and basic prompt rules; use UNKNOWN for per-field rules.
intermediate: Required. Define per-entity rules and merge precedence.
advanced: Required. Add batching and refined prompt UX policy.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, examples, policy rationale, field/precedence/prompt_required, prompt UI/copy, merge precedence, telemetry fields, error ux ref, copy policy, batching, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If policy.default is UNKNOWN → block Completeness Gate.
If telemetry.conflict_metric is UNKNOWN → block Completeness Gate.
If errors.unresolved_conflict_behavior is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.OFS
Pass conditions:
required_fields_present == true
default_policy_defined == true
reconciliation_rules_defined == true
telemetry_defined == true
error_surfacing_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

OFS-05

OFS-05 — Sync Observability (stuck queues, retries, metrics)
Header Block
template_id: OFS-05
title: Sync Observability (stuck queues, retries, metrics)
type: offline_sync_observability
template_version: 1.0.0
output_path: 10_app/offline_sync/OFS-05_Sync_Observability.md
compliance_gate_id: TMP-05.PRIMARY.OFS
upstream_dependencies: ["OFS-02", "SMD-06", "CER-05"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "OFS-02", "SMD-06", "CER-05", "CER-02"]
required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose
Define the canonical observability requirements for offline sync: metrics for queue depth/drain success, stuck queue detection, retry/backoff visibility, conflict rates, and alerts/dashboards for operational monitoring. This template must be consistent with client data-layer observability and logging/redaction policies and must not invent telemetry fields that violate privacy constraints.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
OFS-02 Sync Model: {{ofs.sync_model}}
SMD-06 Client Data Layer Observability: {{smd.observability}} | OPTIONAL
CER-05 Client Logging/Crash Reporting: {{cer.logging}} | OPTIONAL
CER-02 Retry/Recovery Patterns: {{cer.retry_patterns}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Core sync metrics set (queue depth, drain success/fail)
Stuck queue detection rules (time thresholds)
Retry/backoff metrics and visibility
Conflict rate metrics (from OFS-04)
Log field requirements (op_id, attempt_count, error_code)
Dashboards minimum panels
Alerts (stuck queues, drain failure spikes)
Privacy/redaction constraints (no PII)

Optional Fields
Per-entity sync health | OPTIONAL
Sampling policy | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Stuck detection MUST be deterministic and based on measurable time/attempt thresholds.
No PII in sync telemetry/logs.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Core Metrics
   queue_depth: {{metrics.queue_depth}}
   drain_success: {{metrics.drain_success}}
   drain_failure: {{metrics.drain_failure}} | OPTIONAL
   drain_duration_ms: {{metrics.drain_duration_ms}} | OPTIONAL
2. Stuck Queue Detection
   stuck_definition: {{stuck.definition}}
   stuck_threshold_ms: {{stuck.threshold_ms}} | OPTIONAL
   max_attempts_before_stuck: {{stuck.max_attempts}} | OPTIONAL
3. Retry / Backoff Visibility
   retry_attempts_metric: {{retry.retry_attempts_metric}}
   backoff_time_metric: {{retry.backoff_time_metric}} | OPTIONAL
4. Conflict Metrics
   conflict_metric: {{conflicts.metric}} | OPTIONAL
   resolution_metric: {{conflicts.resolution_metric}} | OPTIONAL
5. Logs — Required Fields
   timestamp: {{logs.timestamp}}
   op_id: {{logs.op_id}}
   queue_status: {{logs.queue_status}} | OPTIONAL
   attempt_count: {{logs.attempt_count}} | OPTIONAL
   error_code: {{logs.error_code}} | OPTIONAL
   endpoint_id: {{logs.endpoint_id}} | OPTIONAL
   redaction_applied: {{logs.redaction_applied}} | OPTIONAL
6. Dashboards
   sync_dashboard: {{dashboards.sync_dashboard}}
   minimum_panels: {{dashboards.minimum_panels}}
7. Alerts
   stuck_queue_alert: {{alerts.stuck_queue_alert}}
   drain_failure_spike_alert: {{alerts.drain_failure_spike_alert}} | OPTIONAL
   threshold_model: {{alerts.threshold_model}} (static/baseline/UNKNOWN)
   routing_policy: {{alerts.routing_policy}} | OPTIONAL
8. Privacy / Redaction
   no_pii_rule: {{privacy.no_pii_rule}}
   field_allowlist: {{privacy.field_allowlist}} | OPTIONAL
9. References
   Sync model: {{xref:OFS-02}}

Client observability: {{xref:SMD-06}} | OPTIONAL
Client logging: {{xref:CER-05}} | OPTIONAL
Retry patterns: {{xref:CER-02}} | OPTIONAL

Cross-References
Upstream: {{xref:OFS-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:OPS-OBS}} | OPTIONAL
Standards: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Required. Define queue depth + stuck definition; use UNKNOWN for dashboards/alerts tooling.
intermediate: Required. Define alerts and required log fields.
advanced: Required. Add per-entity health and sampling policies.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, optional metrics, thresholds/max attempts, conflict metrics, optional log fields, field allowlist, routing policy, per-entity/sampling, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If stuck.definition is UNKNOWN → block Completeness Gate.
If logs.op_id is UNKNOWN → block Completeness Gate.
If dashboards.minimum_panels is UNKNOWN → block Completeness Gate.
If privacy.no_pii_rule is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.OFS
Pass conditions:
required_fields_present == true
core_metrics_defined == true
stuck_detection_defined == true
log_fields_defined == true
dashboard_and_alerts_defined == true
privacy_constraints_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

Mobile Performance & Battery (MBAT)

Mobile Performance & Battery (MBAT)
MBAT-01 Background Work Rules (limits, schedules)
MBAT-02 Network Usage Policy (batching, retries, metered)
MBAT-03 Battery Budget & Constraints (targets)
MBAT-04 Perf Profiling Plan (mobile tooling)

MBAT-01

MBAT-01 — Background Work Rules (limits, schedules)
Header Block
template_id: MBAT-01
title: Background Work Rules (limits, schedules)
type: mobile_background_work_rules
template_version: 1.0.0
output_path: 10_app/mobile_perf/MBAT-01_Background_Work_Rules.md
compliance_gate_id: TMP-05.PRIMARY.MBAT
upstream_dependencies: ["MOB-04", "OFS-02", "MPUSH-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "MOB-04", "OFS-02", "MPUSH-04", "MBAT-02"]
required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose
Define the canonical rules for mobile background work: what runs in background, OS constraints, scheduling rules, network/battery constraints, and how background work interacts with offline sync and push delivery. This template must be consistent with mobile lifecycle rules and must not invent background capabilities not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
MOB-04 App Lifecycle + State: {{mob.lifecycle}} | OPTIONAL
OFS-02 Sync Model: {{ofs.sync_model}} | OPTIONAL
MPUSH-04 Delivery/Retry Rules: {{mpush.delivery_retry}} | OPTIONAL
MBAT-02 Network Usage Policy: {{mbat.network_policy}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Background work types allowed (sync, uploads, refresh, notifications)
OS constraints summary (iOS/Android)
Scheduling model (periodic/event-driven)
Network constraints (wifi-only, metered)
Battery constraints (doze/low power)
Max runtime limits (time caps)
Retry/backoff rules for background tasks
User controls (allow background refresh) if applicable
Telemetry requirements (bg task success/fail)

Optional Fields
Per-task priority tiers | OPTIONAL

Foreground service rules (Android) | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Background tasks MUST honor battery/network policies and OS limits.
Background sync MUST align with offline sync model ({{xref:OFS-02}}).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Allowed Background Work
   types: {{bg.types}}
   notes: {{bg.notes}} | OPTIONAL
2. OS Constraints
   ios_constraints: {{os.ios_constraints}}
   android_constraints: {{os.android_constraints}} | OPTIONAL
3. Scheduling
   model: {{schedule.model}} (periodic/event/UNKNOWN)
   triggers: {{schedule.triggers}} | OPTIONAL
   min_interval_minutes: {{schedule.min_interval_minutes}} | OPTIONAL
4. Network Constraints
   wifi_only_rule: {{net.wifi_only_rule}}
   metered_network_rule: {{net.metered_network_rule}} | OPTIONAL
5. Battery Constraints
   low_power_mode_rule: {{battery.low_power_mode_rule}}
   doze_mode_rule: {{battery.doze_mode_rule}} | OPTIONAL
6. Runtime Limits
   max_runtime_seconds: {{runtime.max_runtime_seconds}}
   max_tasks_per_day: {{runtime.max_tasks_per_day}} | OPTIONAL
7. Retry / Backoff
   backoff_policy: {{retry.backoff_policy}}
   max_attempts: {{retry.max_attempts}} | OPTIONAL
8. User Controls
   background_refresh_setting: {{user.background_refresh_setting}} | OPTIONAL
   user_opt_out_supported: {{user.opt_out_supported}} | OPTIONAL
9. Telemetry
   bg_task_success_metric: {{telemetry.success_metric}}
   bg_task_failure_metric: {{telemetry.failure_metric}} | OPTIONAL
   fields: {{telemetry.fields}} | OPTIONAL
10. References
    App lifecycle: {{xref:MOB-04}} | OPTIONAL
    Sync model: {{xref:OFS-02}} | OPTIONAL

Network policy: {{xref:MBAT-02}} | OPTIONAL
Push delivery/retry: {{xref:MPUSH-04}} | OPTIONAL

Cross-References
Upstream: {{xref:MOB-04}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:MBAT-02}}, {{xref:MBAT-05}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Not required.
intermediate: Required. Define allowed types + scheduling + network/battery constraints.
advanced: Required. Add priority tiers and Android foreground service rules if used.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, android constraints, triggers/min interval, metered/doze rules, max tasks per day, max attempts, user controls, telemetry fields, priority tiers/fg service, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If bg.types is UNKNOWN → block Completeness Gate.
If schedule.model is UNKNOWN → block Completeness Gate.
If runtime.max_runtime_seconds is UNKNOWN → block Completeness Gate.
If retry.backoff_policy is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.MBAT
Pass conditions:
required_fields_present == true
allowed_bg_work_defined == true
constraints_defined == true
runtime_and_retry_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

MBAT-02

MBAT-02 — Network Usage Policy (batching, retries, metered)
Header Block
template_id: MBAT-02
title: Network Usage Policy (batching, retries, metered)
type: mobile_network_usage_policy
template_version: 1.0.0
output_path: 10_app/mobile_perf/MBAT-02_Network_Usage_Policy.md
compliance_gate_id: TMP-05.PRIMARY.MBAT
upstream_dependencies: ["CPR-01", "CER-02", "OFS-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "CPR-01", "CER-02", "OFS-02", "SMD-02"]
required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose
Define the canonical policy for mobile network usage: batching rules, metered network behavior, retry/backoff behavior, payload limits, and data-saver mode behavior. This template must be consistent with performance budgets, retry patterns, and sync model and must not invent network policies not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
CPR-01 Performance Budget: {{cpr.budget}} | OPTIONAL
CER-02 Retry/Recovery Patterns: {{cer.retry_patterns}} | OPTIONAL
OFS-02 Sync Model: {{ofs.sync_model}} | OPTIONAL
SMD-02 Cache Strategy: {{smd.cache_strategy}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Batching rules (what can batch)
Request concurrency limits
Payload size limits (per request)
Metered network behavior (restrict, allow, UNKNOWN)
WiFi-only rules for heavy operations
Retry/backoff binding (CER-02)
Timeout policy (mobile)
Data saver mode behavior (if supported)
Telemetry requirements (network errors, bytes)

Optional Fields
Regional network profiles | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Mobile network policy MUST respect retry patterns and avoid aggressive retries on metered networks.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Batching
   batching_enabled: {{batch.enabled}}
   batching_rules: {{batch.rules}}
2. Concurrency Limits
   max_concurrent_requests: {{conc.max_requests}}
   per_host_limits: {{conc.per_host_limits}} | OPTIONAL
3. Payload Limits
   max_payload_kb: {{payload.max_kb}}
   streaming_allowed: {{payload.streaming_allowed}} | OPTIONAL
4. Metered Network
   metered_behavior: {{metered.behavior}} (restrict/allow/UNKNOWN)
   restricted_operations: {{metered.restricted_ops}} | OPTIONAL
5. WiFi-Only Rules
   wifi_only_operations: {{wifi.only_ops}}
   override_allowed: {{wifi.override_allowed}} | OPTIONAL
6. Retry/Backoff
   retry_policy_ref: {{retry.policy_ref}} (expected: {{xref:CER-02}}) | OPTIONAL
   metered_retry_policy: {{retry.metered_policy}} | OPTIONAL
7. Timeout Policy
   connect_timeout_ms: {{timeout.connect_ms}}
   request_timeout_ms: {{timeout.request_ms}} | OPTIONAL
8. Data Saver Mode
   data_saver_supported: {{saver.supported}}
   data_saver_rules: {{saver.rules}} | OPTIONAL
9. Telemetry
   bytes_sent_metric: {{telemetry.bytes_sent_metric}}
   bytes_received_metric: {{telemetry.bytes_received_metric}} | OPTIONAL
   network_error_metric: {{telemetry.network_error_metric}}
10. References
    Performance budget: {{xref:CPR-01}} | OPTIONAL
    Retry patterns: {{xref:CER-02}} | OPTIONAL
    Sync model: {{xref:OFS-02}} | OPTIONAL
    Cache strategy: {{xref:SMD-02}} | OPTIONAL

Cross-References
Upstream: {{xref:CPR-01}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL

Downstream: {{xref:MBAT-03}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Not required.
intermediate: Required. Define batching/concurrency/payload + metered behavior.
advanced: Required. Add data saver profiles and metered retry nuances.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, per-host limits, streaming allowed, restricted ops, override allowed, retry policy ref, metered retry policy, request timeout, saver rules, bytes received metric, regional profiles, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If batch.rules is UNKNOWN → block Completeness Gate.
If payload.max_kb is UNKNOWN → block Completeness Gate.
If metered.behavior is UNKNOWN → block Completeness Gate.
If telemetry.network_error_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.MBAT
Pass conditions:
required_fields_present == true
batching_and_limits_defined == true
metered_and_wifi_rules_defined == true
timeouts_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

MBAT-03

MBAT-03 — Battery Budget & Constraints (targets)
Header Block
template_id: MBAT-03
title: Battery Budget & Constraints (targets)
type: mobile_battery_budget_constraints
template_version: 1.0.0
output_path: 10_app/mobile_perf/MBAT-03_Battery_Budget_Constraints.md
compliance_gate_id: TMP-05.PRIMARY.MBAT
upstream_dependencies: ["MBAT-01", "MBAT-02", "CPR-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "MBAT-01", "MBAT-02", "CPR-01"]
required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose
Define the canonical battery budget and constraints for the mobile app: target usage, background limits, network/CPU constraints, and how to detect and prevent battery regressions. This template must be consistent with background work and network policies and must not invent battery targets not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
MBAT-01 Background Work Rules: {{mbat.bg_work_rules}}
MBAT-02 Network Usage Policy: {{mbat.net_policy}}
CPR-01 Performance Budget: {{cpr.budget}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Battery targets (baseline target definition)
High-cost operations list (camera, GPS, video)
Background battery constraints (limits)
CPU usage constraints (if defined)
Network usage constraints pointer (MBAT-02)
Low power mode behavior
Telemetry requirements (battery impact proxies)
Regression detection plan (profiling/gates)

Optional Fields
Device tier differences | OPTIONAL
User-facing battery saver setting | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Battery rules MUST be enforceable via limits or gating.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Battery Targets
   target_definition: {{battery.target_definition}}
   max_background_drain: {{battery.max_background_drain}} | OPTIONAL
2. High-Cost Operations
   operations: {{costly.operations}}
   mitigations: {{costly.mitigations}} | OPTIONAL
3. Background Constraints
   background_limits: {{bg.limits}}
   schedule_constraints_ref: {{bg.schedule_constraints_ref}} (expected: {{xref:MBAT-01}}) | OPTIONAL
4. CPU Constraints
   cpu_budget_rule: {{cpu.budget_rule}} | OPTIONAL
   thermal_throttle_behavior: {{cpu.thermal_behavior}} | OPTIONAL
5. Network Constraints
   network_policy_ref: {{net.policy_ref}} (expected: {{xref:MBAT-02}}) | OPTIONAL
6. Low Power Mode
   low_power_behavior: {{lpm.behavior}}
   disable_features_in_lpm: {{lpm.disable_features}} | OPTIONAL
7. Telemetry
   battery_proxy_metrics: {{telemetry.proxy_metrics}}
   alerts: {{telemetry.alerts}} | OPTIONAL
8. Regression Detection
   profiling_ref: {{regress.profiling_ref}} (expected: {{xref:CPR-04}}) | OPTIONAL
   gate_ref: {{regress.gate_ref}} (expected: {{xref:CPR-05}}) | OPTIONAL
   fail_criteria: {{regress.fail_criteria}} | OPTIONAL
9. References
   Background work: {{xref:MBAT-01}}
   Network policy: {{xref:MBAT-02}}
   Profiling plan: {{xref:CPR-04}} | OPTIONAL
   Perf gates: {{xref:CPR-05}} | OPTIONAL

Cross-References
Upstream: {{xref:MBAT-01}}, {{xref:MBAT-02}} , {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:MBAT-04}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Not required.
intermediate: Required. Define battery targets and low power mode behavior and telemetry.
advanced: Required. Add regression gates and device tier differences.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, max background drain, mitigations, schedule ref, cpu rules, network policy ref, disable features, alerts, profiling/gate refs, fail criteria, device tiers, user setting, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If battery.target_definition is UNKNOWN → block Completeness Gate.
If bg.limits is UNKNOWN → block Completeness Gate.
If lpm.behavior is UNKNOWN → block Completeness Gate.
If telemetry.proxy_metrics is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.MBAT
Pass conditions:
required_fields_present == true
battery_targets_defined == true
constraints_defined == true
low_power_behavior_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

MBAT-04

MBAT-04 — Perf Profiling Plan (mobile tooling)
Header Block
template_id: MBAT-04
title: Perf Profiling Plan (mobile tooling)
type: mobile_perf_profiling_plan
template_version: 1.0.0
output_path: 10_app/mobile_perf/MBAT-04_Mobile_Perf_Profiling_Plan.md
compliance_gate_id: TMP-05.PRIMARY.MBAT
upstream_dependencies: ["CPR-04", "MBAT-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "CPR-04", "MBAT-03", "MOB-04"]
required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose
Define the canonical mobile performance profiling plan, focused on mobile tooling and scenarios: frame drops, startup time, memory, battery proxies, background work, and network usage under real device constraints. This template must be consistent with overall profiling plan and battery/network constraints and must not invent profiling tooling not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
CPR-04 Profiling Plan: {{cpr.profiling_plan}}
MBAT-03 Battery Budget & Constraints: {{mbat.battery_budget}} | OPTIONAL
MOB-04 App Lifecycle + State: {{mob.lifecycle}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Mobile scenarios to measure (startup, navigation, list scroll, background sync)
Metrics collected (fps, frame time, memory, cpu, network)
Tooling list (RN perf tools, native profilers)
Device list (test devices)
Network profiles (wifi/4g/poor)
Background profiling rules (doze, low power mode)
Reporting format/location
Cadence/ownership

Optional Fields
Automation in CI | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Metrics MUST align to MBAT/CPR budgets.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Scenarios
   scenarios:
   {{scenarios[0]}}
   {{scenarios[1]}}
   {{scenarios[2]}}
2. Metrics
   fps_metrics: {{metrics.fps}}
   frame_time_metrics: {{metrics.frame_time}} | OPTIONAL
   memory_metrics: {{metrics.memory}} | OPTIONAL
   cpu_metrics: {{metrics.cpu}} | OPTIONAL
   network_metrics: {{metrics.network}} | OPTIONAL
   battery_proxy_metrics: {{metrics.battery_proxy}} | OPTIONAL
3. Tools
   tools: {{tools.list}}
4. Devices & Networks
   devices: {{tests.devices}}
   network_profiles: {{tests.network_profiles}} | OPTIONAL
5. Background Profiling
   low_power_mode_tests: {{bg.low_power_mode_tests}}
   doze_mode_tests: {{bg.doze_mode_tests}} | OPTIONAL
6. Reporting
   report_location: {{report.location}}
   report_format: {{report.format}} (md/json/dashboard/UNKNOWN)
7. Cadence & Ownership
   cadence: {{ops.cadence}}
   owner: {{ops.owner}}
   review_process: {{ops.review_process}} | OPTIONAL
8. References
   Overall profiling plan: {{xref:CPR-04}}
   Battery constraints: {{xref:MBAT-03}} | OPTIONAL
   Lifecycle rules: {{xref:MOB-04}} | OPTIONAL

Cross-References
Upstream: {{xref:CPR-04}} , {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:MBAT-05}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Not required.
intermediate: Required. Define scenarios, tools, devices, reporting.
advanced: Required. Add CI automation and background constraint profiling rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, optional metrics, network profiles, doze tests, review process, CI automation, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If tools.list is UNKNOWN → block Completeness Gate.
If report.location is UNKNOWN → block Completeness Gate.
If ops.owner is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.MBAT
Pass conditions:
required_fields_present == true
scenarios_defined == true
tools_defined == true
reporting_defined == true
cadence_and_owner_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

Mobile Deep Links & Universal Links (MDL)

Mobile Deep Links & Universal Links (MDL)
MDL-01 Link Scheme & Domains (app links, universal links)
MDL-02 Routing Rules (link → screen/action mapping)
MDL-03 Auth Gating & Safety (signed links, validation)
MDL-04 Fallback Behavior (not installed, unknown routes)

MDL-01

MDL-01 — Link Scheme & Domains (app links, universal links)
Header Block
template_id: MDL-01
title: Link Scheme & Domains (app links, universal links)
type: mobile_deep_link_scheme_domains
template_version: 1.0.0
output_path: 10_app/mobile_links/MDL-01_Link_Scheme_Domains.md
compliance_gate_id: TMP-05.PRIMARY.MDL
upstream_dependencies: ["ROUTE-03", "CSec-05"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "ROUTE-03", "ROUTE-06", "CSec-05"]
required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose
Define the canonical deep link schemes and domains for mobile: custom URL scheme(s), iOS Universal Links domains, Android App Links domains, and allowlist rules. This template must be consistent with deep link map and secure deep link handling and must not invent schemes/domains not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
ROUTE-03 Deep Link Map: {{route.deep_link_map}} | OPTIONAL
ROUTE-06 Unknown/Invalid Handling: {{route.unknown_handling}} | OPTIONAL
CSec-05 Secure Deep Link Handling: {{csec.deep_link_security}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Custom schemes list (scheme names)
Primary link domains list (https domains)
iOS universal links enabled (true/false/UNKNOWN)
Android app links enabled (true/false/UNKNOWN)
Domain allowlist rules (exact match vs wildcard)
Path allowlist rules (if any)
Environment differences (dev/stage/prod)
Fallback behavior when not installed (web fallback)

Optional Fields
Subdomain policy | OPTIONAL
Link shortener policy | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Domains/schemes MUST be allowlisted and match CSec-05.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Custom Schemes
   schemes:
   {{schemes[0]}}
   {{schemes[1]}} | OPTIONAL
2. Domains
   domains:
   {{domains[0]}}
   {{domains[1]}} | OPTIONAL
   allowlist_rule: {{domains.allowlist_rule}} (exact/wildcard/UNKNOWN)
   subdomain_policy: {{domains.subdomain_policy}} | OPTIONAL
3. Universal/App Links
   ios_universal_links_enabled: {{links.ios_enabled}}
   android_app_links_enabled: {{links.android_enabled}}
   path_allowlist: {{links.path_allowlist}} | OPTIONAL
4. Environment Differences
   dev_domains: {{env.dev_domains}} | OPTIONAL
   stage_domains: {{env.stage_domains}} | OPTIONAL
   prod_domains: {{env.prod_domains}} | OPTIONAL
   scheme_by_env: {{env.scheme_by_env}} | OPTIONAL
5. Fallback When Not Installed
   web_fallback_enabled: {{fallback.web_enabled}}
   fallback_url_policy: {{fallback.url_policy}} | OPTIONAL
6. References
   Deep link map: {{xref:ROUTE-03}} | OPTIONAL
   Secure deep link handling: {{xref:CSec-05}} | OPTIONAL
   Unknown route handling: {{xref:ROUTE-06}} | OPTIONAL

Cross-References
Upstream: {{xref:ROUTE-03}} | OPTIONAL, {{xref:CSec-05}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:MDL-02}}, {{xref:MDL-03}} | OPTIONAL
Standards: {{standards.rules[STD-SECURITY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Not required.

intermediate: Required. Define schemes/domains and whether universal/app links enabled.
advanced: Required. Add env differences, subdomain/link shortener policies.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, secondary schemes/domains, allowlist rule, subdomain policy, path allowlist, env mappings, fallback url policy, shortener policy, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If schemes list is UNKNOWN → block Completeness Gate.
If domains list is UNKNOWN → block Completeness Gate.
If fallback.web_enabled is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.MDL
Pass conditions:
required_fields_present == true
schemes_and_domains_defined == true
platform_link_enablement_defined == true
fallback_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

MDL-02

MDL-02 — Routing Rules (link → screen/action mapping)
Header Block
template_id: MDL-02
title: Routing Rules (link → screen/action mapping)
type: mobile_deep_link_routing_rules
template_version: 1.0.0
output_path: 10_app/mobile_links/MDL-02_Routing_Rules.md
compliance_gate_id: TMP-05.PRIMARY.MDL
upstream_dependencies: ["MDL-01", "ROUTE-03", "MOB-01"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "MDL-01", "ROUTE-03", "MOB-01", "ROUTE-04"]
required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose
Define the canonical routing rules for mobile deep links: how matched URLs/schemes map to screens/actions, how navigation state is constructed, and how to handle cold start vs warm start. This template must be consistent with route/deep link maps and mobile navigation structure and must not invent link targets not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
MDL-01 Scheme & Domains: {{mobile.links}}
ROUTE-03 Deep Link Map: {{route.deep_link_map}}
MOB-01 Mobile Navigation Map: {{mob.nav_map}}
ROUTE-04 Guard Rules: {{route.guard_rules}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Routing evaluation order (scheme/domain/path match)
Match result format (link_id, target_route/screen)
Cold start routing rules (initial stack construction)
Warm start routing rules (push vs reset vs modal)
Param mapping rules (url params → route params)
Guard application (auth/role gating)
Fallback handling (invalid/unknown links)
Telemetry requirements (link routing success/fail)

Optional Fields
Deferred deep links (install then open) | OPTIONAL
Campaign param handling | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Routing MUST only navigate to allowlisted targets defined in ROUTE-03.
Guarding MUST follow ROUTE-04; do not bypass auth/role checks.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Evaluation Order
   order:
   {{eval.order[0]}}
   {{eval.order[1]}}
   {{eval.order[2]}} | OPTIONAL
2. Match Result
   result_fields: {{match.result_fields}} (link_id, target_route_id, params, action/UNKNOWN)
   resolution_rule: {{match.resolution_rule}} | OPTIONAL
3. Cold Start Routing
   cold_start_rule: {{cold.rule}}
   initial_stack_build: {{cold.initial_stack_build}} | OPTIONAL
4. Warm Start Routing
   warm_start_rule: {{warm.rule}}
   navigation_mode: {{warm.navigation_mode}} (push/reset/modal/UNKNOWN)
   preserve_current_state: {{warm.preserve_current_state}} | OPTIONAL
5. Param Mapping
   param_mapping_rule: {{params.mapping_rule}}
   type_validation_rule: {{params.type_validation_rule}} | OPTIONAL
6. Guard Application
   apply_auth_guard: {{guards.apply_auth_guard}}
   apply_role_guard: {{guards.apply_role_guard}} | OPTIONAL
   logged_out_behavior: {{guards.logged_out_behavior}} | OPTIONAL
7. Fallback Handling
   unknown_link_rule: {{fallback.unknown_link_rule}}
   invalid_param_rule: {{fallback.invalid_param_rule}} | OPTIONAL
8. Telemetry
   routing_success_metric: {{telemetry.success_metric}}
   routing_failure_metric: {{telemetry.failure_metric}} | OPTIONAL
   fields: {{telemetry.fields}} | OPTIONAL
9. References
   Scheme/domains: {{xref:MDL-01}}
   Deep link map: {{xref:ROUTE-03}}
   Mobile nav: {{xref:MOB-01}}
   Guard rules: {{xref:ROUTE-04}} | OPTIONAL
   Unknown handling: {{xref:ROUTE-06}} | OPTIONAL

Cross-References
Upstream: {{xref:MDL-01}}, {{xref:ROUTE-03}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:MDL-03}}, {{xref:MDL-04}} | OPTIONAL
Standards: {{standards.rules[STD-SECURITY]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Not required.
intermediate: Required. Define eval order + cold/warm routing rules + guard application.
advanced: Required. Add deferred deep link rules and robust telemetry fields.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, eval order extras, resolution rule, initial stack build, preserve state, type validation, role guard, logged out behavior, invalid param rule, telemetry fields, deferred/campaign handling, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If cold.rule is UNKNOWN → block Completeness Gate.
If params.mapping_rule is UNKNOWN → block Completeness Gate.
If telemetry.success_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.MDL
Pass conditions:
required_fields_present == true
routing_rules_defined == true
guards_applied == true
fallback_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

MDL-03

MDL-03 — Auth Gating & Safety (signed links, validation)
Header Block
template_id: MDL-03
title: Auth Gating & Safety (signed links, validation)
type: mobile_deep_link_auth_safety
template_version: 1.0.0
output_path: 10_app/mobile_links/MDL-03_Auth_Gating_Safety.md
compliance_gate_id: TMP-05.PRIMARY.MDL
upstream_dependencies: ["CSec-05", "ROUTE-04", "MDL-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "CSec-05", "ROUTE-04", "MDL-02", "CER-04"]
required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose
Define the canonical auth gating and safety rules for mobile deep links, including how signed links are validated (if supported), how auth-required targets behave when logged out, and how to prevent unsafe actions via links. This template must be consistent with secure deep link handling and route guard rules and must not invent security mechanics not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
CSec-05 Secure Deep Link Handling: {{csec.deep_link_security}}
ROUTE-04 Guard Rules: {{route.guard_rules}} | OPTIONAL
MDL-02 Routing Rules: {{mobile.routing_rules}} | OPTIONAL
CER-04 Session Expiry Handling: {{cer.session_expiry}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Auth-required target rules (which links require auth)
Logged-out behavior (login prompt + return-to)
Signed link support (true/false/UNKNOWN)
Signature validation rules (if supported)
Expiry rules (if signed/limited-time)
Sensitive action protections (confirmations, rate limits)
Replay protection rules (one-time links) if applicable
Telemetry requirements (invalid signature, denied links)
Error UX rules (safe messaging)

Optional Fields
Device binding rules | OPTIONAL

Multi-tenant context rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Signed link failures MUST not leak verification details.
Auth gating MUST align with {{xref:ROUTE-04}} and session rules with {{xref:CER-04}}.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Auth-Required Targets
   targets_require_auth: {{auth.targets_require_auth}}
   how_to_identify: {{auth.how_to_identify}} | OPTIONAL
2. Logged-Out Behavior
   prompt_login: {{logged_out.prompt_login}}
   return_to_rule: {{logged_out.return_to_rule}}
   fallback_route: {{logged_out.fallback_route}} | OPTIONAL
3. Signed Links
   supported: {{signed.supported}}
   signature_scheme: {{signed.scheme}} | OPTIONAL
   verification_rules: {{signed.verification_rules}} | OPTIONAL
   failure_behavior: {{signed.failure_behavior}} | OPTIONAL
4. Expiry
   expiry_supported: {{expiry.supported}} | OPTIONAL
   expiry_field: {{expiry.field}} | OPTIONAL
   expired_link_behavior: {{expiry.expired_behavior}} | OPTIONAL
5. Sensitive Actions
   confirm_required_for: {{sensitive.confirm_required_for}} | OPTIONAL
   rate_limit_ref: {{sensitive.rate_limit_ref}} | OPTIONAL
6. Replay / One-Time Links
   one_time_links_supported: {{replay.one_time_supported}} | OPTIONAL
   replay_protection_rule: {{replay.protection_rule}} | OPTIONAL
7. Telemetry
   invalid_signature_metric: {{telemetry.invalid_signature_metric}} | OPTIONAL
   auth_denied_metric: {{telemetry.auth_denied_metric}}
   fields: {{telemetry.fields}} | OPTIONAL
8. Error UX
   safe_copy_rules: {{ux.safe_copy_rules}}
   support_contact_policy: {{ux.support_contact_policy}} | OPTIONAL
9. References
   Secure deep link handling: {{xref:CSec-05}}
   Guard rules: {{xref:ROUTE-04}} | OPTIONAL

Routing rules: {{xref:MDL-02}} | OPTIONAL
Session expiry: {{xref:CER-04}} | OPTIONAL

Cross-References
Upstream: {{xref:CSec-05}}, {{xref:MDL-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:MDL-04}} | OPTIONAL
Standards: {{standards.rules[STD-SECURITY]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Not required.
intermediate: Required. Define logged-out behavior + auth denied metrics + safe copy.
advanced: Required. Add signed link validation, expiry, and replay protection rules.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, identify rule, fallback route, signature scheme/verification/failure behavior, expiry details, sensitive action confirms/ratelimits, one-time links, invalid signature metric, telemetry fields, support contact, device/tenant context, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If logged_out.return_to_rule is UNKNOWN → block Completeness Gate.
If signed.supported is UNKNOWN → block Completeness Gate (for apps using signed links).
If telemetry.auth_denied_metric is UNKNOWN → block Completeness Gate.
If ux.safe_copy_rules is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.MDL
Pass conditions:
required_fields_present == true
auth_gating_defined == true
safety_rules_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

MDL-04

MDL-04 — Fallback Behavior (not installed, unknown routes)
Header Block
template_id: MDL-04
title: Fallback Behavior (not installed, unknown routes)
type: mobile_deep_link_fallback_behavior
template_version: 1.0.0
output_path: 10_app/mobile_links/MDL-04_Fallback_Behavior.md
compliance_gate_id: TMP-05.PRIMARY.MDL
upstream_dependencies: ["MDL-01", "ROUTE-06", "ROUTE-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "MDL-01", "ROUTE-06", "ROUTE-03"]
required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose
Define the canonical fallback behaviors for mobile deep links, including what happens when the app is not installed, when a route is unknown/invalid, and when a link cannot be resolved. This template must be consistent with link scheme/domains and unknown route handling and must not invent fallback routes not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
MDL-01 Scheme & Domains: {{mobile.links}}
ROUTE-06 Unknown Handling: {{route.unknown_handling}} | OPTIONAL
ROUTE-03 Deep Link Map: {{route.deep_link_map}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Not-installed behavior (web fallback/store redirect)
Unknown route behavior (not_found handling)
Invalid param behavior (fallback route or error)
Expired link behavior (if applicable)
Store routing rules (iOS/Android store urls)
Telemetry requirements (fallback triggered metrics)
User copy policy (safe messaging)

Optional Fields
Deferred deep link provider notes | OPTIONAL
Per-env differences | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Fallbacks MUST not create open redirects or unsafe navigation.
Unknown routes MUST follow {{xref:ROUTE-06}}.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Not Installed
   web_fallback_enabled: {{not_installed.web_enabled}}
   web_fallback_url: {{not_installed.web_url}} | OPTIONAL
   store_redirect_enabled: {{not_installed.store_redirect_enabled}}
   ios_store_url: {{not_installed.ios_store_url}} | OPTIONAL
   android_store_url: {{not_installed.android_store_url}} | OPTIONAL
2. Unknown Route
   unknown_route_rule: {{unknown.route_rule}}
   not_found_route_id: {{unknown.not_found_route_id}} | OPTIONAL
3. Invalid Params
   invalid_param_rule: {{invalid.param_rule}}
   fallback_route_id: {{invalid.fallback_route_id}} | OPTIONAL
4. Expired Links (Optional)
   expired_link_rule: {{expired.rule}} | OPTIONAL
   expired_link_ux: {{expired.ux}} | OPTIONAL
5. Telemetry
   fallback_metric: {{telemetry.fallback_metric}}
   fields: {{telemetry.fields}} | OPTIONAL
6. Copy Policy
   safe_copy_rules: {{copy.safe_copy_rules}}
   support_contact_policy: {{copy.support_contact_policy}} | OPTIONAL
7. References
   Scheme/domains: {{xref:MDL-01}}
   Deep link map: {{xref:ROUTE-03}} | OPTIONAL
   Unknown handling: {{xref:ROUTE-06}} | OPTIONAL

Cross-References
Upstream: {{xref:MDL-01}}, {{xref:ROUTE-06}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:MPUSH-05}} | OPTIONAL
Standards: {{standards.rules[STD-SECURITY]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Not required.
intermediate: Required. Define not-installed + unknown/invalid rules + telemetry metric.
advanced: Required. Add deferred deep link + per-env differences and expiry UX.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, web fallback url, store URLs, not_found route id, fallback route id, expired rules/ux, telemetry fields, support contact, deferred provider notes, per-env differences, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If not_installed.store_redirect_enabled is UNKNOWN → block Completeness Gate.
If unknown.route_rule is UNKNOWN → block Completeness Gate.
If telemetry.fallback_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.MDL
Pass conditions:
required_fields_present == true
not_installed_defined == true
unknown_invalid_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

Push Notifications (MPUSH)

Push Notifications (MPUSH)
MPUSH-01 Notification Types Catalog (by notif_id)
MPUSH-02 Payload Contract (fields, routing, localization keys)
MPUSH-03 Permission & Opt-In Rules (platform specific)
MPUSH-04 Delivery/Retry Rules (quiet hours, collapse keys)
MPUSH-05 Deep Link Routing (notif tap behavior)
MPUSH-06 Abuse/Spam Controls for Notifications

MPUSH-01

MPUSH-01 — Notification Types Catalog (by notif_id)
Header Block
template_id: MPUSH-01
title: Notification Types Catalog (by notif_id)
type: mobile_push_notification_types_catalog
template_version: 1.0.0
output_path: 10_app/push/MPUSH-01_Notification_Types_Catalog.md
compliance_gate_id: TMP-05.PRIMARY.MPUSH
upstream_dependencies: ["SPEC_INDEX", "ROUTE-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "ROUTE-03", "FE-01"]
required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose
Create the canonical catalog of push notification types, indexed by notif_id, including purpose, triggering events, targeting rules, payload shape reference, and deep link behavior. This template must be consistent with routing/deep links and must not invent notif_ids not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
ROUTE-03 Deep Link Map: {{route.deep_link_map}} | OPTIONAL
FE-01 Route Map + Layout: {{fe.route_layout}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Notification type registry (notif_id list)
notif_id (stable identifier)
name (human-readable)
purpose (why it exists)
trigger source (event_id/job_id/manual)
targeting rules (who receives)
priority level (low/normal/high)
delivery channel (push)
deep link behavior (link_id/route target)
user-facing copy keys (localization keys)
disable/opt-out support (yes/no)

Optional Fields
Quiet hours applicability | OPTIONAL
Collapse/grouping key | OPTIONAL

Sensitive content rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not invent notif_ids; use only {{spec.notifications_by_id}} if present, else mark UNKNOWN and flag.
Deep link targets MUST be allowlisted ({{xref:ROUTE-03}}/{{xref:MDL-02}}).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Catalog Summary
   total_notif_types: {{catalog.total}}
   priority_levels: {{catalog.priority_levels}} | OPTIONAL
2. Notification Types (by notif_id)
   Notification
   notif_id: {{items[0].notif_id}}
   name: {{items[0].name}}
   purpose: {{items[0].purpose}}
   trigger_source: {{items[0].trigger_source}}
   targeting_rules: {{items[0].targeting_rules}}
   priority: {{items[0].priority}}
   deep_link: {{items[0].deep_link}}
   copy_keys:
   title_key: {{items[0].copy_keys.title_key}}
   body_key: {{items[0].copy_keys.body_key}}
   opt_out_supported: {{items[0].opt_out_supported}}
   quiet_hours: {{items[0].quiet_hours}} | OPTIONAL
   collapse_key: {{items[0].collapse_key}} | OPTIONAL
   sensitive_content_rules: {{items[0].sensitive_content_rules}} | OPTIONAL
   open_questions:
   {{items[0].open_questions[0]}} | OPTIONAL

(Repeat per notif_id.)

3. References
   Payload contract: {{xref:MPUSH-02}} | OPTIONAL
   Permission/opt-in rules: {{xref:MPUSH-03}} | OPTIONAL
   Delivery/retry: {{xref:MPUSH-04}} | OPTIONAL
   Deep link routing: {{xref:MPUSH-05}} | OPTIONAL

Cross-References
Upstream: {{xref:SPEC_INDEX}} | OPTIONAL, {{xref:ROUTE-03}} | OPTIONAL

Downstream: {{xref:MPUSH-02}}, {{xref:MPUSH-05}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-SECURITY]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Not required.
intermediate: Required. Populate notif registry and deep link behavior.
advanced: Required. Add quiet hours/collapse keys and sensitive content policy where applicable.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, catalog summary extras, quiet hours, collapse key, sensitive content rules, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If notif registry is UNKNOWN → block Completeness Gate.
If items[*].priority is UNKNOWN → block Completeness Gate.*
*If items[*].copy_keys.title_key/body_key are UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.MPUSH
Pass conditions:
required_fields_present == true
notif_registry_defined == true
deep_links_allowlisted == true
placeholder_resolution == true
no_unapproved_unknowns == true

MPUSH-02

MPUSH-02 — Payload Contract (fields, routing, localization keys)
Header Block
template_id: MPUSH-02
title: Payload Contract (fields, routing, localization keys)
type: mobile_push_payload_contract
template_version: 1.0.0
output_path: 10_app/push/MPUSH-02_Payload_Contract.md
compliance_gate_id: TMP-05.PRIMARY.MPUSH
upstream_dependencies: ["MPUSH-01", "MDL-02", "CSec-05"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "MPUSH-01", "MDL-02", "CSec-05"]
required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose
Define the canonical push notification payload contract: required and optional fields, routing fields to link notifications to in-app screens/actions, localization key requirements, and security constraints. This template must be consistent with notification type catalog and deep link safety rules and must not invent payload fields not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
MPUSH-01 Notification Types Catalog: {{mpush.types}}
MDL-02 Routing Rules: {{mobile.routing_rules}} | OPTIONAL
CSec-05 Secure Deep Link Handling: {{csec.deep_link_security}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Required payload fields (notif_id, title/body keys, routing)
Optional payload fields (collapse key, badge, sound)
Routing contract fields (link_id/route_id/params)
Localization keys contract (title_key/body_key args)
Versioning rules (payload schema version)
Security constraints (no PII, signed link usage if any)
Validation rules (reject unknown fields)
Max payload size policy
Telemetry fields (delivery id, trace id)

Optional Fields
Platform-specific fields (iOS/Android) | OPTIONAL
Encrypted payload section | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
No sensitive data in payload.
Routing targets MUST be allowlisted per deep link rules.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Schema Versioning
   schema_version: {{schema.version}}
   compat_policy: {{schema.compat_policy}} | OPTIONAL
2. Required Fields
   notif_id: {{fields.required.notif_id}}
   title_key: {{fields.required.title_key}}
   body_key: {{fields.required.body_key}}
   routing: {{fields.required.routing}}
   delivery_id: {{fields.required.delivery_id}} | OPTIONAL
3. Optional Fields
   collapse_key: {{fields.optional.collapse_key}} | OPTIONAL
   sound: {{fields.optional.sound}} | OPTIONAL
   badge: {{fields.optional.badge}} | OPTIONAL
   priority: {{fields.optional.priority}} | OPTIONAL
4. Routing Contract
   link_id: {{routing.link_id}} | OPTIONAL
   target_route_id: {{routing.target_route_id}} | OPTIONAL
   params: {{routing.params}} | OPTIONAL
   action: {{routing.action}} | OPTIONAL
   validation_rule: {{routing.validation_rule}}
5. Localization Keys
   args_supported: {{l10n.args_supported}}
   title_args: {{l10n.title_args}} | OPTIONAL
   body_args: {{l10n.body_args}} | OPTIONAL
   missing_key_behavior: {{l10n.missing_key_behavior}} | OPTIONAL
6. Validation
   reject_unknown_fields: {{validate.reject_unknown_fields}}
   required_field_policy: {{validate.required_field_policy}}
7. Size Policy
   max_payload_bytes: {{size.max_payload_bytes}}
   truncate_policy: {{size.truncate_policy}} | OPTIONAL
8. Security Constraints
   no_pii_rule: {{security.no_pii_rule}}
   signed_links_supported: {{security.signed_links_supported}} | OPTIONAL
   signature_field: {{security.signature_field}} | OPTIONAL

9. Telemetry Fields
   trace_id: {{telemetry.trace_id}} | OPTIONAL
   sent_at: {{telemetry.sent_at}} | OPTIONAL
   provider_message_id: {{telemetry.provider_message_id}} | OPTIONAL
10. References
   Notification catalog: {{xref:MPUSH-01}}
   Routing rules: {{xref:MDL-02}} | OPTIONAL
   Secure deep links: {{xref:CSec-05}} | OPTIONAL
   Delivery/retry: {{xref:MPUSH-04}} | OPTIONAL

Cross-References
Upstream: {{xref:MPUSH-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:MPUSH-04}}, {{xref:MPUSH-05}} | OPTIONAL
Standards: {{standards.rules[STD-PII-REDACTION]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Not required.
intermediate: Required. Define required/optional fields and routing validation rule.
advanced: Required. Add versioning/size/security constraints and platform-specific fields.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, compat policy, delivery id, optional fields, routing optional fields, localization args and missing behavior, truncate policy, signed link fields, telemetry fields, platform fields, encrypted section, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If schema.version is UNKNOWN → block Completeness Gate.
If fields.required.notif_id is UNKNOWN → block Completeness Gate.
If routing.validation_rule is UNKNOWN → block Completeness Gate.
If security.no_pii_rule is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.MPUSH
Pass conditions:
required_fields_present == true
payload_contract_defined == true
routing_validation_defined == true
security_constraints_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

MPUSH-03

MPUSH-03 — Permission & Opt-In Rules (platform specific)
Header Block
template_id: MPUSH-03
title: Permission & Opt-In Rules (platform specific)
type: mobile_push_permission_optin_rules
template_version: 1.0.0
output_path: 10_app/push/MPUSH-03_Permission_Opt_In_Rules.md
compliance_gate_id: TMP-05.PRIMARY.MPUSH
upstream_dependencies: ["MPUSH-01", "MDC-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "MPUSH-01", "MDC-02", "MPUSH-06"]
required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose
Define the canonical rules for push notification permission prompts and opt-in flows on iOS/Android, including when to ask, pre-prompt rationale, user preference settings, and handling of denied permissions. This template must be consistent with notification catalog and mobile permission UX rules and must not invent opt-in behaviors not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
MPUSH-01 Notification Types Catalog: {{mpush.types}}
MDC-02 Permissions UX Rules: {{mdc.permissions_ux}} | OPTIONAL
MPUSH-06 Abuse/Spam Controls: {{mpush.abuse_controls}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
iOS permission prompt timing rules
Android permission rules (if applicable)
Pre-prompt rationale rules (what to show before OS prompt)
Opt-in preference model (global + per notif type)
Default opt-in states (per notif type)
Denied handling rules (how to re-enable)
Settings screen rules (where preferences live)
Telemetry requirements (prompt shown, granted/denied)
Quiet hours support (if applicable)

Optional Fields
Provisional authorization (iOS) | OPTIONAL

Topic subscriptions | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Permission prompts SHOULD be triggered by user intent and not shown immediately on first launch unless explicitly required.
Opt-in behavior MUST respect user choices and platform rules.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. iOS Prompt Timing
   ask_when: {{ios.ask_when}}
   pre_prompt_required: {{ios.pre_prompt_required}} | OPTIONAL
2. Android Rules
   runtime_permission_required: {{android.runtime_permission_required}}
   ask_when: {{android.ask_when}} | OPTIONAL
3. Pre-Prompt Rationale
   rationale_copy_policy: {{rationale.copy_policy}}
   rationale_template: {{rationale.template}} | OPTIONAL
4. Preference Model
   global_toggle_supported: {{prefs.global_toggle_supported}}
   per_type_toggles_supported: {{prefs.per_type_supported}}
   default_states_by_type: {{prefs.default_states_by_type}}
5. Denied Handling
   on_denied_behavior: {{deny.on_denied_behavior}}
   settings_route_rule: {{deny.settings_route_rule}} | OPTIONAL
6. Settings Screen
   settings_screen_id: {{settings.screen_id}} | OPTIONAL
   settings_navigation_ref: {{settings.navigation_ref}} | OPTIONAL
7. Telemetry
   prompt_shown_metric: {{telemetry.prompt_shown_metric}}
   permission_granted_metric: {{telemetry.granted_metric}} | OPTIONAL
   permission_denied_metric: {{telemetry.denied_metric}} | OPTIONAL
   fields: {{telemetry.fields}} | OPTIONAL
8. Quiet Hours
   quiet_hours_supported: {{quiet.supported}}
   quiet_hours_rules: {{quiet.rules}} | OPTIONAL
9. References
   Notification types: {{xref:MPUSH-01}}
   Permissions UX rules: {{xref:MDC-02}} | OPTIONAL
   Abuse/spam controls: {{xref:MPUSH-06}} | OPTIONAL

Cross-References
Upstream: {{xref:MPUSH-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:MPUSH-04}}, {{xref:MPUSH-06}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Not required.
intermediate: Required. Define timing + preference model + denied handling.
advanced: Required. Add iOS provisional, topic subs, and telemetry field rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, pre prompt required, android ask when, rationale template, settings routing/screen ids, telemetry fields, quiet hours rules, provisional auth, topic subs, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If ios.ask_when is UNKNOWN → block Completeness Gate.
If prefs.default_states_by_type is UNKNOWN → block Completeness Gate.
If telemetry.prompt_shown_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.MPUSH
Pass conditions:
required_fields_present == true
permission_timing_defined == true
preference_model_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

MPUSh-04

MPUSH-04 — Delivery/Retry Rules (quiet hours, collapse keys)
Header Block
template_id: MPUSH-04
title: Delivery/Retry Rules (quiet hours, collapse keys)
type: mobile_push_delivery_retry_rules
template_version: 1.0.0
output_path: 10_app/push/MPUSH-04_Delivery_Retry_Rules.md
compliance_gate_id: TMP-05.PRIMARY.MPUSH
upstream_dependencies: ["MPUSH-01", "MPUSH-02", "CER-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "MPUSH-01", "MPUSH-02", "CER-02", "FFCFG-01"]
required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose
Define the canonical delivery and retry rules for push notifications, including quiet hours behavior, collapse/grouping behavior, retry semantics, failure handling, and safeguards to prevent notification storms. This template must be consistent with notification types and retry patterns and must not invent delivery behaviors not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
MPUSH-01 Notification Types Catalog: {{mpush.types}}
MPUSH-02 Payload Contract: {{mpush.payload_contract}}
CER-02 Retry/Recovery Patterns: {{cer.retry_patterns}} | OPTIONAL
FFCFG-01 Feature Flag Registry: {{ffcfg.flags}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Quiet hours policy (enabled/behavior)
Collapse/grouping policy (collapse keys)
Delivery priority rules (by notif type)
Retry policy (server/provider retries)
Failure handling (invalid token, provider errors)
Deduplication policy (avoid duplicate sends)
Storm prevention rules (rate caps, batching)
Telemetry requirements (delivery success/fail, drops)
Feature-flag kill switch (if used)

Optional Fields
Per-type overrides | OPTIONAL

Regional delivery constraints | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Delivery rules MUST be consistent with MPUSH-02 payload contract.
Storm prevention MUST exist for high-volume systems.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Quiet Hours
   supported: {{quiet.supported}}
   behavior: {{quiet.behavior}} (delay/drop/UNKNOWN)
   window: {{quiet.window}} | OPTIONAL
2. Collapse / Grouping
   collapse_supported: {{collapse.supported}}
   collapse_key_rule: {{collapse.key_rule}}
   grouping_rule: {{collapse.grouping_rule}} | OPTIONAL
3. Priority Rules
   priority_by_type: {{priority.by_type}}
   provider_priority_mapping: {{priority.provider_mapping}} | OPTIONAL
4. Retry Policy
   retry_supported: {{retry.supported}}
   retry_strategy: {{retry.strategy}} (provider_default/custom/UNKNOWN)
   max_attempts: {{retry.max_attempts}} | OPTIONAL
   backoff_policy: {{retry.backoff_policy}} | OPTIONAL
5. Failure Handling
   invalid_token_handling: {{fail.invalid_token_handling}}
   provider_error_handling: {{fail.provider_error_handling}} | OPTIONAL
6. Deduplication
   dedupe_supported: {{dedupe.supported}}
   dedupe_key_rule: {{dedupe.key_rule}} | OPTIONAL
7. Storm Prevention
   rate_caps: {{storm.rate_caps}}
   batching_rules: {{storm.batching_rules}} | OPTIONAL
   kill_switch_flag_ref: {{storm.kill_switch_flag_ref}} (expected: {{xref:FFCFG-01}}) | OPTIONAL
8. Telemetry
   delivery_success_metric: {{telemetry.success_metric}}
   delivery_failure_metric: {{telemetry.failure_metric}} | OPTIONAL
   dropped_metric: {{telemetry.dropped_metric}} | OPTIONAL
   fields: {{telemetry.fields}} | OPTIONAL

9. References
   Notification types: {{xref:MPUSH-01}}
   Payload contract: {{xref:MPUSH-02}}
   Retry patterns: {{xref:CER-02}} | OPTIONAL
   Feature flags: {{xref:FFCFG-01}} | OPTIONAL

Cross-References
Upstream: {{xref:MPUSH-01}}, {{xref:MPUSH-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:MPUSH-05}}, {{xref:MPUSH-06}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Not required.
intermediate: Required. Define quiet/collapse/priority + telemetry.
advanced: Required. Add storm prevention, kill switch, and retry/backoff rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, quiet window, grouping rule, provider mapping, retry max/backoff, provider error handling, dedupe key rule, batching rules, kill switch ref, telemetry fields, per-type/regional constraints, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If collapse.key_rule is UNKNOWN → block Completeness Gate.
If storm.rate_caps is UNKNOWN → block Completeness Gate.
If telemetry.success_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.MPUSH
Pass conditions:
required_fields_present == true
delivery_rules_defined == true
storm_prevention_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

MPUSH-05

MPUSH-05 — Deep Link Routing (notif tap behavior)
Header Block
template_id: MPUSH-05
title: Deep Link Routing (notif tap behavior)
type: mobile_push_deep_link_routing
template_version: 1.0.0
output_path: 10_app/push/MPUSH-05_Deep_Link_Routing.md
compliance_gate_id: TMP-05.PRIMARY.MPUSH
upstream_dependencies: ["MPUSH-01", "MPUSH-02", "MDL-02"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "MPUSH-01", "MPUSH-02", "MDL-02", "MDL-04"]
required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose
Define the canonical behavior when a user taps a push notification: how the payload routing fields are parsed, how deep link routing is invoked, and how cold start vs warm start behavior differs. This template must be consistent with payload contract and mobile routing rules and must not invent notification targets not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
MPUSH-01 Notification Types Catalog: {{mpush.types}}
MPUSH-02 Payload Contract: {{mpush.payload_contract}}
MDL-02 Routing Rules: {{mobile.routing_rules}}
MDL-04 Fallback Behavior: {{mobile.fallback}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Tap handling entrypoint (handler name / module)
Routing field extraction rules (from payload)
Cold start tap behavior (initialize then route)
Warm start tap behavior (navigate vs reset)
Auth gating on tap (logged out handling)
Invalid routing handling (fallback)
Telemetry (tap events, route success/fail)
Analytics event fields (notif_id, campaign)

Optional Fields
Action buttons behavior (if supported) | OPTIONAL
Badge clearing rules | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Tap routing MUST use MDL routing rules and deep link allowlists.
Invalid/unknown routes MUST follow MDL-04 fallback.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Handler Entrypoint
   handler_module: {{handler.module}}
   handler_function: {{handler.function}}
2. Routing Extraction
   routing_fields_rule: {{extract.routing_fields_rule}}
   campaign_fields_rule: {{extract.campaign_fields_rule}} | OPTIONAL
3. Cold Start Behavior
   cold_start_flow: {{cold.flow}}
   init_order: {{cold.init_order}} | OPTIONAL
4. Warm Start Behavior
   warm_flow: {{warm.flow}}
   navigation_mode: {{warm.navigation_mode}} (push/reset/UNKNOWN)
   preserve_state: {{warm.preserve_state}} | OPTIONAL
5. Auth Gating
   logged_out_behavior: {{auth.logged_out_behavior}}
   return_to_rule: {{auth.return_to_rule}} | OPTIONAL
6. Invalid Routing Fallback
   fallback_rule_ref: {{fallback.rule_ref}} (expected: {{xref:MDL-04}}) | OPTIONAL
   invalid_payload_behavior: {{fallback.invalid_payload_behavior}}
7. Telemetry / Analytics
   tap_metric: {{telemetry.tap_metric}}
   route_success_metric: {{telemetry.route_success_metric}} | OPTIONAL
   route_failure_metric: {{telemetry.route_failure_metric}} | OPTIONAL
   fields: {{telemetry.fields}} | OPTIONAL
8. References
   Notification catalog: {{xref:MPUSH-01}}
   Payload contract: {{xref:MPUSH-02}}
   Mobile routing rules: {{xref:MDL-02}}
   Fallback behavior: {{xref:MDL-04}} | OPTIONAL

Cross-References
Upstream: {{xref:MPUSH-01}}, {{xref:MPUSH-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:MPUSH-06}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Not required.
intermediate: Required. Define handler + cold/warm behavior + telemetry.
advanced: Required. Add action buttons/badge behavior and detailed analytics fields.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, campaign fields, init order, preserve state, return to rule, fallback rule ref, route success/fail metrics, telemetry fields, action buttons, badge clearing, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If handler.function is UNKNOWN → block Completeness Gate.
If extract.routing_fields_rule is UNKNOWN → block Completeness Gate.
If fallback.invalid_payload_behavior is UNKNOWN → block Completeness Gate.
If telemetry.tap_metric is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.MPUSH
Pass conditions:
required_fields_present == true
tap_handling_defined == true
routing_and_fallback_defined == true
telemetry_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

MPUSH-06

MPUSH-06 — Abuse/Spam Controls for Notifications
Header Block
template_id: MPUSH-06
title: Abuse/Spam Controls for Notifications
type: mobile_push_abuse_spam_controls
template_version: 1.0.0
output_path: 10_app/push/MPUSH-06_Abuse_Spam_Controls.md
compliance_gate_id: TMP-05.PRIMARY.MPUSH
upstream_dependencies: ["MPUSH-04", "RLIM-01", "RLIM-03", "RLIM-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "MPUSH-01", "MPUSH-04", "RLIM-01", "RLIM-03", "RLIM-04", "FFCFG-01"]
required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose
Define the canonical abuse/spam controls for notifications: rate caps per user/type, detection signals for abusive senders, enforcement actions, and kill switches. This template must be consistent with rate limit policy and delivery rules and must not invent enforcement mechanisms not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
MPUSH-01 Notification Types Catalog: {{mpush.types}} | OPTIONAL
MPUSH-04 Delivery/Retry Rules: {{mpush.delivery_rules}} | OPTIONAL
RLIM-01 Rate Limit Policy: {{rlim.policy}}
RLIM-03 Abuse Signals & Detection: {{rlim.abuse_signals}} | OPTIONAL
RLIM-04 Enforcement Actions Matrix: {{rlim.actions}} | OPTIONAL
FFCFG-01 Feature Flag Registry: {{ffcfg.flags}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Rate caps (per user, per notif type, per time window)
Abuse signals (high frequency, user reports, invalid tokens)
Detection rules (thresholds or UNKNOWN)
Enforcement actions (throttle/ban/disable type)
Kill switch rules (global + per type)
User experience rules (how to message or silently drop)
Telemetry requirements (blocked sends, caps hit)
Audit logging requirements (who triggered actions)

Optional Fields
Allowlists/exemptions | OPTIONAL

Regional/legal constraints | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Controls MUST bind to RLIM action matrix where possible.
Kill switches MUST be quickly actionable (flag-based) if available.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Rate Caps
   caps: {{caps.definitions}}
   scope_model: {{caps.scope_model}} (per_user/per_type/global/UNKNOWN)
2. Abuse Signals
   signals: {{signals.list}}
   notes: {{signals.notes}} | OPTIONAL
3. Detection Rules
   rules: {{detect.rules}}
   threshold_model: {{detect.threshold_model}} (static/baseline/UNKNOWN) | OPTIONAL
4. Enforcement Actions
   actions_ref: {{enforce.actions_ref}} (expected: {{xref:RLIM-04}}) | OPTIONAL
   action_rules: {{enforce.action_rules}}
5. Kill Switches
   global_kill_switch_flag: {{kill.global_flag}} | OPTIONAL
   per_type_kill_switch_flags: {{kill.per_type_flags}} | OPTIONAL
6. User Experience
   drop_silently_rule: {{ux.drop_silently_rule}}
   user_messaging_rule: {{ux.user_messaging_rule}} | OPTIONAL
7. Telemetry
   caps_hit_metric: {{telemetry.caps_hit_metric}}
   blocked_send_metric: {{telemetry.blocked_send_metric}} | OPTIONAL
   fields: {{telemetry.fields}} | OPTIONAL
8. Audit Logging
   audit_fields: {{audit.fields}}
   audit_retention_policy: {{audit.retention_policy}} | OPTIONAL
9. References
   Delivery rules: {{xref:MPUSH-04}} | OPTIONAL
   Rate limit policy: {{xref:RLIM-01}}
   Abuse signals: {{xref:RLIM-03}} | OPTIONAL
   Enforcement actions: {{xref:RLIM-04}} | OPTIONAL
   Feature flags: {{xref:FFCFG-01}} | OPTIONAL

Cross-References
Upstream: {{xref:RLIM-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:ADMIN-02}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Not required.
intermediate: Required. Define caps + signals + enforcement.
advanced: Required. Add exemptions, legal constraints, and audit/telemetry rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, signals notes, threshold model, actions ref, kill switch flags, user messaging, telemetry fields, retention policy, allowlists, regional constraints, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If caps.definitions is UNKNOWN → block Completeness Gate.
If enforce.action_rules is UNKNOWN → block Completeness Gate.
If telemetry.caps_hit_metric is UNKNOWN → block Completeness Gate.
If audit.fields is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.MPUSH
Pass conditions:
required_fields_present == true
rate_caps_defined == true
enforcement_defined == true
telemetry_defined == true
audit_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

App Store Release & Signing (SIGN)

App Store Release & Signing (SIGN)
SIGN-01 Build Artifact Types (debug/beta/prod)
SIGN-02 Signing Keys & Rotation Policy
SIGN-03 Store Submission Checklist (iOS/Android)
SIGN-04 Versioning Rules (build numbers, semver mapping)
SIGN-05 Release Channel Policy (internal/beta/GA)

SIGN-01

SIGN-01 — Build Artifact Types (debug/beta/prod)
Header Block
template_id: SIGN-01
title: Build Artifact Types (debug/beta/prod)
type: release_build_artifact_types
template_version: 1.0.0
output_path: 10_app/release/SIGN-01_Build_Artifact_Types.md
compliance_gate_id: TMP-05.PRIMARY.SIGN
upstream_dependencies: ["CPR-05", "MBAT-05"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "CPR-05", "MBAT-05"]
required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose
Define the canonical build artifact types produced for the application across environments and release channels (debug/beta/prod), including naming conventions, contents, signing expectations, distribution targets, and validation checks. This template must be consistent with performance/release gates and must not invent artifact types not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
CPR-05 Perf Regression Gates: {{cpr.perf_gates}} | OPTIONAL
MBAT-05 Mobile Perf Regression Gates: {{mbat.perf_gates}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Artifact type registry (artifact_type list)
artifact_type (debug/beta/prod)
platform targets (web/ios/android)
file formats (apk/aab/ipa/web bundle)
build configuration (env flags, debug symbols)
signing requirement (signed/unsigned)
distribution channel (internal/testflight/play internal/GA)
naming/versioning rules binding (SIGN-04)
required validation checks (tests, gates)
retention policy (how long stored)

Optional Fields
Source map / symbol upload rules | OPTIONAL
Feature flag defaults per artifact | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not introduce new artifact types beyond the registry.
Artifact naming/versioning MUST align with {{xref:SIGN-04}} when present.
Artifacts MUST reference required gates/checks (e.g., perf, security) and not bypass them.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Artifact Registry
   artifact_types:
   {{registry.types[0]}}
   {{registry.types[1]}}
   {{registry.types[2]}}
2. Artifact Type Definitions
   Artifact
   artifact_type: {{artifacts[0].artifact_type}}
   platforms: {{artifacts[0].platforms}}
   formats: {{artifacts[0].formats}}
   build_config:
   env: {{artifacts[0].build_config.env}}
   debug_symbols: {{artifacts[0].build_config.debug_symbols}} | OPTIONAL
   minify_obfuscate: {{artifacts[0].build_config.minify_obfuscate}} | OPTIONAL
   feature_flags_default: {{artifacts[0].build_config.feature_flags_default}} | OPTIONAL

signing:
required: {{artifacts[0].signing.required}}
key_profile: {{artifacts[0].signing.key_profile}} | OPTIONAL

distribution:
channel: {{artifacts[0].distribution.channel}}
targets: {{artifacts[0].distribution.targets}} | OPTIONAL

versioning_ref: {{artifacts[0].versioning_ref}} (expected: {{xref:SIGN-04}}) | OPTIONAL

validation_checks:
{{artifacts[0].validation_checks[0]}}
{{artifacts[0].validation_checks[1]}} | OPTIONAL

retention_policy: {{artifacts[0].retention_policy}}
open_questions:
{{artifacts[0].open_questions[0]}} | OPTIONAL

(Repeat per artifact_type.)

3. Common Naming Rules
   artifact_name_pattern: {{naming.pattern}}
   file_naming_conventions: {{naming.file_conventions}} | OPTIONAL
4. Storage / Retention
   storage_location: {{storage.location}}
   retention_default: {{storage.retention_default}} | OPTIONAL
5. References
   Versioning rules: {{xref:SIGN-04}} | OPTIONAL
   Signing keys: {{xref:SIGN-02}} | OPTIONAL
   Submission checklist: {{xref:SIGN-03}} | OPTIONAL
   Release channels: {{xref:SIGN-05}} | OPTIONAL
   Perf gates (web): {{xref:CPR-05}} | OPTIONAL
   Perf gates (mobile): {{xref:MBAT-05}} | OPTIONAL

Cross-References
Upstream: {{xref:CPR-05}} | OPTIONAL, {{xref:MBAT-05}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:SIGN-02}}, {{xref:SIGN-04}}, {{xref:SIGN-05}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-SECURITY]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Not required.
intermediate: Required. Define artifact types, formats, signing, and distribution channels.
advanced: Required. Add symbol/sourcemap rules, retention, and validation gate linkage.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, platform targets, debug symbols, minify/obfuscate, feature flags default, key profile, distribution targets, versioning ref, extra validation checks, retention default, naming file conventions, symbol upload rules, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If registry.types is UNKNOWN → block Completeness Gate.
If artifacts[*].signing.required is UNKNOWN → block Completeness Gate.
If storage.location is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.SIGN
Pass conditions:
required_fields_present == true
artifact_registry_defined == true
platforms_and_formats_defined == true
signing_and_distribution_defined == true
validation_checks_defined == true

placeholder_resolution == true
no_unapproved_unknowns == true

SIGN-02

SIGN-02 — Signing Keys & Rotation Policy
Header Block
template_id: SIGN-02
title: Signing Keys & Rotation Policy
type: signing_keys_rotation_policy
template_version: 1.0.0
output_path: 10_app/release/SIGN-02_Signing_Keys_Rotation_Policy.md
compliance_gate_id: TMP-05.PRIMARY.SIGN
upstream_dependencies: ["SIGN-01", "SEC-KEYS", "ADMIN-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "SIGN-01", "ADMIN-03"]
required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose
Define the canonical policy for signing keys: what keys exist (per platform), where they are stored, who can access them, how they are rotated, how compromises are handled, and audit/logging requirements. This template must be consistent with build artifact types and must not invent key material or processes not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
SIGN-01 Build Artifact Types: {{sign.artifacts}}
ADMIN-03 Audit Trail Spec: {{admin.audit_trail}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Key inventory (key_id list)
Key purpose (ios signing, android signing, etc.)
Storage location policy (HSM/CI secret store/UNKNOWN)
Access control policy (roles, approvals)
Rotation schedule (time-based or event-based)
Revocation/compromise response plan
Backup/recovery policy
Audit logging requirements (who accessed/used keys)
No-commit rule (never commit secrets)

Optional Fields
Per-environment keys (dev/stage/prod) | OPTIONAL
Key ceremony notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-SECURITY]}} | OPTIONAL
Never expose key material in docs/logs.
Rotation MUST be documented and testable.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Key Inventory
   Key
   key_id: {{keys[0].key_id}}
   platform: {{keys[0].platform}} (ios/android/UNKNOWN)
   purpose: {{keys[0].purpose}}
   env_scope: {{keys[0].env_scope}} | OPTIONAL
   storage_location: {{keys[0].storage_location}}
   access_policy: {{keys[0].access_policy}}
   rotation_schedule: {{keys[0].rotation_schedule}}
   revocation_plan: {{keys[0].revocation_plan}} | OPTIONAL
   backup_policy: {{keys[0].backup_policy}} | OPTIONAL
   audit_refs: {{keys[0].audit_refs}} | OPTIONAL
   open_questions:
   {{keys[0].open_questions[0]}} | OPTIONAL

(Repeat per key_id.)

2. Storage & Access Controls (Global)
   no_commit_rule: {{controls.no_commit_rule}}
   secret_store_policy: {{controls.secret_store_policy}}
   approval_required: {{controls.approval_required}} | OPTIONAL
3. Rotation Policy
   rotation_trigger: {{rotation.trigger}} (time/event/UNKNOWN)
   rotation_frequency: {{rotation.frequency}} | OPTIONAL
   rotation_steps: {{rotation.steps}} | OPTIONAL
4. Compromise Response
   detection_signals: {{compromise.detection_signals}} | OPTIONAL
   response_steps: {{compromise.response_steps}}
   communication_policy: {{compromise.communication_policy}} | OPTIONAL
5. Backup / Recovery
   backup_location_policy: {{backup.location_policy}}
   recovery_test_policy: {{backup.recovery_test_policy}} | OPTIONAL
6. Audit Logging
   audit_required_fields: {{audit.required_fields}}
   audit_retention_policy: {{audit.retention_policy}} | OPTIONAL

7. References
   Build artifacts: {{xref:SIGN-01}}
   Audit trail spec: {{xref:ADMIN-03}} | OPTIONAL

Cross-References
Upstream: {{xref:SIGN-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:SIGN-03}}, {{xref:SIGN-05}} | OPTIONAL
Standards: {{standards.rules[STD-SECURITY]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Not required.
intermediate: Required. Define key inventory + storage/access + compromise response.
advanced: Required. Add rotation ceremony and recovery testing rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, env scope, revocation/backup/audit refs, approval required, detection signals, comms policy, retention policy, ceremony notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If keys[*].storage_location is UNKNOWN → block Completeness Gate.
If controls.no_commit_rule is UNKNOWN → block Completeness Gate.
If compromise.response_steps is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.SIGN
Pass conditions:
required_fields_present == true
key_inventory_defined == true
storage_and_access_defined == true
rotation_defined == true
compromise_response_defined == true
audit_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

SIGN-03

SIGN-03 — Store Submission Checklist (iOS/Android)
Header Block
template_id: SIGN-03
title: Store Submission Checklist (iOS/Android)
type: store_submission_checklist
template_version: 1.0.0
output_path: 10_app/release/SIGN-03_Store_Submission_Checklist.md
compliance_gate_id: TMP-05.PRIMARY.SIGN
upstream_dependencies: ["SIGN-01", "SIGN-02", "PRD-06"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "SIGN-01", "SIGN-02", "PRD-06", "CSec-02"]
required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose
Define the canonical store submission checklist for iOS App Store / TestFlight and Android Play Console, including required metadata, privacy declarations, build artifacts, signing validations, compliance checks, and release readiness gates. This template must be consistent with build/signing policies and must not invent store requirements not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
SIGN-01 Build Artifact Types: {{sign.artifacts}}
SIGN-02 Signing Keys Policy: {{sign.keys}} | OPTIONAL
PRD-06 NFRs: {{prd.nfrs}} | OPTIONAL
CSec-02 Data Protection: {{csec.data_protection}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Submission scope (iOS/Android)
Build artifact requirements (what file)
Signing verification steps
Store metadata checklist (name, description, screenshots)
Privacy/security declarations checklist
Permissions declarations checklist (camera/location/etc.)
Testing checklist (smoke/regression/perf)
Compliance gate checklist (required gates passed)
Release notes requirements
Rollback plan reference

Optional Fields
Localization checklist | OPTIONAL

Phased rollout options | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
No submission unless required gates passed (tests, perf, security).
Privacy declarations MUST reflect CSec policies.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. iOS Checklist
   artifact_required: {{ios.artifact_required}}
   signing_check: {{ios.signing_check}}
   metadata:
   app_name: {{ios.meta.app_name}}
   description: {{ios.meta.description}} | OPTIONAL
   screenshots: {{ios.meta.screenshots}}
   privacy_labels: {{ios.meta.privacy_labels}}
   permissions_strings: {{ios.meta.permissions_strings}} | OPTIONAL
   testing: {{ios.testing}}
   gates_required: {{ios.gates_required}}
   release_notes: {{ios.release_notes}}
2. Android Checklist
   artifact_required: {{android.artifact_required}}
   signing_check: {{android.signing_check}}
   metadata:
   app_name: {{android.meta.app_name}}
   description: {{android.meta.description}} | OPTIONAL
   screenshots: {{android.meta.screenshots}}
   data_safety_form: {{android.meta.data_safety_form}}
   permissions: {{android.meta.permissions}} | OPTIONAL
   testing: {{android.testing}}
   gates_required: {{android.gates_required}}
   release_notes: {{android.release_notes}}
3. Compliance Gates
   must_pass: {{gates.must_pass}}
   evidence_required: {{gates.evidence_required}} | OPTIONAL
4. Rollback Plan
   rollback_ref: {{rollback.ref}}
   rollback_steps: {{rollback.steps}} | OPTIONAL
5. References
   Build artifacts: {{xref:SIGN-01}}
   Signing keys: {{xref:SIGN-02}} | OPTIONAL

Data protection: {{xref:CSec-02}} | OPTIONAL
Release channel policy: {{xref:SIGN-05}} | OPTIONAL

Cross-References
Upstream: {{xref:SIGN-01}}, {{xref:SIGN-02}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:SIGN-05}} | OPTIONAL
Standards: {{standards.rules[STD-SECURITY]}} | OPTIONAL,
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Not required.
intermediate: Required. Define iOS + Android checklists and required gates.
advanced: Required. Add phased rollout/localization and evidence capture rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, optional descriptions/permissions strings, evidence required, rollback steps, localization/phased rollout, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If ios.meta.screenshots is UNKNOWN → block Completeness Gate.
If android.meta.data_safety_form is UNKNOWN → block Completeness Gate.
If gates.must_pass is UNKNOWN → block Completeness Gate.
If rollback.ref is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.SIGN
Pass conditions:
required_fields_present == true
ios_and_android_checklists_defined == true
gates_required_defined == true
rollback_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

SIGN-04

SIGN-04 — Versioning Rules (build numbers, semver mapping)
Header Block
template_id: SIGN-04
title: Versioning Rules (build numbers, semver mapping)
type: release_versioning_rules
template_version: 1.0.0
output_path: 10_app/release/SIGN-04_Versioning_Rules.md
compliance_gate_id: TMP-05.PRIMARY.SIGN
upstream_dependencies: ["SIGN-01", "SIGN-03"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "SIGN-01", "SIGN-03"]
required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose
Define the canonical versioning rules across platforms: semantic versioning policy, build number generation, mapping rules per platform, and how version changes relate to releases and rollbacks. This template must be consistent with build artifact types and store submission requirements and must not invent versioning schemes not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
SIGN-01 Build Artifact Types: {{sign.artifacts}} | OPTIONAL
SIGN-03 Store Submission Checklist: {{sign.submission}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
SemVer policy (MAJOR.MINOR.PATCH)
What triggers major/minor/patch bumps
Build number rule (monotonic)
Platform mapping (iOS CFBundleVersion/ShortVersion; Android versionCode/versionName)
Pre-release channel version labeling (beta/rc)
Hotfix versioning rules
Rollback versioning rules
Version source of truth (where set)
Telemetry version fields (app_version, build_number)

Optional Fields
Date-based versioning notes | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Build numbers MUST be monotonically increasing per store rules.
Versioning MUST be deterministic and derived from a single source of truth.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. SemVer Policy
   semver_enabled: {{semver.enabled}}
   bump_rules:
   major: {{semver.major_rule}}
   minor: {{semver.minor_rule}}
   patch: {{semver.patch_rule}}
2. Build Number Rule
   build_number_source: {{build.source}}
   monotonic_rule: {{build.monotonic_rule}}
   format: {{build.format}} | OPTIONAL
3. Platform Mapping
   ios_short_version: {{platform.ios.short_version}}
   ios_build_version: {{platform.ios.build_version}}
   android_version_name: {{platform.android.version_name}}
   android_version_code: {{platform.android.version_code}}
4. Pre-Release Labeling
   beta_label_rule: {{pre.beta_label_rule}} | OPTIONAL
   rc_label_rule: {{pre.rc_label_rule}} | OPTIONAL
5. Hotfix Rules
   hotfix_version_rule: {{hotfix.version_rule}}
   hotfix_branch_rule: {{hotfix.branch_rule}} | OPTIONAL
6. Rollback Rules
   rollback_version_rule: {{rollback.version_rule}}
   rollback_constraints: {{rollback.constraints}} | OPTIONAL
7. Source of Truth
   version_file_or_config: {{sot.location}}
   update_process: {{sot.update_process}} | OPTIONAL
8. Telemetry Fields
   app_version_field: {{telemetry.app_version_field}}
   build_number_field: {{telemetry.build_number_field}} | OPTIONAL
9. References
   Build artifacts: {{xref:SIGN-01}} | OPTIONAL
   Submission checklist: {{xref:SIGN-03}} | OPTIONAL

Cross-References
Upstream: {{xref:SIGN-01}} | OPTIONAL, {{xref:SIGN-03}} | OPTIONAL, {{xref:SPEC_INDEX}} |

OPTIONAL
Downstream: {{xref:SIGN-05}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Not required.
intermediate: Required. Define semver + build number + platform mapping.
advanced: Required. Add pre-release/hotfix/rollback rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, build format, pre-release labels, hotfix branch rules, rollback constraints, update process, build number field, date-based notes, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If semver.enabled is UNKNOWN → block Completeness Gate.
If build.monotonic_rule is UNKNOWN → block Completeness Gate.
If platform mapping fields are UNKNOWN → block Completeness Gate.
If sot.location is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.SIGN
Pass conditions:
required_fields_present == true
semver_and_build_rules_defined == true
platform_mapping_defined == true
source_of_truth_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

SIGN-05

SIGN-05 — Release Channel Policy (internal/beta/GA)
Header Block
template_id: SIGN-05
title: Release Channel Policy (internal/beta/GA)
type: release_channel_policy
template_version: 1.0.0
output_path: 10_app/release/SIGN-05_Release_Channel_Policy.md
compliance_gate_id: TMP-05.PRIMARY.SIGN
upstream_dependencies: ["SIGN-01", "SIGN-03", "SIGN-04"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "SIGN-01", "SIGN-03", "SIGN-04", "CPR-05", "MBAT-05"]
required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose
Define the canonical release channel policy: what channels exist (internal/beta/GA), promotion criteria, gating requirements, rollout strategies, rollback procedures, and roles/responsibilities. This template must be consistent with versioning, store submission, and perf gates and must not invent channels not present in upstream inputs.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
SIGN-01 Build Artifact Types: {{sign.artifacts}}
SIGN-03 Store Submission Checklist: {{sign.submission}} | OPTIONAL
SIGN-04 Versioning Rules: {{sign.versioning}} | OPTIONAL
CPR-05 Perf Regression Gates: {{cpr.perf_gates}} | OPTIONAL
MBAT-05 Mobile Perf Gates: {{mbat.perf_gates}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Channel registry (internal/beta/ga)
Eligibility (who gets each channel)
Promotion criteria (gates, tests)
Approval requirements (roles)
Rollout strategy (phased/canary)
Monitoring window (time)
Rollback criteria and steps
Hotfix lane rules
Telemetry/health checks required (crash rate, ANR, etc.)

Optional Fields
Feature flag default policies per channel | OPTIONAL

Regional phased rollout | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
No promotion unless required gates passed.
Rollouts must be reversible (rollback plan).
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Channels
   channels: {{channels.list}}
   notes: {{channels.notes}} | OPTIONAL
2. Eligibility
   internal_users: {{elig.internal_users}}
   beta_users: {{elig.beta_users}} | OPTIONAL
   ga_users: {{elig.ga_users}} | OPTIONAL
3. Promotion Criteria
   internal_to_beta: {{promote.internal_to_beta}}
   beta_to_ga: {{promote.beta_to_ga}}
   required_gates: {{promote.required_gates}} | OPTIONAL
4. Approvals
   approvers: {{approve.approvers}}
   required_approvals: {{approve.required_approvals}} | OPTIONAL
5. Rollout Strategy
   strategy: {{rollout.strategy}} (phased/canary/UNKNOWN)
   phases: {{rollout.phases}} | OPTIONAL
   kill_switch_policy: {{rollout.kill_switch_policy}} | OPTIONAL
6. Monitoring Window
   monitoring_window: {{monitor.window}}
   signals: {{monitor.signals}} | OPTIONAL
7. Rollback
   rollback_criteria: {{rollback.criteria}}
   rollback_steps: {{rollback.steps}} | OPTIONAL
8. Hotfix Lane
   hotfix_allowed: {{hotfix.allowed}}
   hotfix_rules: {{hotfix.rules}} | OPTIONAL
9. Health Checks
   crash_rate_gate_ref: {{health.crash_rate_gate_ref}} (expected: {{xref:MBAT-05}}/{{xref:CPR-05}}) | OPTIONAL
   anr_gate_ref: {{health.anr_gate_ref}} | OPTIONAL
10. References
    Build artifacts: {{xref:SIGN-01}}

Store submission checklist: {{xref:SIGN-03}} | OPTIONAL
Versioning rules: {{xref:SIGN-04}} | OPTIONAL
Perf gates (web): {{xref:CPR-05}} | OPTIONAL
Perf gates (mobile): {{xref:MBAT-05}} | OPTIONAL

Cross-References
Upstream: {{xref:SIGN-01}}, {{xref:SIGN-04}}, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:RELEASE-GATE}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Not required.
intermediate: Required. Define promotion criteria, approvals, rollout, rollback.
advanced: Required. Add hotfix lane and feature-flag per-channel policies and regional rollout.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, notes, beta/ga eligibility, required gates, approvals counts, rollout phases, kill switch policy, monitor signals, rollback steps, hotfix rules, gate refs, regional rollout, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If channels.list is UNKNOWN → block Completeness Gate.
If promote.beta_to_ga is UNKNOWN → block Completeness Gate.
If rollback.criteria is UNKNOWN → block Completeness Gate.
If monitor.window is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.SIGN
Pass conditions:
required_fields_present == true
channels_defined == true
promotion_and_approvals_defined == true
rollout_and_rollback_defined == true
monitoring_defined == true
placeholder_resolution == true
no_unapproved_unknowns == true

SIGN-06

SIGN-06 — Release Readiness & Go/No-Go Gate
Header Block
template_id: SIGN-06
title: Release Readiness & Go/No-Go Gate
type: release_readiness_gate
template_version: 1.0.0
output_path: 10_app/release/SIGN-06_Release_Readiness_Go_No_Go.md
compliance_gate_id: TMP-05.PRIMARY.SIGN
upstream_dependencies: ["SIGN-01", "SIGN-03", "SIGN-04", "SIGN-05", "CPR-05", "MBAT-05"]
inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX", "SIGN-01", "SIGN-03", "SIGN-04", "SIGN-05", "CPR-05", "MBAT-05", "CER-05"]
required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose
Define the canonical release readiness checklist and the single go/no-go decision gate for promoting a build to the next release channel. This document collects required evidence (tests/gates/compliance/monitoring plan) and defines pass/fail conditions. This template must be consistent with existing signing, versioning, submission, and perf gates and must not invent new gates beyond upstream references.

Inputs Required
SPEC_INDEX: {{spec.index}}
DOMAIN_MAP: {{domain.map}} | OPTIONAL
GLOSSARY: {{glossary.terms}} | OPTIONAL
STANDARDS_INDEX: {{standards.index}} | OPTIONAL
SIGN-01 Build Artifact Types: {{sign.artifacts}} | OPTIONAL
SIGN-03 Store Submission Checklist: {{sign.submission}} | OPTIONAL
SIGN-04 Versioning Rules: {{sign.versioning}} | OPTIONAL
SIGN-05 Release Channel Policy: {{sign.channels}} | OPTIONAL
CPR-05 Perf Regression Gates (web): {{cpr.perf_gates}} | OPTIONAL
MBAT-05 Perf Regression Gates (mobile): {{mbat.perf_gates}} | OPTIONAL
CER-05 Client Logging/Crash Reporting: {{cer.logging}} | OPTIONAL
Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields
Release target (channel + platform)
Build identifiers (app_version, build_number, artifact_id)
Required gates checklist (tests/perf/security)
Evidence bundle pointers (reports/artifacts)
Known issues list (must be empty or explicitly approved)
Rollout plan (phased %, timing)
Monitoring plan (signals + window)
Rollback plan (criteria + steps)

Approvals (who approved, timestamp)
Final go/no-go decision (GO/NO-GO)

Optional Fields
Canary cohort definition | OPTIONAL
Customer/support comms plan | OPTIONAL
Open questions | OPTIONAL

Rules
Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
Do not introduce new gates; reference existing gate IDs/docs only.
If any required evidence is missing, decision MUST be NO-GO unless explicitly allowed by standards.
If a required field is not provided by inputs, write UNKNOWN unless explicitly disallowed.
Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1. Release Target
   channel: {{release.channel}} (internal/beta/ga/UNKNOWN)
   platforms: {{release.platforms}} (web/ios/android/UNKNOWN)
   region_scope: {{release.region_scope}} | OPTIONAL
   rollout_type: {{release.rollout_type}} (all_at_once/phased/canary/UNKNOWN)
2. Build Identification
   app_version: {{build.app_version}}
   build_number: {{build.build_number}}
   artifact_ref: {{build.artifact_ref}} (expected: {{xref:SIGN-01}}) | OPTIONAL
   commit_sha: {{build.commit_sha}} | OPTIONAL
3. Required Gates Checklist
   GATE
   name: {{gates[0].name}}
   ref: {{gates[0].ref}} (e.g., {{xref:CPR-05}} / {{xref:MBAT-05}} / test suite ref)
   status: {{gates[0].status}} (pass/fail/UNKNOWN)
   evidence_ref: {{gates[0].evidence_ref}}
   notes: {{gates[0].notes}} | OPTIONAL

(Repeat per required gate.)

4. Evidence Bundle
   test_reports: {{evidence.test_reports}}
   perf_reports: {{evidence.perf_reports}} | OPTIONAL
   security_checks: {{evidence.security_checks}} | OPTIONAL
   store_submission_evidence: {{evidence.store_submission}} | OPTIONAL
5. Known Issues
   issues:

{{issues[0]}}
{{issues[1]}} | OPTIONAL
approval_to_ship_with_issues: {{issues.approval_to_ship_with_issues}} | OPTIONAL

6. Rollout Plan
   phases: {{rollout.phases}} | OPTIONAL
   start_time: {{rollout.start_time}} | OPTIONAL
   halt_conditions: {{rollout.halt_conditions}}
7. Monitoring Plan
   monitoring_window: {{monitor.window}}
   signals: {{monitor.signals}}
   dashboards: {{monitor.dashboards}} | OPTIONAL
   oncall_owner: {{monitor.oncall_owner}} | OPTIONAL
8. Rollback Plan
   rollback_criteria: {{rollback.criteria}}
   rollback_steps: {{rollback.steps}}
   rollback_owner: {{rollback.owner}} | OPTIONAL
9. Approvals
   approvers: {{approvals.approvers}}
   approval_timestamps: {{approvals.timestamps}} | OPTIONAL
10. Final Decision
    decision: {{decision.go_no_go}} (GO/NO_GO/UNKNOWN)
    decision_time: {{decision.time}} | OPTIONAL
    decision_notes: {{decision.notes}} | OPTIONAL
11. References
    Build artifacts: {{xref:SIGN-01}} | OPTIONAL
    Submission checklist: {{xref:SIGN-03}} | OPTIONAL
    Versioning rules: {{xref:SIGN-04}} | OPTIONAL
    Release channel policy: {{xref:SIGN-05}} | OPTIONAL
    Web perf gates: {{xref:CPR-05}} | OPTIONAL
    Mobile perf gates: {{xref:MBAT-05}} | OPTIONAL
    Logging/crash reporting: {{xref:CER-05}} | OPTIONAL

Cross-References
Upstream: {{xref:SIGN-01}} | OPTIONAL, {{xref:SIGN-05}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
Downstream: {{xref:POST-RELEASE}} | OPTIONAL
Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules
beginner: Not required.
intermediate: Required. Populate build IDs, gate statuses, rollout + monitoring + rollback, and a final GO/NO-GO.
advanced: Required. Add canary cohorts, explicit halt conditions, and evidence traceability rigor.

Unknown Handling
UNKNOWN_ALLOWED: domain.map, glossary.terms, region scope, artifact ref, commit sha, optional evidence, extra issues, issue approval, phased rollout details, dashboards, oncall owner, rollback owner, approval timestamps, decision notes, canary/support comms, open_questions
If any Required Field is UNKNOWN, allow only if:
{{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
If build.app_version is UNKNOWN → block Completeness Gate.
If any required gate status != pass → decision MUST be NO_GO.
If decision.go_no_go is UNKNOWN → block Completeness Gate.

Completeness Gate
Gate ID: TMP-05.PRIMARY.SIGN
Pass conditions:
required_fields_present == true
all_required_gates_passed == true
evidence_refs_present == true
monitoring_and_rollback_defined == true
decision_is_go_or_no_go == true
placeholder_resolution == true
no_unapproved_unknowns == true